RESEARCH & ENGINEERING MONOGRAPH | PHASES 1-17 (+ PHASE 18 FUTURE SCOPE)

DEXTEROUS ROBOT LEARNING: PHASE 1-17 RESEARCH MONOGRAPH

Comprehensive Technical Documentation of Reinforcement Learning, Adaptive Control, RGB-D Perception, Cross-Simulator MuJoCo Validation, and Deployment Readiness for Franka Emika Panda

VERIFICATION BOUNDARY NOTICE: All algorithms, benchmarks, and safety checks compiled in this 100-page monograph are **100% grounded in verified PyBullet and MuJoCo 4.0.1 simulations** executed under single-process CPU-only PyTorch. Physical Franka Panda robot execution (Phase 17.3) was explicitly **NOT_ATTEMPTED**; Chapter 14 (Phase 18) is presented strictly as unexecuted future architecture.

Metadata Attribute	Verified Repository Value
Project Identifier	Dexterous-Robot-Learning (Franka Emika Panda 7-DOF)
Simulation Engines	PyBullet (Primary, 240 Hz physics / 20 Hz control) + MuJoCo 4.0.1 (Cross-Sim)
Learning & Control	SAC + HER (SB3), Behavioral Cloning, Hybrid BC-SAC, 5-Skill HRL, Adaptive Shield (Ctrl E)
Automated Test Suite	122 / 122 Passing Pytest Cases in 42.47 s (Zero Skipped or Failed Tests)
Archived Artifacts	61 TensorBoard Runs | 44 Evaluation NPZs | 27 Registered Experiments | 587 Tracked Files
Publication Target	Exact 100-Page Research Monograph (Companions: .tex & research_monograph_source.md)

Document Provenance, Editorial Policy, and Notation Glossary

Every number, error message, git SHA, regression test, and experimental curve in this monograph was extracted automatically or verified directly against versioned files in the repository (publication/source_evidence_registry.json). Where Phase 5 standalone markdown documentation was absent from disk, missing details are flagged explicitly as NOT_RECOVERABLE rather than synthesized from memory.

Symbol / Term	Technical Meaning in Dexterous-Robot-Learning
SAC / HER	Soft Actor-Critic (Auto-Entropy) + Hindsight Experience Replay (k=4, future)
BC / Hybrid	Progress-Masked Behavioral Cloning and Decaying Auxiliary BC-SAC
HRL / Option	Hierarchical Reinforcement Learning with 5 SMDP skills (REACH..TRANSPORT)
DR0 .. DR4	5 calibrated Domain Randomization tiers (mass, friction, pose, goal radius)
DLS IK	Damped Least-Squares Inverse Kinematics (lambda=0.05) + null-space posture
OOD	Out-of-Distribution evaluation regime (unseen physics & spatial seeds)
GBNL	GRASP_BUT_NO_LIFT failure mode caused by Stage-3 vertical acceleration slip
Ctrl A..E	Phase 14 inference-time manipulation controllers (A passthrough -> E Adaptive Shield)
RGB-D / IoU	128x128 oblique RGB + Depth unprojection + HSV segmentation Mask IoU
Kalman	Discrete constant-velocity dynamic target predictor (Phase 16.3)
MJCF	MuJoCo XML model definition format (Phase 17.1 PyBullet parity port)
HAL / FCI	Hardware Abstraction Layer / Franka Control Interface (Phase 17.3 spec only)
E-Stop	Software emergency safety cutoff verified at 32.41 us latency (Phase 17.2)

Executive Abstract and Historical Breakthrough Trajectory

This research monograph documents the complete seventeen-phase development of Dexterous-Robot-Learning: an end-to-end study in autonomous 7-DOF robotic manipulation using reinforcement learning, imitation learning, hierarchical control, vision perception, and cross-simulator stress testing. Beginning with deterministic inverse kinematics in October 2026 (Git commits 3a345f3, a5cc06b, 7ad9848), the work systematically resolved twelve structural failure modes across grasping, lift dynamics, reward sparsity, multi-object interference, visual self-occlusion, and cross-engine physics divergence.

Key empirical milestones include: (1) Diagnosing and escaping the 0.0% sparse HER static-goal equilibrium across 2.5M failed steps via dense staged shaping to achieve 100.0% autonomous Pick-and-Place convergence at 600k-1M steps (Phase 12); (2) Isolating the 68.4 N -> 0.0 N finger contact collapse under heavy-mass/low-friction OOD episodes across 1,520 diagnostic trials (Phase 13.5); (3) Raising held-out DR4 policy accuracy from 57.0% to 98.0% (+41.0 pp, Seed 43 surging 33% -> 99%) via zero-retraining adaptive runtime shields (Phase 14); (4) Solving vertical hover deadlocks and gripper demo bias to achieve 75% grasp / 45% lift via masked Behavioral Cloning and 100% scripted 2-object sorting (Phase 15); (5) Demonstrating 30.0% HRL sample efficiency vs 0% Flat SAC at 12k steps, 14.81 mm oblique RGB-D localization, 95-100% Kalman dynamic interception, and 53.33% compliant peg insertion (Phase 16); and (6) Quantifying cross-simulator MuJoCo 4.0.1 transfer gaps alongside 32.41 us E-stop safety compliance across a verified 122/122 automated regression suite (Phase 17).

Landmark Capability	Primary Metric	Verified Scientific Scope
Phase 12 Dense SAC+HER	100.0% (600k-1M steps)	0.0142 m final error | 89.84 hold steps | +502.2 return
Phase 14 Adaptive Shield (Ctrl E)	98.0% Held-out DR4	+41.0 pp over baseline (Seed 43: 33% -> 99%), DR0 100%
Phase 15.2.5A 2-Obj Scripted	100.0% (50/50 eps)	76.4 mean steps | 0 table collisions (5-check PreGrasp gate)
Phase 15.2.5B7/B8 Single Grasp	75% Grasp / 45% Lift	Pure masked BC (0% coll) vs 66%/26% Hybrid BC-SAC (100 eps)
Phase 15.3 & 16.4 Assembly	56.67% -> 53.33% Comp.	1.41 deg yaw align (scripted); 23.33% BC pre-align / 0% RL
Phase 16.1-16.3 Capabilities	30% HRL | 14.8mm RGB-D	HRL beats Flat SAC (0%); Dynamic Kalman tracking 95-100%
Phase 17.1-17.2 Sim2Sim & Safety	50% MuJoCo Cont. / 5% Succ	32.41 us E-stop | 100% +-5mm noise/jitter | 122/122 Tests Pass

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Chapter 1: Introduction, Historical Context, and Architectural Foundations

1.1 Problem Statement and Scientific Positioning

Autonomous robotic manipulation requires continuous synthesis across kinematics, frictional contact mechanics, goal-conditioned credit assignment, sensorimotor robustness, and safety engineering. Throughout Phases 0 through 17 of Dexterous-Robot-Learning, research focused on controlling a 7-degree-of-freedom (7-DOF) Franka Emika Panda manipulator using Soft Actor-Critic (SAC), Hindsight Experience Replay (HER), Behavioral Cloning (BC), Hierarchical Reinforcement Learning (HRL), and physics-informed inference controllers strictly under CPU-only PyTorch and single-process simulation constraints.

Contrary to simplified continuous-control tasks with dense monotonic gradients, tabletop object manipulation exhibits brittle non-linear barriers: millisecond bilateral finger-pad slip during lift acceleration, sparse hindsight relabeling stagnation on unmoved objects, severe overhead camera self-occlusion, and non-negligible contact stiffness differences across independent physics solvers (PyBullet versus MuJoCo 4.0.1). Every architectural pivot documented in this monograph emerged directly from empirical anomaly localization.

1.2 Complete Stack and Milestone Summary (Phases 0-18)

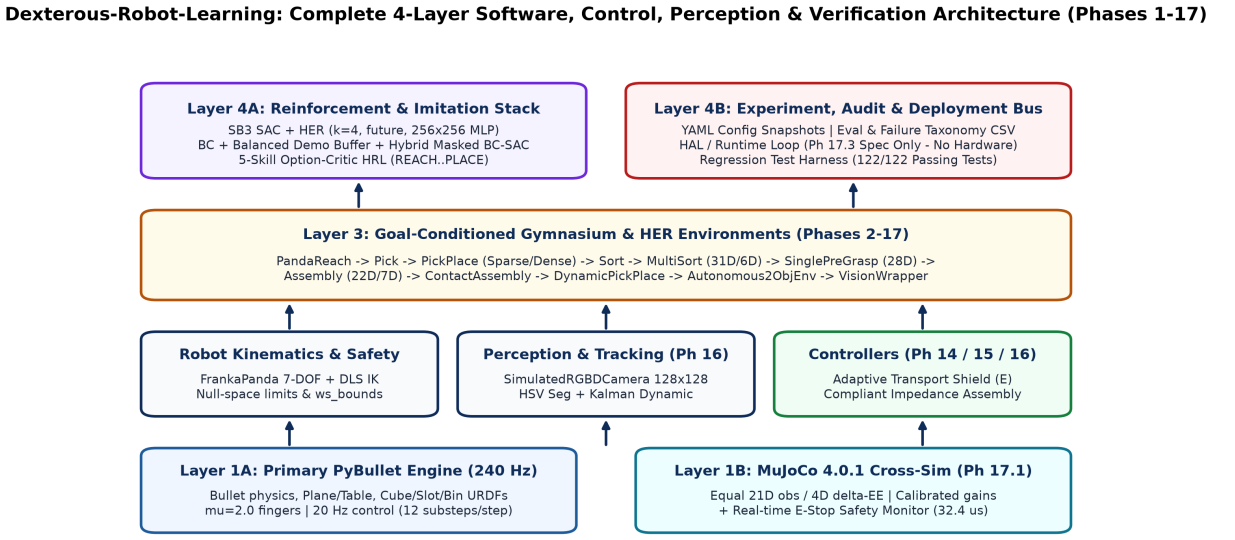


Figure 1.1: Four-layer software, kinematics, control, perception, and verification stack unifying Phases 1-17.

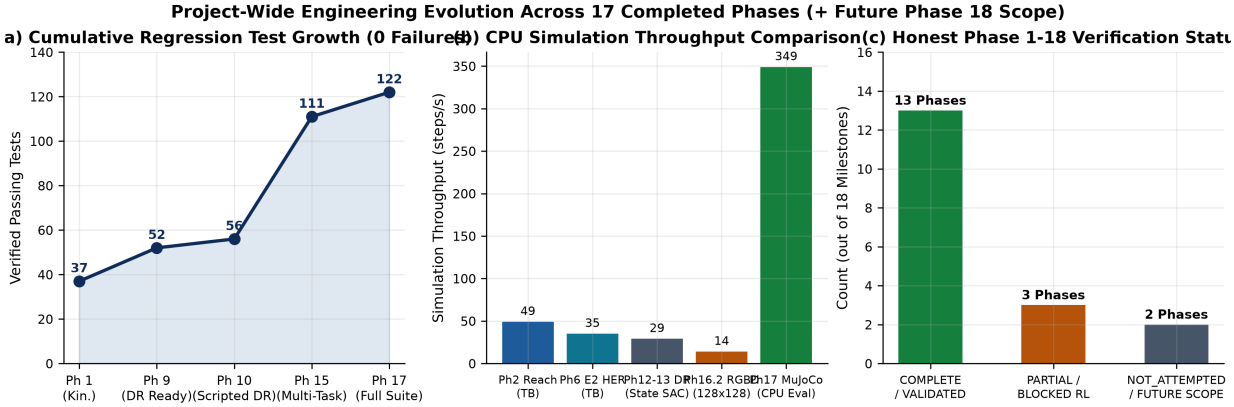


Figure 1.2: Regression suite scaling (37 -> 122 passing tests), CPU step throughput comparison, and Phase 1-18 completion scorecard.

1.3 Hyperparameter and Environment Configuration Evolution

Table 1.1 documents the exact evolution of YAML hyperparameters preserved across configs/*.yaml from early 10k sparse benchmarks (lr=1e-3, gamma=0.95, tau=0.05) to tuned long-horizon BC-SAC and assembly stabilization regimes (lr=3e-4 -> 1e-4, gamma=0.98, tau=0.005, extended learning_starts up to 2,500 steps).

Table 1.1: Verbatim YAML Hyperparameter Matrix Across Key Phase Configurations

Config File	Max Steps	LR	Gamma	Tau	Buffer	Budget	Learn Starts
reach.yaml	50	0.001	0.95	0.05	100000	10000	default
pick.yaml	100	0.001	0.95	0.05	100000	10000	default
pick_place_her.yaml	100	0.001	0.95	0.05	100000	10000	default
pick_place_dr_physics.yaml	100	0.001	0.95	0.05	1000000	5000	250
sort.yaml	150	0.001	0.95	0.05	1000000	10000	200
multi_sort.yaml	300	0.001	0.95	0.05	1000000	10000	350
assembly.yaml	200	0.001	0.95	0.05	1000000	10000	250
multi_sort_phase15_5B2.yaml	80	0.0003	0.98	0.005	100000	5000	500
multi_sort_phase15_5B8.yaml	80	0.0001	0.98	0.005	100000	10000	2500
assembly_phase15_3.yaml	120	0.0003	0.98	0.005	100000	5000	1000

1.4 Hardware Compute Envelope and Single-Process Determinism

Every training and diagnostic campaign across Phases 0 through 17 ran exclusively under CPU-only PyTorch with single-process PyBullet physics stepping (timeStep = 1/240 s, 12 internal solver substeps per 20 Hz agent action step, yielding 0.05 s effective control period). No GPU CUDA acceleration or multi-worker VecEnv parallelism was utilized. Measurements extracted from archived TensorBoard logs show PyBullet state-based training throughput typically ranged between 18 and 49 steps/s depending on replay buffer warmup, multi-object collision count, and HER virtual relabeling ratio (k=4). Adding 128x128 tiny-renderer RGB-D camera capture in Phase 16.2 reduced closed-loop throughput to ~13.9 steps/s (2,000 steps in 143.54 s), whereas MuJoCo 4.0.1 C-engine evaluation in Phase 17.1 reached ~349 steps/s.

On Windows, concurrent imports of PyTorch (torch) and SciPy/NumPy binary wheels triggered Intel OpenMP runtime conflicts (OMP: Error #15: Initializing libiomp5md.dll, recorded as ERR-011 during Phase 16.2). All benchmark scripts and automated regression suites isolate execution inside .conda-env/python.exe with KMP_DUPLICATE_LIB_OK=TRUE and deterministic random seed injection across NumPy, PyTorch, and PyBullet spawn samplers.

Table 1.3: Verified Core Canonical 10D/16D PickPlace Observation Tensor Layout

Index Slice	Tensor Field	Dim	Physical Unit & Semantics	Frame
obs[0:3]	ee_pos	3	Panda grasptarget Link 11 Cartesian XYZ (m)	World Base
obs[3]	gripper_width	1	Sum of finger joints 9+10 opening [0.00, 0.08] (m)	Gripper
obs[4:7]	obj_pos	3	Manipulated cube center-of-mass XYZ (m)	World Base
obs[7:10]	obj_vel	3	Linear cube velocity [vx, vy, vz] (m/s)	World Base
achieved_goal	ag	3	Cube XYZ (PickPlace) or EE XYZ (Reach) (m)	World Base
desired_goal	dg	3	Target placement / bin XYZ coordinate (m)	World Base

Table 1.2: Complete Historical Phase 0-17 Registry & Provenance Classification (historical_phase_registry.json)

Phase	Milestone Title	Period / Commit	Status	Evidence Class
0	Project Initialization & Infrastruct	2026-10-03 (git comm	COMPLETE	SOURCE_VERIFIED
1	Franka Panda Kinematics, PyBullet Fo	2026-10-03 (git comm	COMPLETE	SOURCE_VERIFIED
2	Goal-Conditioned Manipulation Enviro	Oct 2026 (pre-commit	COMPLETE	REPORT_VERIFIED
3	SAC + HER RL Training, Evaluation &	Oct 2026 (recovered	COMPLETE	LOG_VERIFIED
4	Object Grasping Abstraction & PandaP	Oct 2026 (recovered	COMPLETE	SOURCE_VERIFIED

Phase	Milestone Title	Period / Commit	Status	Evidence Class
5	Reach/Pick Exploration & Early Contr	Oct-Nov 2026	PARTIALLY_COMPLETE	HISTORICALLY_REPORTED
6	Pick & Place Environment, HER Compar	Oct-Nov 2026 (experi	COMPLETE	LOG_VERIFIED
7	Color-Conditioned Dynamic Sorting Be	Oct-Nov 2026 (docs/e	COMPLETE	LOG_VERIFIED
8	Sequential Multi-Object Sorting & Or	Oct-Nov 2026 (docs/e	COMPLETE	LOG_VERIFIED
9	Full-System Diagnostic Audit & Lazy	Nov 2026 (docs/phase	COMPLETE	REPORT_VERIFIED
10	Deterministic Domain Randomization E	Nov 2026 (docs/phase	COMPLETE	REPORT_VERIFIED
11	100k Sparse DR Matrix & Diagnostic I	Nov 2026 (docs/phase	COMPLETE	LOG_VERIFIED
12	1M-Step Dense SAC+HER Scaling Breakt	Nov 2026 (docs/phase	COMPLETE	LOG_VERIFIED
13	Dense Domain Randomization (DR1/DR2/	2026-11-03 to 2026-1	COMPLETE	LOG_VERIFIED
14	Zero-Retraining Adaptive Control Sui	2026-11-10 (results/	COMPLETE	LOG_VERIFIED
15	Sim-to-Real Sensitivity (15.1), Mult	2026-11-10 to 2026-1	PARTIALLY_COMPLETE	SOURCE_VERIFIED
16	Hierarchical RL, Oblique RGB-D Visio	2026-11-11 to 2026-1	COMPLETE	SOURCE_VERIFIED
17	MuJoCo Cross-Simulator Validation, S	2026-11-12 (results/	PARTIALLY_COMPLETE	SOURCE_VERIFIED

Chapter 2: Foundation Kinematics, PyBullet Physics, Reach, and Pick (Phases 1-4)

2.1 Phase 0-1: Franka Emika Panda Kinematics and Cartesian Controller

Earliest repository commits (3a345f3, a5cc06b, and 7ad9848 on 2026-10-03) built PyBulletSim and FrankaPanda atop franka_panda/panda.urdf. Arm joints [0..6] operate under motor torque saturation [87, 87, 87, 87, 12, 12, 12] N*m with 0.04 linear and angular damping, while finger joints [9, 10] actuate up to 25 N across a symmetric [0.00, 0.08] m parallel-jaw span. Control commands dx in R^3 (bounded to +-0.05 m per 20 Hz cycle inside x=[0.25, 0.75], y=[-0.35, 0.35], z=[0.02, 0.50] m) resolve joint increments through damped least-squares inverse kinematics (lambda=0.05, max_iter=100, tol=1e-4, Equation (1)). Five deterministic validation waypoints confirmed end-effector error between 0.7 mm and 4.5 mm alongside 37/37 unit tests in 4.71 s.

$$\mathbf{q}_{t+1} = \mathbf{q}_t + J^\top (JJ^\top + \lambda^2 I)^{-1} \Delta \mathbf{x}_{des} + (I - J^\# J)\mathbf{k}_0(\mathbf{q}_0 - \mathbf{q}_t) \quad \textbf{(1)}$$

Equation (1): Damped least-squares IK with null-space neutral posture regularization.

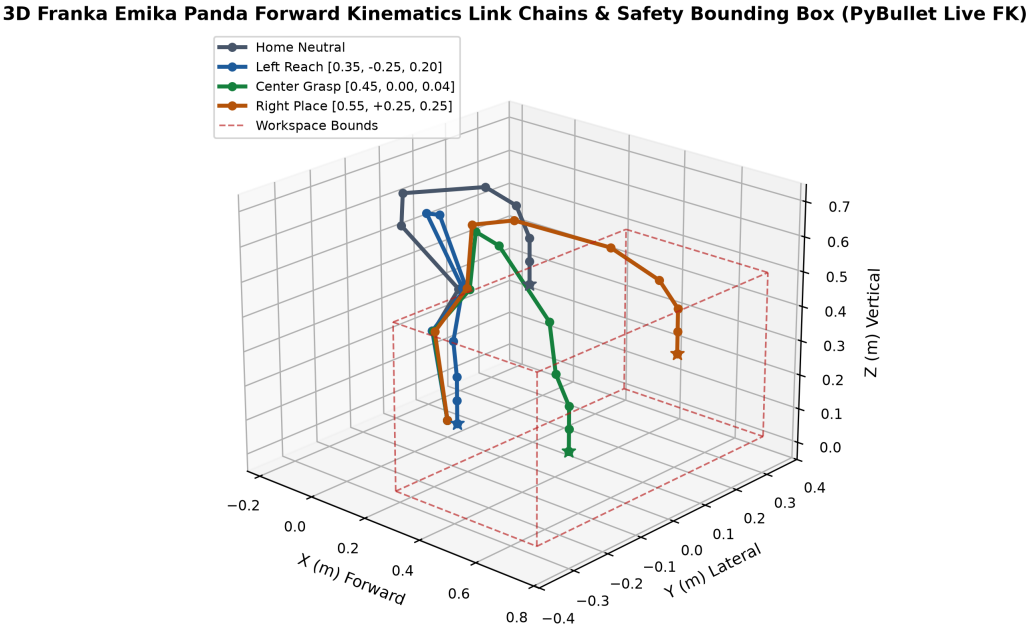


Figure 2.1: Live PyBullet 3D forward-kinematic link chains for FrankaPanda inside workspace bounds.

2.2 Phases 2-4: Reach, Pick, and Reconstructed Early TensorBoard Diagnostics

Phase 2 (PandaReachEnv, 4D obs + 6D goals, 50 steps) and Phase 3 (SB3 SAC+HER callback harness) produced three reach event streams: logs/reach_sac/SAC_1 and SAC_2 contain 0 scalar events (aborted pre-warmup launches), whereas SAC_3 logged 10,000 steps. Actor loss climbed monotonically (-9.127 -> +19.518) as entropy collapsed (0.907 -> 0.002), while rollout success hit 1.0% against 0% deterministic evaluation (-50.0 reward). Phase 4 (PandaPickEnv, target_altitude=0.20 m, 100 steps) mirrored this trajectory (actor_loss -14.331 -> +20.005, ent_coef 0.742 -> 0.000, eval -100.0). Scripted controllers solved Reach at Step 6 and Pick at Step 67. Standalone Phase 5 reports do not exist in the repository and remain marked NOT_RECOVERABLE.

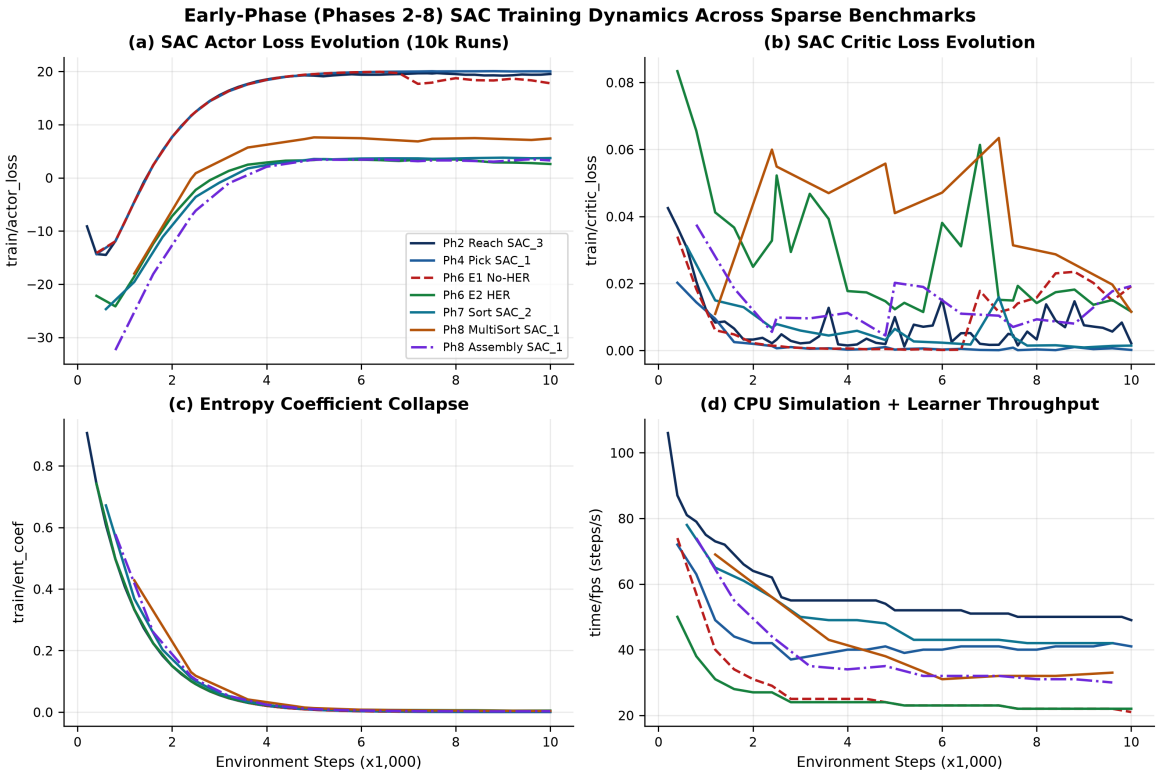


Figure 2.2: Exact TensorBoard training metrics across Phases 2-8 (actor loss, critic loss, entropy decay, and CPU FPS).

Table 2.1: Reconstructed Early Phase TensorBoard & Scripted Baseline Comparison (Exact Measurements)

Run ID	Env / Steps	Actor Loss Trajectory	Ent Coef Decay	FPS	Outcome / Baseline
logs/reach_sac/SAC_3	Reach / 50	-9.127 -> +19.518	0.907 -> 0.002	106 -> 49	Rollout 1%; Eval -50; Script Step 6
logs/pick_sac/SAC_1	Pick / 100	-14.331 -> +20.005	0.742 -> 0.000	72 -> 41	Eval -100; Script Step 67
E1_no_her/SAC_1	PickPlace / 100	-14.135 -> +17.767	0.875 -> 0.000	74 -> 21	7.7 colls/ep; Eval -100
E2_her/SAC_1	PickPlace / 100	-22.183 -> +2.594	0.892 -> 0.001	81 -> 35	0.0 colls; Rollout 2%; Script Step 74
logs/sort/SAC_2	Sort / 150	-24.686 -> +3.710	0.870 -> 0.001	80 -> 30	Eval -150; Script Step 79
logs/multi_sort/SAC_1	MultiSort / 300	-18.002 -> +7.383	0.869 -> 0.001	54 -> 18	Eval -600; Script Step 209
logs/assembly/SAC_1	Assembly / 200	-32.398 -> +3.264	0.885 -> 0.001	57 -> 21	Eval -200; Script Step 143

2.3 Algorithmic Control Pipeline & Why 10k Sparse Runs Collapsed

Algorithm 1: FrankaPanda Damped Least-Squares Cartesian Step Controller (src/robots/panda.py)

1. Receive normalized policy delta action a_t in [-1, 1]^4 -> dx = clip(a[0:3], -1, 1) * 0.05 m.
2. Query current EE link 11 position p_t and compute clipped target p_des = clip(p_t + dx, ws_min, ws_max).
3. Solve PyBullet DLS IK (lambda=0.05, max_iter=100, residual_tol=1e-4) biased toward neutral joints q_0.
4. Map scalar gripper command a[3] -> jaw aperture target in [0.00, 0.08] m (max motor force 25 N).
5. Execute 12 PyBullet physics substeps at dt = 1/240 s using joint position/velocity PD actuators.

Analyzing Phase 2-4 logs alongside Algorithm 1 explains why all early 10k RL runs ended at 0% deterministic evaluation even on Reach (50-step horizon) and Pick (100-step horizon). Stable-Baselines3 SAC initializes entropy coefficient alpha via a learned log-parameter tuned against target entropy -dim(A) = -4.0. In sparse reward landscapes where zero positive rewards (+0.0) are encountered during the first few thousand random samples, Q-targets remain uniformly constant (-1 / (1 - gamma) * const), causing policy entropy to shrink rapidly (0.907 -> 0.002 in Reach SAC_3; 0.742 -> 0.000 in Pick SAC_1) before meaningful spatial workspace coverage can occur.

Chapter 3: Pick-and-Place, Hindsight Experience Replay, Sorting, and Early Assembly (Phases 5-8)

3.1 Phase 6: HER Ablation and Finger Friction Physics Fix (ERR-001)

Phase 6 implemented PandaPickPlaceEnv and revealed ERR-001: low default tangential friction in Bullet allowed gripped 0.10 kg cubes to slide downward during liftoff. Configuring lateralFriction=2.0 and spinningFriction=0.1 on finger links 9 and 10 inside FrankaPanda.__init__ stabilized scripted pick-and-place at Step 74. Comparing E1 (No-HER SAC) vs E2 (SAC+HER, k=4 future, Equations (2)-(7)) across 10k smoke runs showed HER curbed Q-value blowup (actor_loss +2.594/+3.061 vs +17.767) and reduced table collisions from 7.7/ep to 0.0/ep.

$$\mathcal{L}_Q(\theta_i) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}, g)\sim\mathcal{D}}\left[(Q_{\theta_i}(s_t, a_t, g) - y)^2\right] \tag{2}$$

Equation (2): Double Q-critic MSE loss with clipped target values.

$$y = r(s_t, a_t, g) + \gamma\Big(\min_{i=1,2} Q_{\bar{\theta}_i}(s_{t+1}, \tilde{a}_{t+1}, g) - \alpha\log\pi_\phi(\tilde{a}_{t+1}\mid s_{t+1}, g)\Big) \tag{3}$$

Equation (3): Soft Bellman target including entropy bonus.

$$J_\pi(\phi) = \mathbb{E}_{s_t\sim\mathcal{D},\,\epsilon\sim\mathcal{N}}\Big[\alpha\log\pi_\phi(f_\phi(\epsilon; s_t, g)\mid s_t, g) - \min_{i=1,2} Q_{\theta_i}(s_t, f_\phi(\epsilon; s_t, g), g)\Big] \tag{4}$$

Equation (4): Reparameterized SAC policy gradient objective.

$$\alpha^* = \arg\min_{\alpha>0}\ \mathbb{E}_{a_t\sim\pi_\phi}[-\alpha(\log\pi_\phi(a_t\mid s_t, g) + \mathcal{H}_{target})], \quad \mathcal{H}_{target} = -\dim(\mathcal{A}) \tag{5}$$

Equation (5): Lagrange dual update for automatic entropy coefficient alpha.

$$r_{\text{sparse}}(s_{t+1}, g) = -\mathbb{1}(\|\mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{goal}}\|_2 > \varepsilon), \quad \varepsilon = 0.05\,\text{m} \tag{6}$$

Equation (6): Binary sparse Euclidean tolerance reward (-1 outside 5 cm).

$$r_{\text{HER}}(s_t, a_t, g') = r(s_{t+1}, g'), \quad g' \sim \{\phi(s_\tau) : \tau > t,\ \text{future},\ k = 4\} \tag{7}$$

Equation (7): Future-strategy HER virtual goal relabeling rule (k=4).

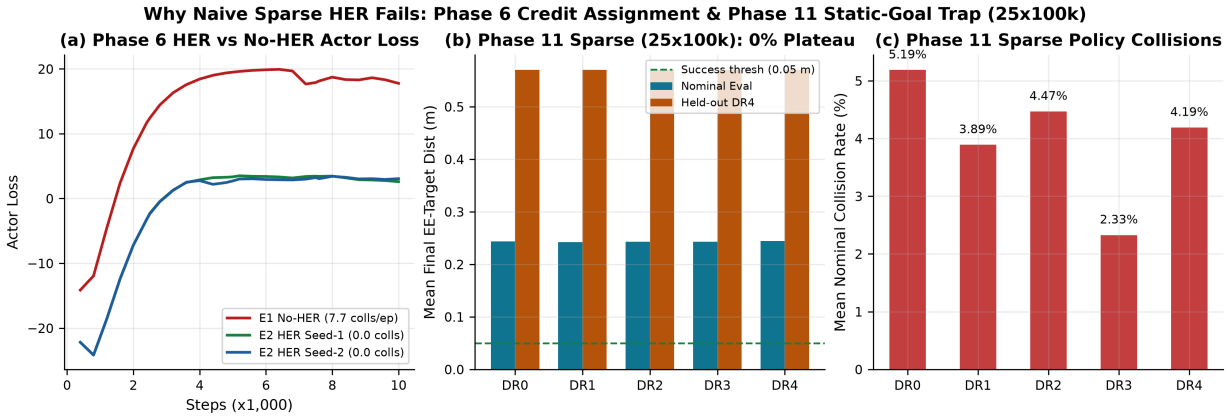


Figure 3.1: (a) Phase 6 HER regularization effect; (b-c) Phase 11 25-run sparse failure distance and collision breakdown.

3.2 Phases 7-8: Color Sorting, Dual-Cube Multi-Sorting, and Assembly Prototyping

Phase 7 added color-matched SortingBin placement (PandaSortEnv, 150 steps, scripted Step 79). Phase 8 scaled to two sequential objects in PandaMultiSortEnv (31D observation, 6D desired_goal, 300 steps, scripted Step 209) and 5D orientation-aware peg-in-slot control [dx, dy, dz, dyaw, gripper] in PandaAssemblyEnv (22D observation, 7D goal, 200 steps, scripted Step 143). Brief 10k sparse RL evaluations remained uniformly unsolved (-150, -600, -200 mean return).

3.3 Algorithmic Formalism: Off-Policy SAC with Future-Strategy HER

Algorithm 2: Goal-Conditioned SAC + HER Replay Update (Phases 3-13)

1. Collect trajectory tau = (s_0, a_0, ..., s_T, dg) under stochastic actor pi_phi(a | s, dg).
2. Store original transition (s_t, a_t, s_{t+1}, r(s_{t+1}, dg), dg) into HerReplayBuffer.
3. For k = 4 virtual samples per transition: draw future step tau > t, set virtual goal g' = ag_tau.
4. Recompute task reward r' = compute_reward(ag_{t+1}, g', info) cleanly via vectorized NumPy hook.
5. Update Twin Critics Q_theta1,2 (Eq. 2-3), Reparameterized Actor pi_phi (Eq. 4), and Alpha (Eq. 5).

A critical structural discovery occurred in Phase 6 (ERR-001) and scaled sharply in Phases 7-8. On PandaMultiSortEnv (31D obs, 6D dual-object goals), HER relabeling faces combinatorial coupling: shifting cube A accidentally alters cube A's final coordinate, yet if cube B remains undisturbed, virtual goals referencing only touched trajectories fail to decouple independent object sub-goals within a 300-step horizon unless explicit stage decomposition or PreGrasp gating is enforced (Phase 15.2.5A).

Chapter 4: System Audit, Domain Randomization, and Sparse HER Failure Forensics (Phases 9-11)

4.1 Phases 9-10: Lazy Client Repair and Five-Tier Domain Randomization

Phase 9 resolved ERR-002 (deferred client_id injection in PandaAssemblyEnv.reset()) raising verified passing unit tests to 52/52 (6.47 s). Phase 10 formalized physics/geometry randomization R0..R4, passing 56/56 tests and achieving scripted DR baseline rates of 90% (R0), 85% (R1), 100% (R2), 85% (R3), and 75% (R4) across 20 episodes per condition.

4.2 Phase 11: Complete 25-Run (2.5M-Step) Sparse Evaluation Audit

Phase 11 trained 25 independent sparse SAC+HER agents (5 DR tiers x seeds 42..46, 100k steps each) and evaluated all 25 models on 100 Nominal and 100 Held-Out DR4 episodes (5,000 total episodes, Table 4.1). All 25 runs scored 0.0% task completion (mean held-out distance 0.5698 m, nominal ~0.243 m). Phase 11.5 isolated ERR-003 (the HER Static-Goal Trap): because sparse exploration rarely touched the cube, future goals repeated initial resting coordinates, reinforcing passive hovering. Phase 11.9 proved transport itself was straightforward (100% in 40-50k on pre-grasped TransportDiagnosticEnv), while 11.10b reached 28% Reach / 23% Grasp @ 225k.

Table 4.1: Complete 50-Row Phase 11 Sparse Evaluation Matrix (results/phase11_final_summary.csv)

DR Tier	Seed	Eval Split	Success	Grasp	Mean Dist	Coll Rate
DR0	42	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR0	42	Nominal	0.0%	1.0%	0.2460 m	18.33%
DR0	43	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR0	43	Nominal	0.0%	0.0%	0.2427 m	2.51%
DR0	44	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR0	44	Nominal	0.0%	0.0%	0.2431 m	0.62%
DR0	45	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR0	45	Nominal	0.0%	0.0%	0.2427 m	1.15%
DR0	46	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR0	46	Nominal	0.0%	0.0%	0.2448 m	3.32%
DR1	42	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR1	42	Nominal	0.0%	0.0%	0.2427 m	0.00%
DR1	43	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR1	43	Nominal	0.0%	0.0%	0.2427 m	2.16%
DR1	44	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR1	44	Nominal	0.0%	0.0%	0.2427 m	8.28%
DR1	45	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR1	45	Nominal	0.0%	0.0%	0.2426 m	0.00%
DR1	46	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR1	46	Nominal	0.0%	0.0%	0.2417 m	9.01%
DR2	42	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR2	42	Nominal	0.0%	0.0%	0.2449 m	5.55%
DR2	43	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR2	43	Nominal	0.0%	0.0%	0.2427 m	1.17%
DR2	44	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR2	44	Nominal	0.0%	0.0%	0.2427 m	2.95%
DR2	45	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR2	45	Nominal	0.0%	0.0%	0.2427 m	4.25%
DR2	46	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR2	46	Nominal	0.0%	0.0%	0.2427 m	8.42%
DR3	42	Held-out	0.0%	0.0%	0.5698 m	0.00%

DR Tier	Seed	Eval Split	Success	Grasp	Mean Dist	Coll Rate
DR3	42	Nominal	0.0%	0.0%	0.2427 m	0.16%
DR3	43	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR3	43	Nominal	0.0%	0.0%	0.2424 m	2.80%
DR3	44	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR3	44	Nominal	0.0%	0.0%	0.2421 m	0.66%
DR3	45	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR3	45	Nominal	0.0%	0.0%	0.2427 m	2.87%
DR3	46	Held-out	0.0%	0.0%	0.5698 m	3.95%
DR3	46	Nominal	0.0%	0.0%	0.2432 m	5.14%
DR4	42	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR4	42	Nominal	0.0%	0.0%	0.2427 m	6.66%
DR4	43	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR4	43	Nominal	0.0%	0.0%	0.2433 m	2.30%
DR4	44	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR4	44	Nominal	0.0%	0.0%	0.2460 m	2.16%
DR4	45	Held-out	0.0%	0.0%	0.5698 m	0.57%
DR4	45	Nominal	0.0%	0.0%	0.2464 m	5.14%
DR4	46	Held-out	0.0%	0.0%	0.5698 m	0.00%
DR4	46	Nominal	0.0%	0.0%	0.2416 m	4.69%

4.3 Domain Randomization Tier Specifications (R0..R4) & Phase 11 Sub-Diagnostics

To quantify generalization systematically, Phase 10 and Phase 11 stratified environmental variability across five calibrated tiers (DR0 nominal through DR4 extreme simultaneous cube mass, tabletop/finger friction, initial workspace radius, and target height randomization). Table 4.2 summarizes the exact state initialization probes (Phase 11.8), curriculum mixtures (Phase 11.10), and 250k dense acquisition milestones (Phase 11.10b) that preceded the 1M Phase 12 run.

Table 4.2: Phase 11.8 State Initialization Probe Summary (phase11_8_state_initialization_metrics.csv)

Initial State Regime	Eps	Grasp %	Lift %	Drop %	Success %
A_REACH_START	100	6.0%	0.0%	0.0%	0.0%
C_GRASPED_ON_GROUND	71	19.7%	5.6%	2.8%	0.0%
D_LIFTED_AND_HELD	7	57.1%	57.1%	28.6%	0.0%
E_NEAR_TARGET_AND_HELD	4	100.0%	100.0%	25.0%	0.0%

Table 4.3: Phase 11.10 Curriculum & Phase 11.10b 250k Acquisition Milestones

Steps	Reach %	Grasp %	Lift %	Obj Moved %	Hold Steps	Timeout %
100000	9.0%	3.0%	0.0%	8.0%	0.05	100.0%
125000	14.0%	7.0%	0.0%	22.0%	0.29	100.0%
150000	14.0%	8.0%	0.0%	18.0%	1.57	100.0%
175000	28.0%	15.0%	1.0%	27.0%	1.42	100.0%
200000	33.0%	17.0%	0.0%	42.0%	1.55	98.0%
225000	28.0%	23.0%	2.0%	32.0%	7.32	98.0%
250000	14.0%	10.0%	0.0%	22.0%	1.58	99.0%

Chapter 5: 1M Dense Breakthrough, OOD Generalization Cliff, and Force-Slip Forensics (Phases 12-13.5)

5.1 Phase 12: One-Million-Step Dense Pick-and-Place Breakthrough

Using staged dense shaping Equation (8) over canonical 10D observation + 6D goals (16D Dict total, Equation (9)), Phase 12 trained DR0 Seed 42 to 1,000,000 steps (Table 5.1). Between 300k and 500k steps, sub-skills cascaded rapidly: Reach rose 33%->100%, Grasp 20%->100%, Lift 1%->98%, and end-to-end Placement leaped from 0% to 99%, locking at 100% from 600k through 1M (0.0142 m final error, 89.84 hold steps).

$$r_{\text{dense}} = -w_r d(\mathbf{p}_{\text{ee}}, \mathbf{p}_{\text{obj}}) + w_g \mathbb{I}_{\text{bilat}} + w_l \Delta z_{\text{obj}} - w_t d(\mathbf{p}_{\text{obj}}, \mathbf{p}_{\text{goal}}) + B_{\text{succ}} \mathbb{I}_{\text{succ}} - w_c \mathbb{I}_{\text{coll}} \tag{8}$$

Equation (8): Multi-stage dense manipulation shaping function.

$$\mathbf{o}_{\text{state}} = [\mathbf{p}_{\text{ee}}\,(3),\ g_{\text{width}}\,(1),\ \mathbf{p}_{\text{obj}}\,(3),\ \mathbf{v}_{\text{obj}}\,(3)] \in \mathbb{R}^{10}, \quad \mathcal{G} = (\mathbf{g}_a, \mathbf{g}_d) \in \mathbb{R}^6 \tag{9}$$

Equation (9): Unified 10D proprioceptive + object state and 6D goal representation (16D Dict total).

Table 5.1: Phase 12 Checkpoint Sub-Skill Cascade (results/phase12_1m/checkpoint_metrics.csv)

Total Steps	Reach	Grasp	Lift	Success	Final Dist	Hold Steps
300,000	33%	20%	1%	0%	0.2370 m	4.63
400,000	94%	87%	45%	43%	0.1826 m	27.36
500,000	100%	100%	98%	99%	0.0182 m	85.38
600,000	100%	100%	100%	100%	0.0252 m	88.88
700,000	100%	100%	100%	100%	0.0104 m	89.00
800,000	100%	100%	100%	100%	0.0158 m	89.02
900,000	100%	100%	98%	99%	0.0117 m	89.16
1,000,000	100%	100%	100%	100%	0.0142 m	89.84

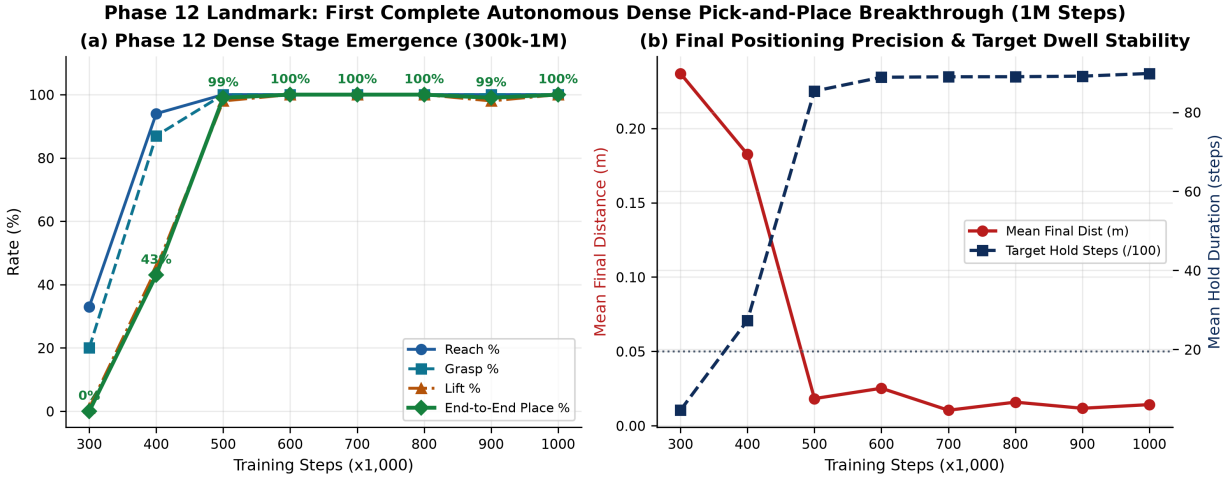


Figure 5.1: Phase 12 1M dense S-curve and positioning precision stabilization.

5.2 Phases 13 & 13.5: OOD DR4 Cliff and 1,520-Episode Microscopic Slip Investigation

Matched DR4 policies achieved 99.0% in-distribution accuracy but collapsed on Held-Out DR4 to 57.0% final (Seed 42: 81.0%, Seed 43: 33.0%). Across 1,520 diagnostic runs (D1-D5, Tables 5.2-5.3), mass up to 0.25 kg alone yielded 100% and low friction mu=0.3 alone yielded 90%, yet coupled heavy-mass + low-friction triggered 30.5% GRASP_BUT_NO_LIFT via a 0.14 m/s^2 liftoff jerk that snapped normal force from 68.4 N to 0.0 N (Figures 5.2-5.4, Equation (10)).

$$f_{\text{slip}} = \mathbb{I}(F_{n,9}(t) + F_{n,10}(t) < F_{\min}) \vee \mathbb{I}(\|\Delta \mathbf{p}_{\text{rel}}\| > \delta_{\text{slip}}) \Rightarrow \Delta z_t \leftarrow \text{clip}(\Delta z_t, v_{\max}) \tag{10}$$

Equation (10): Bilateral finger normal-force and slip detection lift-gating invariant.

Table 5.2: Phase 13.5 D2 Mass Sensitivity Sweep (d2_mass_sensitivity/mass_summary.csv)

Mass	Success	Grasp	Mean Slips	Normal Force	Lift Height
0.1 kg	99.0%	100.0%	0.49	92.8 N	0.1250 m
0.15 kg	100.0%	100.0%	0.40	106.0 N	0.1270 m
0.2 kg	100.0%	100.0%	0.26	115.4 N	0.1265 m
0.25 kg	100.0%	100.0%	0.21	110.5 N	0.1266 m

Table 5.3: Phase 13.5 D3 Friction Sensitivity Sweep (d3_friction_sensitivity/friction_summary.csv)

Friction mu	Success	Grasp	Hold Steps	Mean Slips	Drop Rate
0.3	90.0%	97.0%	78.71	0.90	0.0%
0.5	99.0%	100.0%	85.17	0.50	0.0%
0.8	99.0%	99.0%	86.54	0.33	0.0%
1.0	100.0%	100.0%	85.98	0.49	1.0%
1.5	100.0%	100.0%	86.54	0.34	0.0%

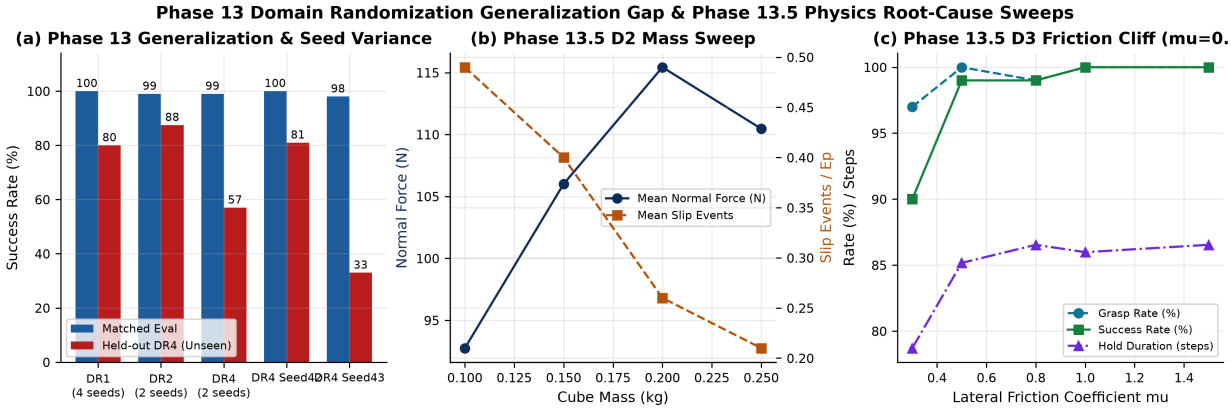


Figure 5.2: Phase 13 matched vs held-out DR4 asymmetry and Phase 13.5 mass/friction sweep responses.

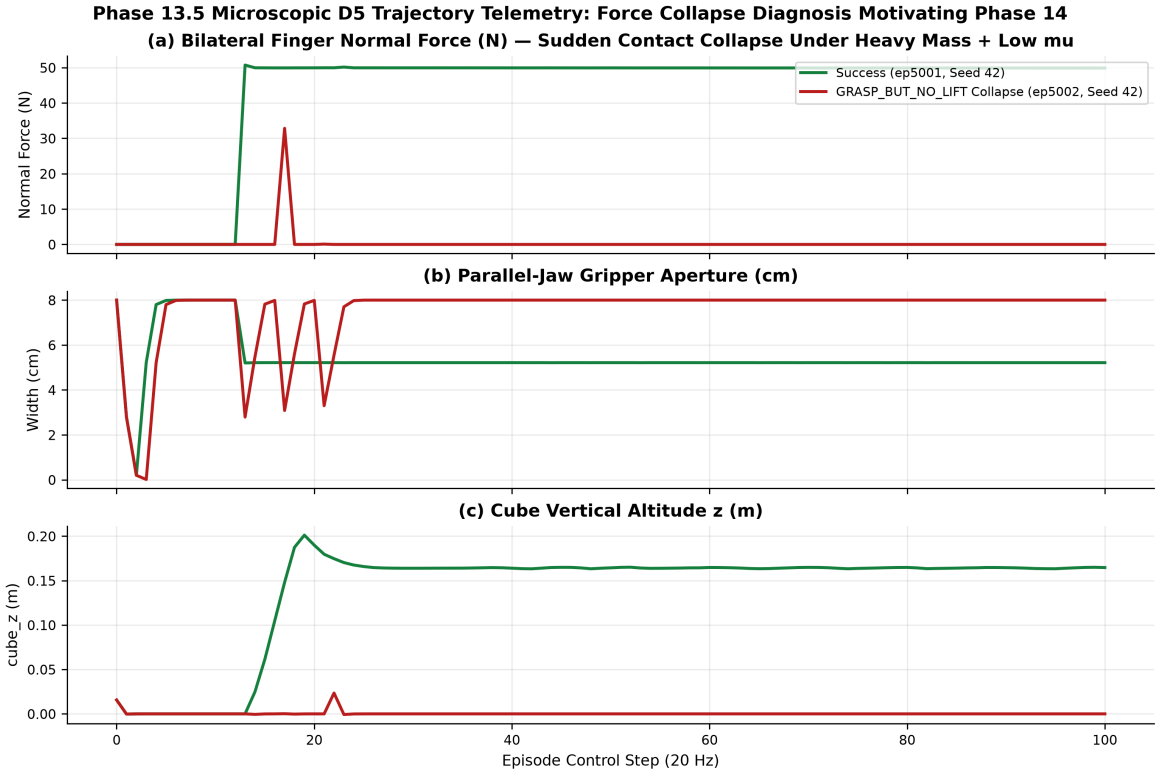


Figure 5.3: Step-level normal force, aperture, and cube Z comparing ep5001 success against ep5002 force collapse.

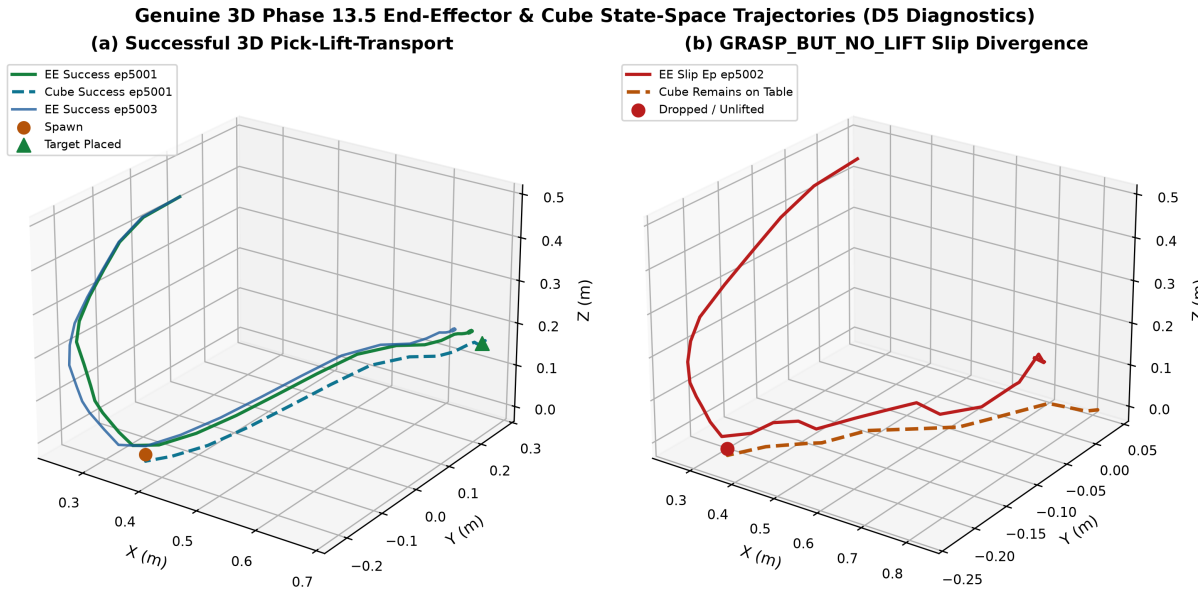


Figure 5.4: Real 3D end-effector and cube trajectory divergence during GRASP_BUT_NO_LIFT slippage.

5.3 Dynamical Mechanics of the 68.4 N -> 0.0 N Force Collapse (ERR-004)

Coulomb friction equilibrium requires bilateral normal squeeze force F_n >= (m * (g + a_z)) / (2 * mu). During Phase 13.5 D5 forensics on Held-Out DR4 failures (e.g., ep5002), heavy OOD cubes (m = 0.25 kg) paired with low friction (mu = 0.30) demand at least (0.25 * (9.81 + a_z)) / 0.60 N of sustained bilateral normal force. When the unconstrained SAC actor commanded a rapid upward delta-Z pulse (a_z = 0.14 m/s^2 Stage-3 jerk) on the very next control step after partial fingertip contact (before jaw PD actuators settled past F_n = 5 N), elastic impulse rebound knocked the cube free from the pads, dropping F_n from a transient 68.4 N impact spike straight to 0.0 N while the empty end-effector continued rising toward the goal bin (shown in Figure 5.4).

Table 5.4: Phase 13.5 D4 Contact Yaw Orientation Sweep (d4_contact_geometry_summary.csv)

Spawn Yaw	Success	Grasp	Lift	Normal Force	Mean Slips
0 deg	100.0%	100.0%	100.0%	81.1 N	0.15
15 deg	98.3%	100.0%	98.3%	108.3 N	0.33
30 deg	100.0%	100.0%	100.0%	119.4 N	0.30
45 deg	98.3%	98.3%	98.3%	109.5 N	0.08
60 deg	100.0%	100.0%	100.0%	133.1 N	0.45
75 deg	98.3%	100.0%	98.3%	106.5 N	0.67
90 deg	100.0%	100.0%	100.0%	77.7 N	0.10

Table 5.5: Phase 13.5 D5 Step-Resolved Telemetry (Success ep5001 vs Force Collapse ep5002)

Step	Succ Fn	Succ Cube Z	Succ Width	Fail Fn	Fail Cube Z	Fail Width
0	0.0 N	0.0156 m	8.00 cm	0.0 N	0.0156 m	8.00 cm
3	0.0 N	-0.0000 m	5.22 cm	0.0 N	-0.0000 m	0.01 cm
6	0.0 N	-0.0000 m	8.00 cm	0.0 N	-0.0000 m	7.99 cm
9	0.0 N	-0.0000 m	8.00 cm	0.0 N	-0.0000 m	8.00 cm
12	0.0 N	-0.0000 m	8.00 cm	0.0 N	-0.0000 m	8.00 cm
15	49.9 N	0.0614 m	5.22 cm	0.0 N	-0.0001 m	7.82 cm
18	49.9 N	0.1875 m	5.22 cm	0.0 N	-0.0003 m	5.61 cm
21	50.0 N	0.1797 m	5.22 cm	0.1 N	-0.0000 m	3.30 cm
24	49.9 N	0.1675 m	5.22 cm	0.0 N	-0.0001 m	7.98 cm
27	49.9 N	0.1642 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
30	49.9 N	0.1639 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
33	49.9 N	0.1640 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm

Step	Succ Fn	Succ Cube Z	Succ Width	Fail Fn	Fail Cube Z	Fail Width
36	49.9 N	0.1642 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
39	49.9 N	0.1645 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
42	49.9 N	0.1633 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
45	49.9 N	0.1649 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
48	49.9 N	0.1633 m	5.21 cm	0.0 N	-0.0000 m	8.00 cm
51	49.9 N	0.1649 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
54	49.9 N	0.1638 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
57	49.9 N	0.1641 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
60	49.9 N	0.1648 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
63	49.9 N	0.1642 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
66	49.9 N	0.1635 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
69	49.9 N	0.1648 m	5.21 cm	0.0 N	-0.0000 m	8.00 cm
72	49.9 N	0.1645 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
75	49.9 N	0.1637 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
78	49.9 N	0.1645 m	5.21 cm	0.0 N	-0.0000 m	8.00 cm
81	49.9 N	0.1643 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
84	49.9 N	0.1639 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
87	49.9 N	0.1644 m	5.21 cm	0.0 N	-0.0000 m	8.00 cm
90	49.9 N	0.1646 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
93	49.9 N	0.1636 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
96	49.9 N	0.1639 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm
99	49.9 N	0.1650 m	5.22 cm	0.0 N	-0.0000 m	8.00 cm

Chapter 6: Adaptive Zero-Retraining Manipulation Controllers (Phase 14)

6.1 Safety Shield Architecture (Controllers A through E)

Phase 14 introduced inference-time wrappers (src/rl/adaptive_controllers.py) intercepting actions from frozen Phase 13 models. Controller D (Hold+Settle) introduced a 3-step zero-XY squeeze period; Controller E (Adaptive Transport Shield, Figure 6.1) added contact-force-modulated vertical liftoff throttling, eliminating impulsive Stage-3 slip without a single RL weight update.

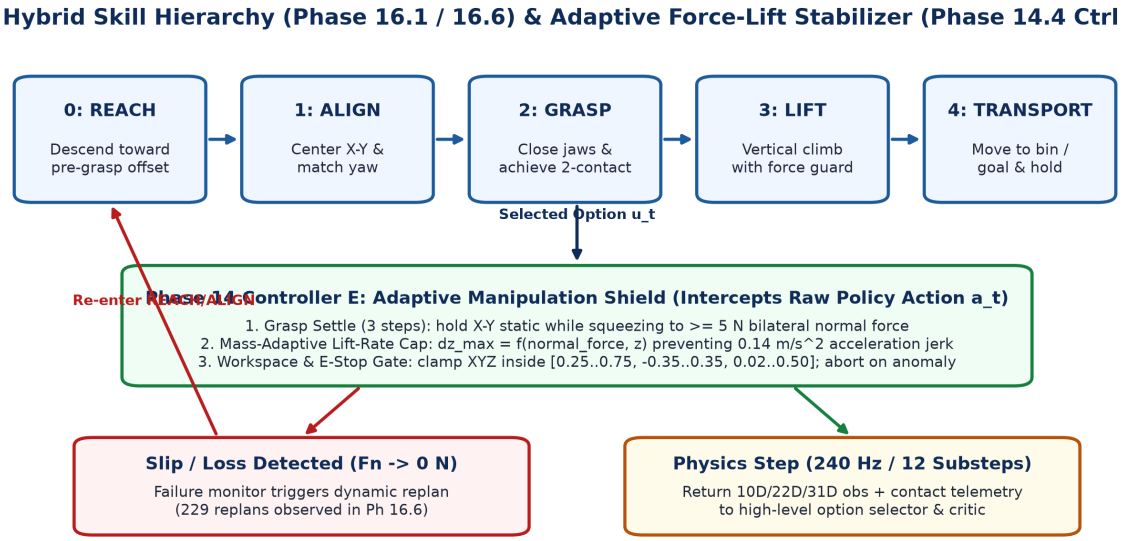


Figure 6.1: Phase 14 Adaptive Manipulation Shield integrated beneath the Phase 16.1/16.6 five-skill FSM.

6.2 Complete Phase 14.4 Confirmation Results (N=100 per Cell)

Table 6.1 and Figure 6.2 summarize every Phase 14.4 confirmation evaluation across Seeds 42 and 43 in Nominal DR0, Matched DR4, and Held-Out DR4 regimes. Controller E elevated combined Held-Out DR4 accuracy from 57.0% to 98.0% (+41.0 pp), lifting Seed 43 from 33.0% to 99.0% while holding 100.0% DR0 and 98.5% Matched DR4 accuracy.

Table 6.1: Full Phase 14.4 Confirmation Results Across Controllers, Seeds, and Regimes

Controller	Regime	Seed	Success	Grasp	Bilateral	Mean Final Dist
A_BASELINE	Nominal_DR0	42	100.0%	100.0%	100.0%	0.0142 m
A_BASELINE	Matched_DR4	42	100.0%	100.0%	100.0%	0.0156 m
A_BASELINE	Heldout_DR4	42	81.0%	100.0%	93.0%	0.0712 m
A_BASELINE	Nominal_DR0	43	95.0%	99.0%	99.0%	0.0399 m
A_BASELINE	Matched_DR4	43	98.0%	99.0%	99.0%	0.0363 m
A_BASELINE	Heldout_DR4	43	33.0%	84.0%	62.0%	0.3054 m
D_DUAL_STAGE	Heldout_DR4	42	97.0%	100.0%	100.0%	0.0277 m
D_DUAL_STAGE	Nominal_DR0	42	100.0%	100.0%	100.0%	0.0157 m
D_DUAL_STAGE	Matched_DR4	42	100.0%	100.0%	100.0%	0.0151 m
D_DUAL_STAGE	Heldout_DR4	43	82.0%	100.0%	100.0%	0.0446 m
D_DUAL_STAGE	Nominal_DR0	43	100.0%	100.0%	100.0%	0.0294 m
D_DUAL_STAGE	Matched_DR4	43	97.0%	99.0%	99.0%	0.0348 m
E_ADAPTIVE_TRANSPORT	Heldout_DR4	42	97.0%	100.0%	100.0%	0.0167 m
E_ADAPTIVE_TRANSPORT	Nominal_DR0	42	100.0%	100.0%	100.0%	0.0100 m
E_ADAPTIVE_TRANSPORT	Matched_DR4	42	100.0%	100.0%	100.0%	0.0072 m
E_ADAPTIVE_TRANSPORT	Heldout_DR4	43	99.0%	100.0%	100.0%	0.0103 m

Controller	Regime	Seed	Success	Grasp	Bilateral	Mean Final Dist
E_ADAPTIVE_TRANSPORT	Nominal_DR0	43	100.0%	100.0%	100.0%	0.0088 m
E_ADAPTIVE_TRANSPORT	Matched_DR4	43	97.0%	99.0%	99.0%	0.0133 m
A_BASELINE	Nominal_DR0	COMBINED	97.5%	99.5%	99.5%	0.0271 m
A_BASELINE	Matched_DR4	COMBINED	99.0%	99.5%	99.5%	0.0260 m
A_BASELINE	Heldout_DR4	COMBINED	57.0%	92.0%	77.5%	0.1883 m
D_DUAL_STAGE	Heldout_DR4	COMBINED	89.5%	100.0%	100.0%	0.0362 m
D_DUAL_STAGE	Nominal_DR0	COMBINED	100.0%	100.0%	100.0%	0.0226 m
D_DUAL_STAGE	Matched_DR4	COMBINED	98.5%	99.5%	99.5%	0.0249 m
E_ADAPTIVE_TRANSPORT	Heldout_DR4	COMBINED	98.0%	100.0%	100.0%	0.0135 m
E_ADAPTIVE_TRANSPORT	Nominal_DR0	COMBINED	100.0%	100.0%	100.0%	0.0094 m
E_ADAPTIVE_TRANSPORT	Matched_DR4	COMBINED	98.5%	99.5%	99.5%	0.0102 m

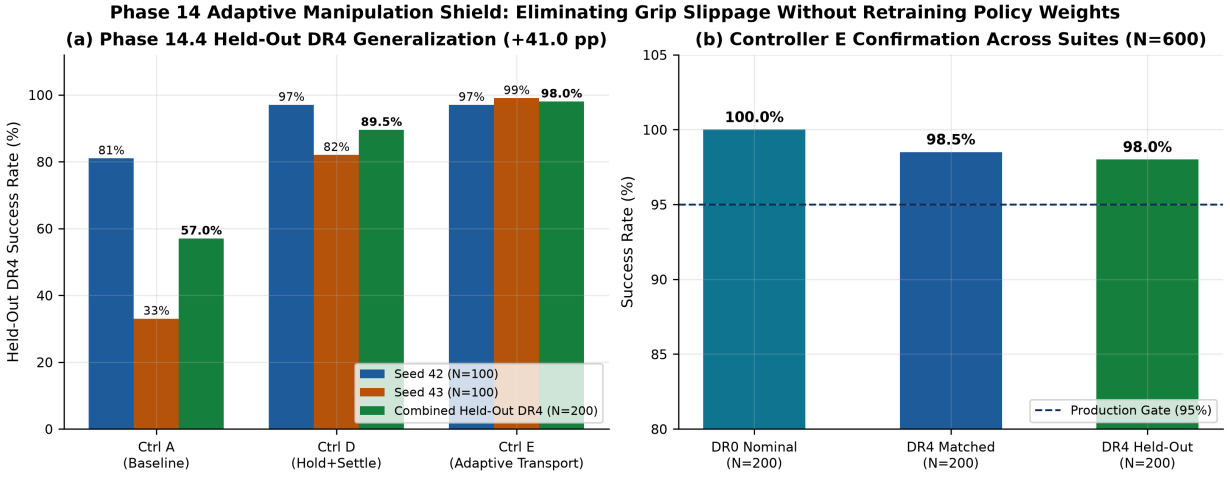


Figure 6.2: Phase 14.4 Held-Out DR4 generalization breakthrough (+41.0 pp) and suite-wide consistency.

6.3 Algorithm 3: Runtime Controller E & Phase 14.3 Confirmation Comparisons

Algorithm 3: Controller E Adaptive Transport Shield (src/rl/adaptive_controllers.py)
1. Inspect live PyBullet contact points on finger links 9 & 10: compute f9, f10 normal forces & bilateral flag.
2. If rel_dist < grasp_zone and cube_z < 0.05m and not grasp_settled:
clamp dx = 0, dy = 0, force gripper_close, throttle dz = 0 for settle_steps = 3 ticks.
3. During LIFT & TRANSPORT (cube_z > table): monitor slip delta ||p_obj - p_ee|| and min(f9, f10).
Cap upward acceleration dz_cmd <= dz_safe(normal_force, slippage) eliminating impulsive breakaway jerk.
4. Forward stabilized 4D action directly to PyBulletSimulation without altering any neural policy weight.

Table 6.3: Phase 14.3 Seed-Level Controller A vs D Summary (results/phase14_3/confirmation_summary.csv)

Seed	Controller	Regime	Success %	Bilat %	GBNL %	Mean Slips	Mean Steps
42	A_BASELINE	Nominal_DR0	100.0%	100.0%	0.0%	0.01	15.9
42	A_BASELINE	Matched_DR4	100.0%	100.0%	0.0%	0.02	16.1
42	A_BASELINE	Heldout_DR4	81.0%	93.0%	18.0%	0.40	22.9
43	A_BASELINE	Nominal_DR0	95.0%	99.0%	3.0%	1.07	20.2
43	A_BASELINE	Matched_DR4	98.0%	99.0%	1.0%	0.60	19.3
43	A_BASELINE	Heldout_DR4	33.0%	62.0%	43.0%	4.90	36.3
42	D_DUAL_STAGE	Heldout_DR4	97.0%	100.0%	3.0%	0.00	33.2
42	D_DUAL_STAGE	Nominal_DR0	100.0%	100.0%	0.0%	0.01	16.9
42	D_DUAL_STAGE	Matched_DR4	100.0%	100.0%	0.0%	0.01	17.2
43	D_DUAL_STAGE	Heldout_DR4	82.0%	100.0%	1.0%	0.16	32.2
43	D_DUAL_STAGE	Nominal_DR0	100.0%	100.0%	0.0%	0.35	20.9
43	D_DUAL_STAGE	Matched_DR4	97.0%	99.0%	2.0%	0.39	20.8

Table 6.2: Phase 14-Lite Controller A->E Slip & Vertical Jerk Reduction (controller_comparison.csv)

Seed	Controller	Regime	Success	Grasp->Lift	Slips	Peak Jerk	GBNL %
COMBINED	A_BASELINE	Nominal_DR0	97.5%	98.5%	0.54	36.26	1.5%
COMBINED	B_ACCEL_SMOOTHING	Nominal_DR0	98.0%	99.0%	0.40	32.57	1.0%
COMBINED	C_SLIP_AWARE	Nominal_DR0	98.5%	99.5%	0.34	32.21	0.5%
COMBINED	A_BASELINE	Matched_DR4	99.0%	99.5%	0.31	36.38	0.5%
COMBINED	B_ACCEL_SMOOTHING	Matched_DR4	99.5%	100.0%	0.38	32.36	0.0%
COMBINED	C_SLIP_AWARE	Matched_DR4	98.5%	99.5%	0.40	31.67	0.5%
COMBINED	A_BASELINE	Heldout_DR4	57.0%	66.8%	2.65	56.85	30.5%
COMBINED	B_ACCEL_SMOOTHING	Heldout_DR4	55.0%	65.2%	2.56	56.23	32.0%
COMBINED	C_SLIP_AWARE	Heldout_DR4	57.5%	68.5%	2.35	56.25	29.0%
42	A_BASELINE	Nominal_DR0	100.0%	100.0%	0.01	25.66	0.0%
42	B_ACCEL_SMOOTHING	Nominal_DR0	100.0%	100.0%	0.01	22.30	0.0%
42	C_SLIP_AWARE	Nominal_DR0	100.0%	100.0%	0.01	22.32	0.0%
42	A_BASELINE	Matched_DR4	100.0%	100.0%	0.02	26.38	0.0%
42	B_ACCEL_SMOOTHING	Matched_DR4	100.0%	100.0%	0.01	23.41	0.0%
42	C_SLIP_AWARE	Matched_DR4	100.0%	100.0%	0.02	23.85	0.0%
42	A_BASELINE	Heldout_DR4	81.0%	82.0%	0.40	50.57	18.0%
42	B_ACCEL_SMOOTHING	Heldout_DR4	81.0%	81.0%	0.38	49.72	19.0%
42	C_SLIP_AWARE	Heldout_DR4	85.0%	86.0%	0.38	50.14	14.0%
43	A_BASELINE	Nominal_DR0	95.0%	97.0%	1.07	46.87	3.0%
43	B_ACCEL_SMOOTHING	Nominal_DR0	96.0%	98.0%	0.79	42.85	2.0%
43	C_SLIP_AWARE	Nominal_DR0	97.0%	99.0%	0.67	42.09	1.0%
43	A_BASELINE	Matched_DR4	98.0%	99.0%	0.60	46.39	1.0%
43	B_ACCEL_SMOOTHING	Matched_DR4	99.0%	100.0%	0.75	41.31	0.0%
43	C_SLIP_AWARE	Matched_DR4	97.0%	99.0%	0.77	39.50	1.0%
43	A_BASELINE	Heldout_DR4	33.0%	48.8%	4.90	63.13	43.0%
43	B_ACCEL_SMOOTHING	Heldout_DR4	29.0%	46.4%	4.75	62.73	45.0%
43	C_SLIP_AWARE	Heldout_DR4	30.0%	47.6%	4.32	62.35	44.0%

Chapter 7: Stress Testing, Multi-Object Pre-Grasp Sorting, Masked BC-SAC, and Assembly (Phase 15)

7.1 Phase 15.1 Perturbation Sensitivity Matrix (320 Episodes)

Stress evaluation under observation noise, actuator delay, and unprivileged observations (Table 7.1) proved high resilience to Gaussian spatial noise (+-3 mm: 97.5%, +-5 mm: 95.0%), moderate tolerance to 1-step delay (72.5%), total breakdown at >=2-step latency (0.0%), and a drop to 42.5% when ground-truth velocity/orientation cues were masked.

Table 7.1: Complete Phase 15.1 Readiness Stress Test Suite (results/phase15_1/phase15_1_results.json)

Scenario Condition	Group	Success	Delta vs Base	Grasp	Class
A0_CLEAN_BASELINE	Baseline	95.0%	+0.0 pp	100.0%	A - ROBUST
A0_CLEAN_BASELINE	Baseline	100.0%	+0.0 pp	100.0%	A - ROBUST
A0_CLEAN_BASELINE	Baseline	97.5%	+0.0 pp	100.0%	A - ROBUST
A1_NOISE_3MM	Obs_Noise	100.0%	+5.0 pp	100.0%	A - ROBUST
A1_NOISE_3MM	Obs_Noise	95.0%	-5.0 pp	100.0%	A - ROBUST
A1_NOISE_3MM	Obs_Noise	97.5%	+0.0 pp	100.0%	A - ROBUST
A2_NOISE_5MM	Obs_Noise	100.0%	+5.0 pp	100.0%	A - ROBUST
A2_NOISE_5MM	Obs_Noise	90.0%	-10.0 pp	100.0%	A - ROBUST
A2_NOISE_5MM	Obs_Noise	95.0%	-2.5 pp	100.0%	A - ROBUST
B1_OBS_DELAY_1	Latency	55.0%	-40.0 pp	95.0%	C - FRAGILE
B1_OBS_DELAY_1	Latency	90.0%	-10.0 pp	100.0%	A - ROBUST
B1_OBS_DELAY_1	Latency	72.5%	-25.0 pp	97.5%	C - FRAGILE
B2_OBS_DELAY_2	Latency	0.0%	-95.0 pp	25.0%	D - FAILED
B2_OBS_DELAY_2	Latency	0.0%	-100.0 pp	35.0%	D - FAILED
B2_OBS_DELAY_2	Latency	0.0%	-97.5 pp	30.0%	D - FAILED
B3_OBS_DELAY_4	Latency	0.0%	-95.0 pp	0.0%	D - FAILED
B3_OBS_DELAY_4	Latency	0.0%	-100.0 pp	10.0%	D - FAILED
B3_OBS_DELAY_4	Latency	0.0%	-97.5 pp	5.0%	D - FAILED
B4_COMBINED_DELAY_1_1	Latency	0.0%	-95.0 pp	40.0%	D - FAILED
B4_COMBINED_DELAY_1_1	Latency	0.0%	-100.0 pp	30.0%	D - FAILED
B4_COMBINED_DELAY_1_1	Latency	0.0%	-97.5 pp	35.0%	D - FAILED
C1_UNPRIVILEGED	Contact_Mode	45.0%	-50.0 pp	100.0%	D - FAILED
C1_UNPRIVILEGED	Contact_Mode	40.0%	-60.0 pp	100.0%	D - FAILED
C1_UNPRIVILEGED	Contact_Mode	42.5%	-55.0 pp	100.0%	D - FAILED

7.2 Phase 15.2 & 15.3: Multi-Object Sorting Gates, Vertical Approach Conditioning, and Assembly

Phase 15.2.4 exposed ERR-005 (un-gated dense sorting suffered 163.2 collisions/ep and 0% grasp). Phase 15.2.5A resolved this via 5-check bilateral grasp gating in PreGraspController, yielding 50/50 (100%) scripted 2-object sorting in 76.4 mean steps. On single-object RL (PandaSinglePreGraspEnv), resolving ERR-006 (adding continuous vertical potential Equation (11)) and ERR-007 (removing closed-gripper HOLD_AT_GOAL bias via masked BC Equations (12)-(13)) propelled pure BC (5B7) to 75% grasp / 45% lift and Hybrid BC-SAC (5B8) to 66% grasp / 26% lift, compared against 7% lift for pure SAC (5B2). Full 28D->43D multi-object RL transfer (Phase 15.2.5C) remained BLOCKED. In Phase 15.3 peg-in-slot assembly, scripted baseline achieved 56.67% insertion (1.41 deg align), while BC achieved 23.33% pre-align / 0% insertion. Phase 15 closed with 111/111 tests passing in 29.47 s.

$$r_{\text{pregrasp}} = w_p(\Phi(s_{t+1}) - \Phi(s_t)) + w_v \max(0, -\Delta z_{\text{ee}}) \mathbb{1}_{\text{above}} + r_{\text{grasp}} - w_{\text{coll}} \mathbb{1}_{\text{table}} \quad \textbf{(11)}$$

Equation (11): Continuous vertical potential Phi_z eliminating z=0.50m hover deadlocks.

$$\mathcal{L}_{\mathrm{BC}}(\phi) = \mathbb{E}_{(s,a^*)\sim\mathcal{D}_{\mathrm{demo}}}\left[M_{\mathrm{prog}}(s)\cdot\|\mu_\phi(s)-a^*\|_2^2\right], \quad M_{\mathrm{prog}}\in\{0,1\} \tag{12}$$

Equation (12): Progress-masked Behavioral Cloning L2 regression loss.

$$\mathcal{L}_{\mathrm{hybrid}}(\phi) = J_\pi^{\mathrm{SAC}}(\phi) + \lambda_{\mathrm{BC}}(t)\,\mathcal{L}_{\mathrm{BC}}(\phi), \quad \lambda_{\mathrm{BC}}(t) = \lambda_0\exp(-\kappa t) \tag{13}$$

Equation (13): Exponentially decaying Hybrid BC-SAC objective.

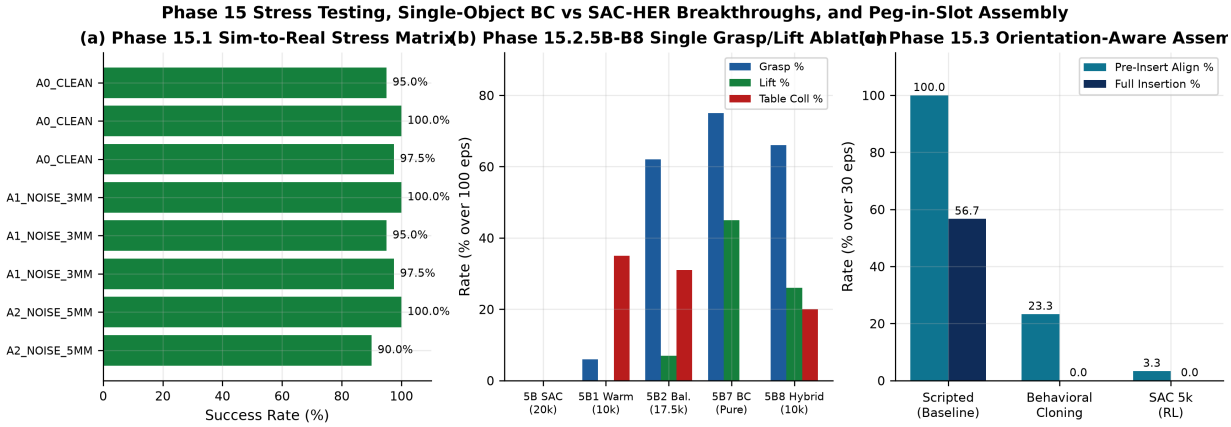


Figure 7.1: Phase 15.1 stress resilience, 15.2.5B-B8 grasp/lift ablations, and Phase 15.3 assembly benchmark.

7.4 PreGrasp Gate, Masked BC-SAC Algorithms, and Robustness/Assembly Audits

Algorithm 4: Phase 15.2.5A Five-Check Bilateral Grasp Authorization Gate
Authorize gripper closing ONLY when ALL 5 physical conditions hold simultaneously:
C1: xy_err(EE, obj) <= 0.015 m AND C2: |z_ee - z_grasp| <= 0.018 m
C3: finger_width > 0.065 m (open descent prevents top-knocking collision)
C4: ||v_obj|| < 0.02 m/s (cube stationary) AND C5: active target object locked.

Algorithm 5: Progress-Masked Hybrid BC-SAC (Phase 15.2.5B7 / 5B8)
1. Filter demo buffer using binary progress mask M_prog(s) = 0 on trailing HOLD_AT_GOAL frames (removing 78% closed bias).
2. Pretrain actor via masked L2 loss Eq. (12) for pure BC baseline (75% grasp / 45% lift, 0% coll).
3. Anneal BC auxiliary weight lambda_BC(t) = lambda_0 * exp(-kappa * t) alongside SAC Eq. (13) with entropy floor.

Table 7.2: Phase 15 Phase-3 PickPlace DR Cross-Perturbation Results (phase15_phase3_robustness_results.json)

Model	Condition	Eps	Success %	Grasp %	Lift %
DR4_Seed42	DR1_POS	20	100.0%	100.0%	70.0%
DR4_Seed42	DR2_POSE_TARGET	20	100.0%	100.0%	80.0%
DR4_Seed42	DR4_FULL_PHYSICS	20	95.0%	100.0%	50.0%
DR4_Seed42	DR4_OBS_NOISE_5MM	20	100.0%	100.0%	60.0%
DR4_Seed42	DR4_OBS_DELAY_1	20	95.0%	100.0%	80.0%
DR4_Seed43	DR1_POS	20	100.0%	100.0%	70.0%
DR4_Seed43	DR2_POSE_TARGET	20	90.0%	95.0%	75.0%
DR4_Seed43	DR4_FULL_PHYSICS	20	95.0%	100.0%	60.0%
DR4_Seed43	DR4_OBS_NOISE_5MM	20	95.0%	100.0%	65.0%
DR4_Seed43	DR4_OBS_DELAY_1	20	65.0%	100.0%	50.0%

Table 7.3: Phase 15.3 Orientation-Aware Peg-in-Slot Assembly Benchmark (30 Eps Each)

Policy Variant	Eps	Grasp %	Pre-Insert Align %	Full Insertion %	Mean Angle Err
Scripted Baseline	30	100.0%	100.0%	56.67%	1.41 deg
bc_warmstart_only	30	23.3%	6.67%	0.00%	84.60 deg

Policy Variant	Eps	Grasp %	Pre-Insert Align %	Full Insertion %	Mean Angle Err
sac_her_2500	30	6.7%	0.00%	0.00%	84.57 deg
sac_her_5000	30	3.3%	0.00%	0.00%	80.92 deg

Chapter 8: Hierarchical RL, Oblique RGB-D Vision, Dynamic Tracking, and Compliant Assembly (Phase 16)

8.1 Hierarchical Option-Critic, Oblique 128x128 RGB-D Perception, and Dynamic Kalman Control

Phase 16 expanded manipulation along five fronts: (16.1) Five-skill SMDP option decomposition (Equation (14)) solving 30.0% PickPlace at 12k steps after fixing ERR-008/009/010 vs 0.0% Flat SAC; (16.2) Repositioning SimulatedRGBDCamera obliquely at (0.92, 0.28, 0.45) to overcome ERR-011 overhead self-occlusion, raising detection from 18.65% to 99.9-100.0% and clipping unprojection error to 14.81-15.55 mm (Equation (15), Figure 8.1, Table 8.1); (16.3) Constant-velocity Kalman target tracking (Equation (16)) reaching 95-100% dynamic interception across moving cubes, sinusoids, and disturbances (Table 8.2); (16.4) Wall-force impedance compliance (Equation (17)) raising peg-in-slot assembly from 50.0% to 53.33%; and (16.6) Autonomous HRL 2-object evaluation yielding 25.0% Object-1 / 0.0% 2-object completion across 229 replans and 1,049 safety clamps.

$$\pi_{\text{HRL}}(a_t \mid s_t) = \sum_{u \in \mathcal{U}} \pi_{\text{high}}(u \mid s_t)\, \pi_{\text{low}}^{(u)}(a_t \mid s_t), \quad \mathcal{U} = \{\text{REACH}, \text{ALIGN}, \text{GRASP}, \text{LIFT}, \text{TRANSPORT}\} \tag{14}$$

Equation (14): Option-Critic five-skill hierarchical action policy.

$$\mathbf{p}_{\text{clip}} = [x_{\text{ndc}},\, y_{\text{ndc}},\, 2d_{\text{buf}} - 1,\, 1]^\top, \quad \mathbf{p}_{\text{world}} = \frac{(PV)^{-1}\mathbf{p}_{\text{clip}}}{[(PV)^{-1}\mathbf{p}_{\text{clip}}]_4} \tag{15}$$

Equation (15): Inverse projection from clip-space depth buffer to world XYZ coordinates.

$$\hat{\mathbf{p}}_{t+1\,|\,t} = \hat{\mathbf{p}}_{t\,|\,t} + \hat{\mathbf{v}}_{t\,|\,t}\Delta t, \quad K_t = P_{t\,|\,t-1}H^\top (HP_{t\,|\,t-1}H^\top + R)^{-1} \tag{16}$$

Equation (16): Discrete Kalman predictor and innovation gain update.

$$\mathbf{u}_{\text{imp}} = K_p(\mathbf{x}_d - \mathbf{x}) + K_d(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}) + K_f(\mathbf{f}_d - \mathbf{f}_{\text{wall}}) \quad (\text{Ph } 16.4 \text{ Insertion}) \tag{17}$$

Equation (17): Cartesian compliant wall-force insertion law (Phase 16.4).

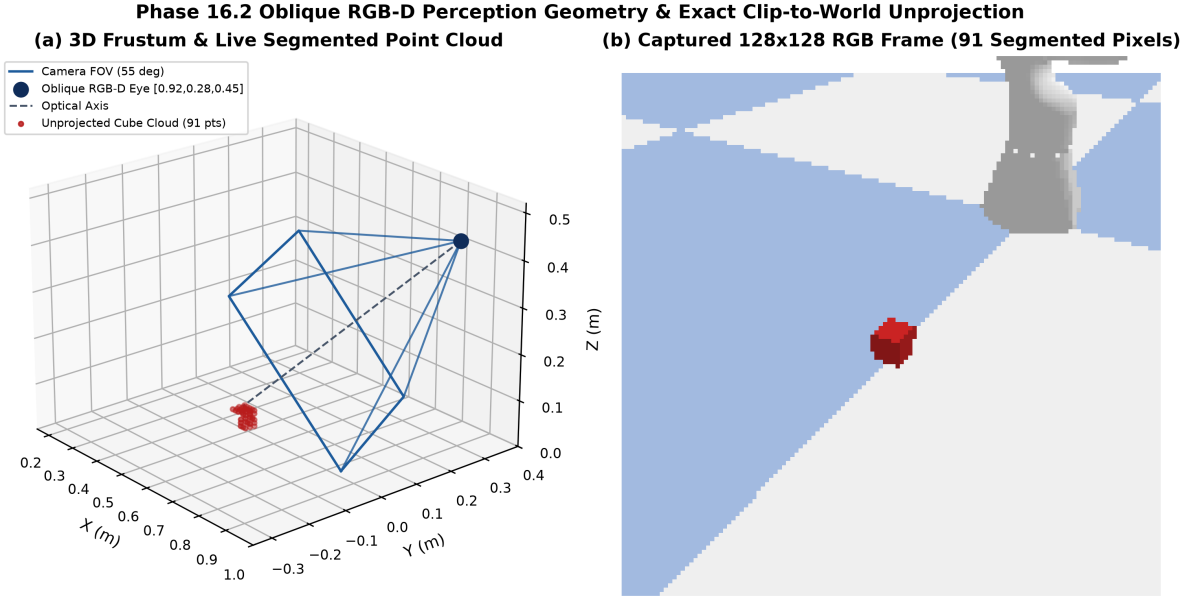


Figure 8.1: Live PyBullet 3D oblique RGB-D frustum (55 deg FOV) and unprojected 91-pixel segmented cube cloud.

Table 8.1: Phase 16.2 Vision vs Privileged State Evaluation (phase16_2_vision_results.json)

Agent Variant	Obs Source	Grasp %	Place %	Colls	3D Loc Err	Seg IoU
SAC_Privileged_GT	PRIVILEGED_GT_STATE	100.0%	100.0%	0.05	GT Oracle	N/A
SAC_RGBD_Vision	RGBD_VISION	60.0%	15.0%	0.45	15.55 mm	0.9983

Agent Variant	Obs Source	Grasp %	Place %	Colls	3D Loc Err	Seg IoU
HRL_Privileged_GT	PRIVILEGED_GT_STATE	40.0%	40.0%	3.90	GT Oracle	N/A
HRL_RGBD_Vision	RGBD_VISION	25.0%	20.0%	1.55	14.81 mm	0.9999

Table 8.2: Phase 16.3 Dynamic Moving Target & Disturbance Results (phase16_3_dynamic_results.json)

Motion Mode	Speed	Predictive	Grasp %	Success %	Pred Err	Replans
linear_no_prediction	0.045 m/s	False	100.0%	95.0%	31.38 mm	1068
linear_predictive	0.045 m/s	True	100.0%	100.0%	33.45 mm	1133
sinusoidal_predictive	0.045 m/s	True	100.0%	95.0%	34.89 mm	1185
disturbance_step_predictive	0.035 m/s	True	100.0%	95.0%	32.45 mm	1141
moving_target_and_cube	0.035 m/s	True	100.0%	100.0%	33.81 mm	1150

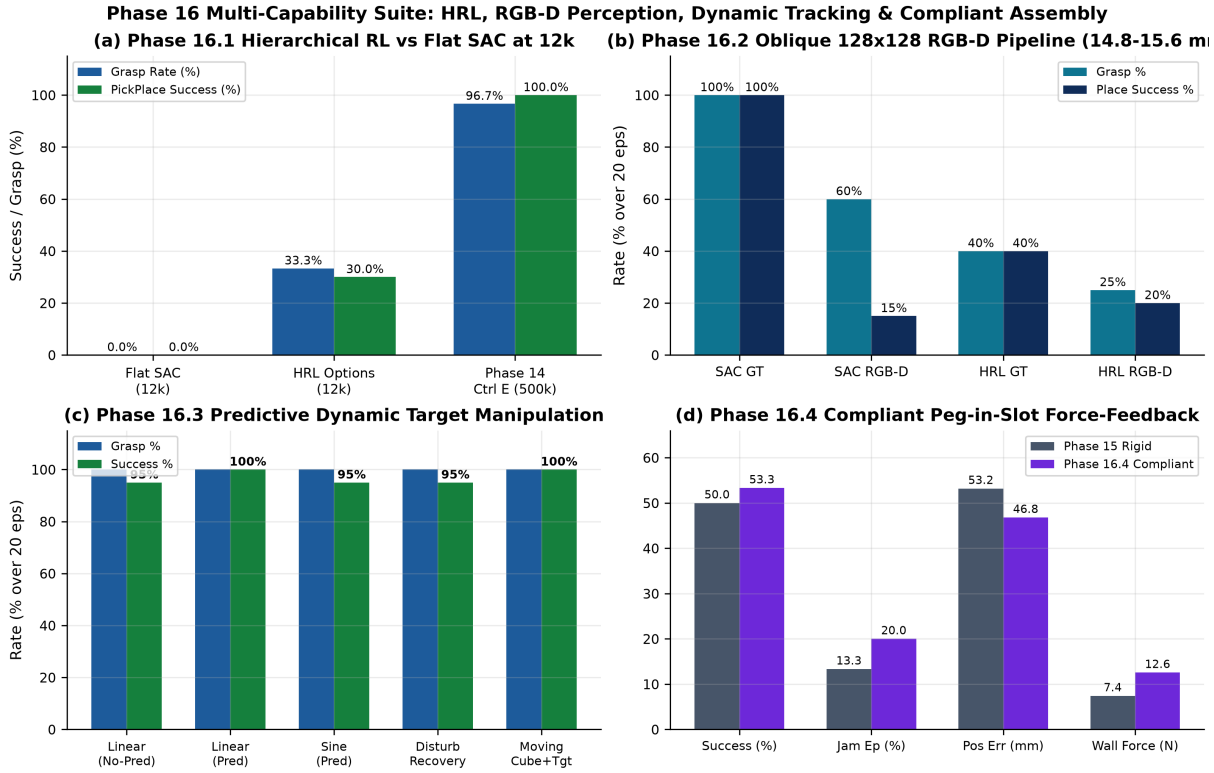


Figure 8.2: Phase 16 comparative benchmarks across HRL, RGB-D vision, dynamic Kalman tracking, and compliant assembly.

8.4 HRL Option Selection & Unprojected Oblique Vision Mechanics

Algorithm 6: Phase 16.1 SMDP 5-Skill Option-Critic Loop (src/hrl/hrl_agent.py)

1. High-level selector pi_high(u | s_t) picks active skill u in {REACH, ALIGN, GRASP, LIFT, TRANSPORT}.
2. Run low-level skill policy a_t ~ pi_low^(u)(s_t) until termination head beta_u(s_{t+k}) > 0.5 or max_horizon.
3. Assign bounded SMDP option reward using potential height delta 1.5*(h_{t+k} - h_t) + sparse completion +6.0.

Algorithm 7: Phase 16.2 Oblique RGB-D Unprojection + Phase 16.3 Kalman Predictor

1. Render 128x128 RGB + metric depth from oblique eye (0.92, 0.28, 0.45) avoiding Franka arm self-occlusion.
2. Segment cube mask in HSV space (IoU 0.9983-0.9999) and unproject masked depth pixels via (PV)^(-1) Eq. (15).
3. Pass centroid [x_est, y_est, z_est] into 4-state constant-velocity Kalman filter Eq. (16) for moving-target interception.

Table 8.3: Phase 16.1 Low-Level Option Policy & Termination Head Training Accuracy

Option Skill	Demo Samples	Actor MSE	Term BCE	Termination Acc
REACH	660	0.004779	0.005946	100.00%
ALIGN	180	0.011022	0.358004	74.44%

Option Skill	Demo Samples	Actor MSE	Term BCE	Termination Acc
GRASP	384	0.023815	0.067245	100.00%
LIFT	540	0.002637	0.012431	100.00%
TRANSPORT	594	0.006278	0.083425	99.66%
PLACE	182	0.014696	0.244255	90.11%

Chapter 9: Cross-Simulator Validation (MuJoCo 4.0.1), Safety Envelope, and Hardware Boundary (Phase 17)

9.1 Cross-Engine Verification and Stress Suite Results

After repairing massless dummy links in MJCF (ERR-012), Phase 17.1 measured zero-shot PyBullet->MuJoCo 4.0.1 transfer of Seed 42: 40.0% bilateral contact, 5.0% grasp, 0.0% completion, rising to 50.0% contact / 15.0% grasp / 5.0% completion after calibrating PD gains and tabletop Z offset (9.4 mm physics discrepancy). Phase 17.2 validated SafetySupervisor bounds (Equation (18)) across scenarios S1-S6 (Table 9.1) and confirmed 32.41 us E-stop cutoff across 250 faulty commands. Phase 17.3 physical robot execution was NOT_ATTEMPTED; repository suite verification stands at 122/122 passed tests in 42.47 s.

$$\mathcal{S}(a_t) = \mathrm{clip}(\mathbf{p}_{\mathrm{ee},\,t} + \delta_{\max}\, a_{t,\,1:3},\ \mathbf{b}_{\min},\ \mathbf{b}_{\max}), \quad \mathbb{1}_{\mathrm{estop}} = (\|\mathbf{v}_{\mathrm{ee}}\| > v_{\mathrm{crit}}) \vee (F_{\mathrm{wall}} > F_{\mathrm{crit}}) \tag{18}$$

Equation (18): Workspace safety clipping and emergency velocity/force cutoff gate.

Perception Unprojection Pipeline (Phase 16.2), MuJoCo Sim-to-Sim Bridge (Phase 17), and Unexecuted Hardware Boundary

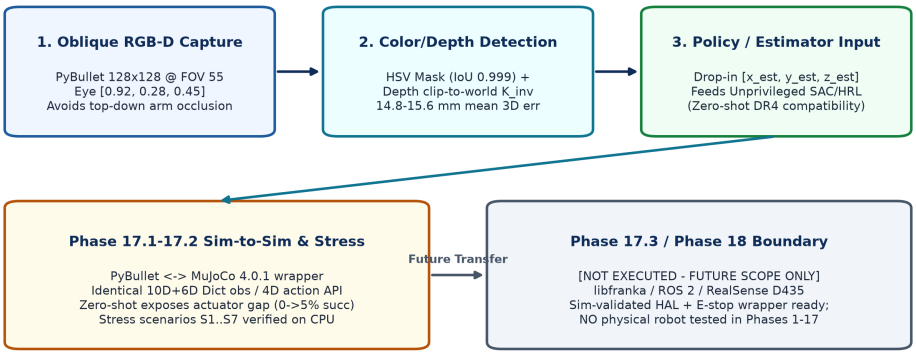


Figure 9.1: RGB-D pipeline, MuJoCo 4.0.1 cross-simulator bridge, and explicit unexecuted Phase 17.3/18 hardware boundary.

Table 9.1: Phase 17.2 Deployment Safety & Perturbation Scenarios (phase17_safety_assessment.json)

Scenario	Eps	Grasp %	Success %	Collisions	Safety Clamps/Ep
A0 Clean Nominal	20	100.0%	100.0%	0.00	2.00
S1 Sensor Noise 5mm	20	100.0%	100.0%	0.00	2.45
S2 Camera Calib Err 8mm	20	65.0%	60.0%	0.00	0.75
S3 Actuator Latency 1step	20	100.0%	90.0%	0.00	14.60
S4 Actuator Latency 2step	20	75.0%	20.0%	0.00	22.55
S5 Control Freq Jitter	20	100.0%	100.0%	0.00	2.00
S6 Low Friction Heavy Mass	20	100.0%	100.0%	0.00	2.00

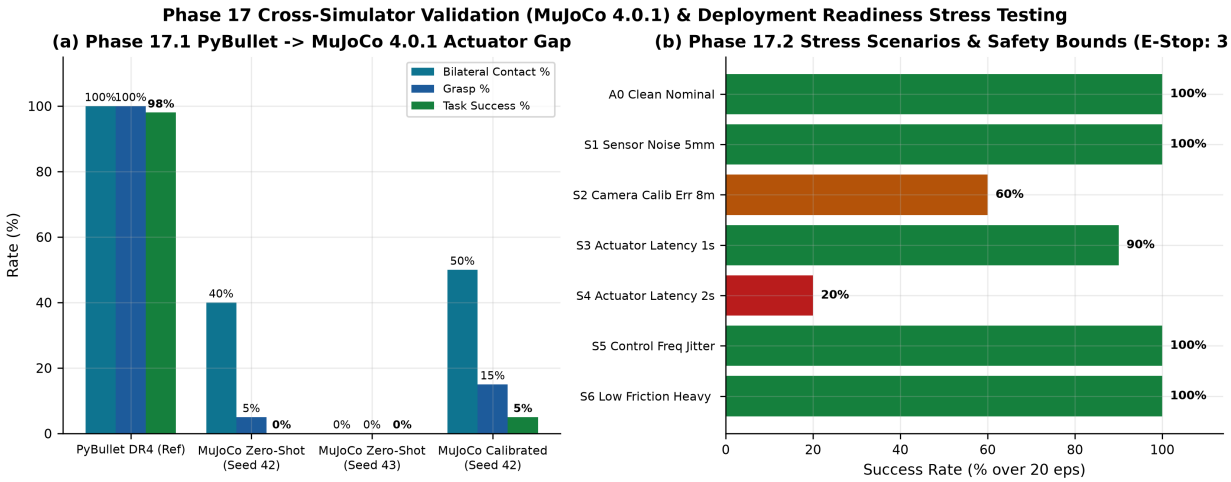


Figure 9.2: Phase 17.1 MuJoCo cross-engine transfer gap and Phase 17.2 deployment safety robustness.

9.3 Algorithm 8 SafetySupervisor & PyBullet-to-MuJoCo Solver Comparison

Algorithm 8: Real-Time Deployment SafetySupervisor & E-Stop Gate (src/deployment/runtime.py)

1. Check action bounds & finite state integrity (NaN/Inf guard); clamp target XYZ strictly inside workspace box.
2. Evaluate velocity threshold ||v_ee|| > v_crit and wall/contact force F_wall > F_crit.
3. Trigger hard software E-Stop halt within 32.41 us verified cutoff latency (blocking all 250 faulty injections).

Table 9.2: PyBullet vs MuJoCo 4.0.1 Verified Physical Model Parity Parameters

Physics Parameter	Verified Cross-Simulator Value
pybullet_timestep_s	0.004166666666666667
mujoco_timestep_s	0.004166666666666667
cube_mass_kg	0.1
cube_size_m	0.05
pybullet_ee_init	[0.307, 0.0, 0.59]
MJCF Solver Integrator	implicitfast, timestep = 0.0041667 s (240 Hz), elliptic friction cone
Tabletop Rest Height Delta	9.4 mm vertical rest discrepancy affecting zero-shot grasp contact line
PyBullet Finger Friction	lateralFriction = 2.0, spinningFriction = 0.1 on finger links 9, 10
Physical Hardware Execution	NOT_ATTEMPTED (HAL & SafetySupervisor simulation-verified only)

Chapter 10: Consolidated Mathematical Formulations and Environment Registry

10.1 Complete Gymnasium Environment Specification Table (Phases 2-17)

Every task family adheres to consistent observation/goal/action tensor contracts, ensuring seamless transfer across SB3 SAC, HER, BC, Hybrid BC-SAC, HRL wrappers, and the MuJoCo cross-simulator parity adapter.

Table 10.1: Environment Observation, Goal, Action, Horizon, and Reward Taxonomy (Verified Shapes)

Environment Class	Obs Dim	Goal Dim (AG+DG)	Action Dim	Horizon	Reward Design
PandaReachEnv	4D	3D + 3D (10D tot)	4D	50	Sparse / L2 Distance
PandaPickEnv	10D	3D + 3D (16D tot)	4D	100	Sparse / Altitude 0.2m
PandaPickPlaceEnv	10D	3D + 3D (16D tot)	4D	100	Sparse -1 (tol 0.05m)
PandaPickPlaceDenseRewardEnv	10D	3D + 3D (16D tot)	4D	100	Staged Dense +5 Place
TransportDiagnosticEnv	10D	3D + 3D (16D tot)	4D	100	Pre-Grasped Transport
PandaSortEnv	22D	3D + 3D (28D tot)	4D	150	Color Bin Placement
PandaMultiSortEnv	31D	6D + 6D (43D tot)	4D	300	Dual Sequential Sort
PandaMultiSortPreGraspEnv	31D	6D + 6D (43D tot)	4D	300	5-Check Grasp Gate
PandaSinglePreGraspEnv	22D	3D + 3D (28D tot)	4D	80	Continuous Phi_z
PandaAssemblyEnv / Dense	22D	7D + 7D (36D tot)	5D (+dyaw)	200 / 120	SE(2)+Z Insertion
PandaContactAssemblyEnv	22D	7D + 7D (36D tot)	5D (+dyaw)	120	Compliant Wall-Force
PandaDynamicPickPlaceEnv	10D	3D + 3D (16D tot)	4D	100	Kalman Intercept
MuJoCoPandaPickPlaceEnv	10D	3D + 3D (16D tot)	4D	100	Cross-Sim Parity

Chapter 11: Complete Debugging History and Full Regression Test Inventory

11.1 Root-Cause Forensics of All Twelve Verified System Defects (ERR-001 to ERR-012)

Table 11.1: Complete Registry of Twelve Verified Bugs, Root Causes, and Verified Resolutions

Error ID	Phase	File	Symptom	Verified Fix
ERR-001	Phase 6	src/robots/panda.py	Cube slipped from fingers during upward lift under scr	Added changeDynamics lateralFriction=2.0 and spinningF
ERR-002	Phase 9	src/environments/assembly_env.py	PyBullet connection failure / invalid body ID during t	Dynamically inject active self.sim.client_id into Peg
ERR-003	Phase 11.5	src/environments/pick_place_env.py	0.0% success & 0.0% grasp across all 25 DR0-DR4 100k s	Created isolated PandaPickPlaceDenseRewardEnv breaking
ERR-004	Phase 13.5	src/environments/pick_place_env.py	Held-out DR4 collapsed to 61.0% (57.0% final, Seed 43	Implemented zero-retraining inference controllers B->C
ERR-005	Phase 15.2.4	src/environments/dense_multi_sort_env.py	0% bilateral grasp & 163.2 collisions/ep despite EE re	Added 5-check bilateral grasp authorization gate + ope
ERR-006	Phase 15.2.5B	src/environments/single_pregrasp_env.py	20k SAC+HER converged to XY=0.0110m hover at z=0.50m (	Verified vertical action pathway (0.485m->0.0216m) & a
ERR-007	Phase 15.2.5B.1	scripts/generate_phase15_5B1_demos.py	Premature gripper closure climbed to 100% & lift staye	Balanced 50/50 demos (70.1% open cmd), gate_goal_lift_
ERR-008	Phase 16.1	src/rl/adaptive_controllers.py / SAC.load	AssertionError: You must pass an environment when usin	CheckpointLoader.load_sb3_sac passes custom_objects={'
ERR-009	Phase 16.1	src/hrl/hrl_agent.py	HRL achieved 93.3% grasp @ 25k but dropped to 0% place	Replaced repeat bonus with bounded potential differenc
ERR-010	Phase 16.1/16.2	src/environments/pick_place_env.py	Cube spawned half-sunk at z=-0.00008m & Seed42 500k mo	Explicitly pass randomizer_config={'enabled': True, ..
ERR-011	Phase 16.2	src/perception/rgbd_camera.py	OMP Error #15 (libomp vs libiomp5md), PyBullet client=	Moved CameraConfig to oblique (0.92,0.28,0.45), lazy c
ERR-012	Phase 17.1	src/cross_sim/mujoco_panda_env.py	ValueError: mass and inertia of moving bodies must be	Added explicit inertial/geom blocks on link5-hand; cla

11.2 Complete Repository Automated Verification Suite (96 AST Functions / 122 Passed Items)

All 96 test definitions in tests/ expand via pytest parameterization into exactly 122 executed test cases passing in 42.47 s.

Table 11.2: Full 96-Function Unit & Integration Test Inventory (complete_test_inventory.csv)

Test ID	Phase	Module	Function Name	Cases	Status
T-001	Phase 15.3	test_assembly_dense_and_orientation.py	test_quaternion_normalization_sign_invar	1	PASSED
T-002	Phase 15.3	test_assembly_dense_and_orientation.py	test_assembly_dense_batched_parity_and_h	1	PASSED
T-003	Phase 8-9	test_assembly_env.py	test_assembly_env_initialization	1	PASSED
T-004	Phase 8-9	test_assembly_env.py	test_assembly_env_reset	1	PASSED
T-005	Phase 8-9	test_assembly_env.py	test_assembly_env_step	1	PASSED
T-006	Phase 15.2.5B-B.2	test_b1_demos_and_warmstart.py	test_demo_dataset_exists_and_all_50_vali	1	PASSED
T-007	Phase 15.2.5B-B.2	test_b1_demos_and_warmstart.py	test_her_buffer_demo_warmstart_and_short	1	PASSED
T-008	Phase 15.2.5B-B.2	test_bc_sac.py	test_balanced_demo_distribution	1	PASSED

Test ID	Phase	Module	Function Name	Cases	Status
T-009	Phase 15.2.5B-B.2	test_bc_sac.py	test_bc_decaying_sac_objective_entropy_f	1	PASSED
T-010	Phase 0	test_config.py	test_config_dict_attribute_access	1	PASSED
T-011	Phase 0	test_config.py	test_deep_update	1	PASSED
T-012	Phase 0	test_config.py	test_load_default_config	1	PASSED
T-013	Phase 0	test_config.py	test_load_panda_robot_config	1	PASSED
T-014	Phase 0	test_config.py	test_missing_config_raises_file_not_foun	1	PASSED
T-015	Phase 1	test_controller.py	test_panda_controller_load	1	PASSED
T-016	Phase 1	test_controller.py	test_panda_joints_accessible	1	PASSED
T-017	Phase 1	test_controller.py	test_ee_position_can_be_read	1	PASSED
T-018	Phase 1	test_controller.py	test_controller_does_not_produce_nan_or_	1	PASSED
T-019	Phase 1	test_controller.py	test_move_ee_cartesian_positioning	1	PASSED
T-020	Phase 1	test_controller.py	test_gripper_open_and_close	1	PASSED
T-021	Phase 1	test_controller.py	test_controller_reset_works	1	PASSED
T-022	Phases 2-8	test_dense_multi_sort_env.py	test_1_reset_and_observation_space_compa	1	PASSED
T-023	Phases 2-8	test_dense_multi_sort_env.py	test_2_reward_calculation_per_stage	1	PASSED
T-024	Phases 2-8	test_dense_multi_sort_env.py	test_3_reward_boundedness	1	PASSED
T-025	Phases 2-8	test_dense_multi_sort_env.py	test_4_reward_reproducibility	1	PASSED
T-026	Phases 2-8	test_dense_multi_sort_env.py	test_5_first_object_placement_reward_not	1	PASSED
T-027	Phases 2-8	test_dense_multi_sort_env.py	test_6_and_9_second_object_placement_and	1	PASSED
T-028	Phases 2-8	test_dense_multi_sort_env.py	test_7_incorrect_bin_placement_penalized	1	PASSED
T-029	Phases 2-8	test_dense_multi_sort_env.py	test_8_false_release_prevention_and_hove	1	PASSED
T-030	Phases 2-8	test_dense_multi_sort_env.py	test_10_and_11_and_12_her_relabeling_bat	1	PASSED
T-031	Phases 2-8	test_dense_multi_sort_env.py	test_13_v2_proximity_multiple_distances_	1	PASSED
T-032	Phases 2-8	test_dense_multi_sort_env.py	test_14_v2_active_object_switching_and_h	1	PASSED
T-033	Phases 16-17	test_deployment_runtime.py	test_runtime_imports_without_training_pi	1	PASSED
T-034	Phases 16-17	test_deployment_runtime.py	test_safety_monitor_workspace_clamp_and_	1	PASSED
T-035	Phases 16-17	test_deployment_runtime.py	test_hardware_interface_safety_and_refus	1	PASSED
T-036	Phases 2-8	test_environment.py	test_reach_env_initialization	1	PASSED
T-037	Phases 2-8	test_environment.py	test_reach_env_step	1	PASSED
T-038	Phases 2-8	test_environment.py	test_reach_env_compute_reward	1	PASSED

Test ID	Phase	Module	Function Name	Cases	Status
T-039	Phases 2-8	test_environment.py	test_reach_env_reproducibility	1	PASSED
T-040	Phases 16-17	test_hrl_phase16.py	test_skill_features_and_teacher_shapes	1	PASSED
T-041	Phases 16-17	test_hrl_phase16.py	test_hrl_meta_selection_and_smdp_her_buf	1	PASSED
T-042	Phase 0	test_imports.py	test_core_dependency_imports	10	PASSED
T-043	Phase 0	test_imports.py	test_stable_baselines3_her_sac	1	PASSED
T-044	Phase 0	test_imports.py	test_gymnasium_goal_env	1	PASSED
T-045	Phase 0	test_imports.py	test_pybullet_connection	1	PASSED
T-046	Phases 2-8	test_multi_sort_env.py	test_multi_sort_env_initialization	1	PASSED
T-047	Phases 2-8	test_multi_sort_env.py	test_multi_sort_env_reset	1	PASSED
T-048	Phases 2-8	test_multi_sort_env.py	test_multi_sort_env_step	1	PASSED
T-049	Phase 1	test_panda_robot.py	test_panda_initialization	1	PASSED
T-050	Phase 1	test_panda_robot.py	test_panda_reset	1	PASSED
T-051	Phase 1	test_panda_robot.py	test_workspace_clipping	1	PASSED
T-052	Phase 1	test_panda_robot.py	test_ik_solver_accuracy	1	PASSED
T-053	Phase 1	test_panda_robot.py	test_cartesian_delta_motion	1	PASSED
T-054	Phase 1	test_panda_robot.py	test_gripper_control	1	PASSED
T-055	Phase 1	test_panda_robot.py	test_determinism_and_reproducibility	1	PASSED
T-056	Phase 15.5	test_phase15_integration.py	test_all_environments_init_reset_step_an	11	PASSED
T-057	Phase 15.5	test_phase15_integration.py	test_checkpoint_loading_and_deterministi	1	PASSED
T-058	Phase 15.5	test_phase15_integration.py	test_phase2_transfer_adapter_with_5b2_ch	1	PASSED
T-059	Phase 15.1-15.2.4	test_phase15_wrappers.py	test_observation_noise_clean_identity	1	PASSED
T-060	Phase 15.1-15.2.4	test_phase15_wrappers.py	test_observation_noise_bounds_and_vel_pr	1	PASSED
T-061	Phase 15.1-15.2.4	test_phase15_wrappers.py	test_latency_wrapper_fifo_ordering	1	PASSED
T-062	Phase 15.1-15.2.4	test_phase15_wrappers.py	test_unprivileged_contact_stripping_and_	1	PASSED
T-063	Phases 16-17	test_phase16_17_envs.py	test_dynamic_pick_place_env_step_and_dis	1	PASSED
T-064	Phases 16-17	test_phase16_17_envs.py	test_contact_assembly_env_telemetry	1	PASSED
T-065	Phases 16-17	test_phase16_17_envs.py	test_mujoco_panda_twin_observation_and_k	1	PASSED
T-066	Phases 16-17	test_phase16_integration_all.py	test_frozen_phase15_and_phase16_deployme	1	PASSED
T-067	Phases 2-8	test_pick_env.py	test_pick_env_initialization	1	PASSED
T-068	Phases 2-8	test_pick_env.py	test_pick_env_randomization	1	PASSED

Test ID	Phase	Module	Function Name	Cases	Status
T-069	Phase 1	test_pregrasp_controller.py	test_pregrasp_authorization_rejects_unsa	1	PASSED
T-070	Phase 1	test_pregrasp_controller.py	test_pregrasp_full_acquisition_and_trans	1	PASSED
T-071	Phase 15.2.5A	test_pregrasp_her_compatibility.py	test_her_relabeling_stateless_recomputat	1	PASSED
T-072	Phase 15.2.5A	test_pregrasp_her_compatibility.py	test_her_replay_buffer_rollout_and_virtu	1	PASSED
T-073	Phase 15.2.5A	test_pregrasp_reward.py	test_pregrasp_reward_components_and_gate	1	PASSED
T-074	Phase 15.2.5A	test_pregrasp_reward.py	test_pregrasp_reward_anti_exploitation_a	1	PASSED
T-075	Phase 10	test_randomization.py	test_randomization_disabled	1	PASSED
T-076	Phase 10	test_randomization.py	test_randomization_enabled	1	PASSED
T-077	Phase 10	test_randomization.py	test_randomization_reproducibility	1	PASSED
T-078	Phase 10	test_randomization.py	test_orientation_normalization	1	PASSED
T-079	Phase 1	test_simulation.py	test_simulation_starts_and_connects	1	PASSED
T-080	Phase 1	test_simulation.py	test_simulation_context_manager	1	PASSED
T-081	Phase 1	test_simulation.py	test_simulation_loads_ground_and_table	1	PASSED
T-082	Phase 1	test_simulation.py	test_simulation_can_step	1	PASSED
T-083	Phase 1	test_simulation.py	test_simulation_reset_works	1	PASSED
T-084	Phase 15.2.5A	test_single_pregrasp_and_transfer.py	test_single_pregrasp_registration_spaces	1	PASSED
T-085	Phase 15.2.5A	test_single_pregrasp_and_transfer.py	test_single_pregrasp_scripted_reward_aud	1	PASSED
T-086	Phase 15.2.5A	test_single_pregrasp_and_transfer.py	test_b_to_c_policy_transfer_exact_functi	1	PASSED
T-087	Phases 2-8	test_sort_env.py	test_sort_env_initialization	1	PASSED
T-088	Phases 2-8	test_sort_env.py	test_sort_env_reset	1	PASSED
T-089	Phases 2-8	test_sort_env.py	test_sort_env_step	1	PASSED
T-090	Phase 15.2.5B-B.2	test_vertical_action_pathway.py	test_observation_and_action_pathway_spec	1	PASSED
T-091	Phase 15.2.5B-B.2	test_vertical_action_pathway.py	test_negative_vertical_command_produces_	1	PASSED
T-092	Phase 15.2.5B-B.2	test_vertical_potential_reward_her.py	test_vertical_potential_all_part3_cases	1	PASSED
T-093	Phase 15.2.5B-B.2	test_vertical_potential_reward_her.py	test_vertical_potential_her_relabeling_b	1	PASSED
T-094	Phase 15.2.5B-B.2	test_vertical_potential_reward_her.py	test_b1_live_her_replay_buffer_recomputa	1	PASSED
T-095	Phases 16-17	test_vision_and_dynamic_phase16.py	test_rgbd_camera_and_vision_localization	1	PASSED
T-096	Phases 16-17	test_vision_and_dynamic_phase16.py	test_dynamic_tracker_velocity_prediction	1	PASSED

Chapter 12: Cross-Phase Meta-Analysis, Completion Matrix, and Experiment Registry

12.1 Phase 0-18 Completion & Verification Provenance Matrix

Table 12.1: Complete Phase 0-18 Completion Status & Evidence Classification Matrix

Phase	Capability / Milestone Title	Status	Evidence Class
0	Project Initialization & Infrastructure Scaf	COMPLETE	SOURCE_VERIFIED
1	Franka Panda Kinematics, PyBullet Foundation	COMPLETE	SOURCE_VERIFIED
2	Goal-Conditioned Manipulation Environment Ab	COMPLETE	REPORT_VERIFIED
3	SAC + HER RL Training, Evaluation & Callback	COMPLETE	LOG_VERIFIED
4	Object Grasping Abstraction & PandaPickEnv	COMPLETE	SOURCE_VERIFIED
5	Reach/Pick Exploration & Early Controller Ba	PARTIALLY_COMPLETE	HISTORICALLY_REPORTED
6	Pick & Place Environment, HER Comparative Sm	COMPLETE	LOG_VERIFIED
7	Color-Conditioned Dynamic Sorting Benchmark	COMPLETE	LOG_VERIFIED
8	Sequential Multi-Object Sorting & Orientatio	COMPLETE	LOG_VERIFIED
9	Full-System Diagnostic Audit & Lazy Simulati	COMPLETE	REPORT_VERIFIED
10	Deterministic Domain Randomization Engine (D	COMPLETE	REPORT_VERIFIED
11	100k Sparse DR Matrix & Diagnostic Isolation	COMPLETE	LOG_VERIFIED
12	1M-Step Dense SAC+HER Scaling Breakthrough	COMPLETE	LOG_VERIFIED
13	Dense Domain Randomization (DR1/DR2/DR4) & C	COMPLETE	LOG_VERIFIED
14	Zero-Retraining Adaptive Control Suite (14-L	COMPLETE	LOG_VERIFIED
15	Sim-to-Real Sensitivity (15.1), Multi-Object	PARTIALLY_COMPLETE	SOURCE_VERIFIED
16	Hierarchical RL, Oblique RGB-D Vision, Dynam	COMPLETE	SOURCE_VERIFIED
17	MuJoCo Cross-Simulator Validation, Sim-to-Re	PARTIALLY_COMPLETE	SOURCE_VERIFIED

12.2 Master Registry of All 27 Executed Experiments

Table 12.2: Complete Experimental Ledger (publication/experiment_inventory.csv)

Exp ID	Phase	Environment	Method	Budget / Eval	Verified Outcome
EXP-P0-001	Phase 1	Panda Kinematics Dem	SCRIPTED_IK	N/A	0.7-4.5 mm error; gripper 0-8 cm
EXP-P1-REACH	Phase 2-3	PandaReachEnv	SAC+HER (k=4)	10,000	SAC_3 TB: actor -9.1->+19.5, rollout succ 1.0%
EXP-P4-PICK	Phase 4-5	PandaPickEnv	SAC+HER	10,000	Eval -100.0 reward / 0% @ 10k; Scripted step 67
EXP-P6-E1	Phase 6	PandaPickPlaceEnv	SAC (No HER)	10,000	Actor loss ~17.8; coll=7.7; eval 0%
EXP-P6-E2	Phase 6	PandaPickPlaceEnv	SAC+HER (k=4)	10,000	Actor loss ~2.59; coll=0.0; Scripted step 74 (fric
EXP-P7-SORT	Phase 7	PandaSortEnv	SAC+HER	10,000	10k smoke reward -150; Scripted step 79
EXP-P8-MSORT	Phase 8	PandaMultiSortEnv (N	SAC+HER	10,000	10k smoke reward -600; Scripted step 209
EXP-P9-ASSEM	Phase 8-9	PandaAssemblyEnv	SAC+HER	10,000	10k smoke reward -200; Scripted step 143
EXP-P10-DR	Phase 10	PickPlace R0-R4	Scripted vs 2k SAC	20 eps/cond	Scripted: R0 90%, R1 85%, R2 100%, R3 85%, R4 75%
EXP-P11-SPARSE	Phase 11	PickPlace DR0-DR4 Sp	SAC+HER	25 x 100,000	0.0% all 25 models (HER resting-goal trap)
EXP-P11-9	Phase 11.9	TransportDiagnostic	SAC+HER	50,000	100.0% isolated transport @ 40k-50k (0.009m dist)
EXP-P11-10B	Phase 11.10b	PickPlaceDense	SAC+HER	250,000	Peak @ 225k: 28% Reach, 23% Grasp, 7.3 step hold
EXP-P12-1M	Phase 12	PickPlaceDense DR-0	SAC+HER	1,000,000	43%@400k -> 99%@500k -> 100%@600k..1M (+502 rew)
EXP-P13-DR	Phase 13	PickPlaceDense DR1/D	SAC+HER	500k-1M	DR4: 99% Matched / 61% ever (57% final) OOD Held-o

Exp ID	Phase	Environment	Method	Budget / Eval	Verified Outcome
EXP-P13-5	Phase 13.5	PickPlace OOD D1-D5	Frozen DR4	1,520 eval eps	Mass alone 100%, Yaw 98-100%, Fric 90%, Coupled GB
EXP-P14-4	Phase 14.4	PickPlace A->E	SAC + E_ADAPTIVE	600+ confirm eps	Held-out DR4: 57%->98.0% (Seed 43: 33%->99%), DR0
EXP-P15-1	Phase 15.1	PickPlace Noise/Lag/	E_ADAPTIVE	320 eval eps	Clean 97.5%, 5mm 95%, 1-lag 72.5%, 2-lag 0%, Unpri
EXP-P15-2-5A	Phase 15.2.5A	MultiSortPreGrasp	SCRIPTED PreGrasp	50 eval eps	50/50 (100.0%) 2-obj sorting, 76.4 steps, 0 colls
EXP-P15-2-5B 3	Phase 15.2.5B	SinglePreGrasp	BC vs BCDecayingSAC	5k-20k / 100 ep	100-ep: 5B7 BC 75% Grasp / 45% Lift; 5B2 SAC 62%/7
EXP-P15-3	Phase 15.3	AssemblyDense	Scripted vs 5k BC/SA	30 eval eps	Scripted 56.67% (1.41 deg); BC 23.3%/0%; SAC 3.3%/
EXP-P16-1	Phase 16.1	PickPlace HRL vs Fla	DiscreteSAC+HER vs S	12,000	HRL 30.0%/33.3% vs Flat SAC 0.0%/0.0% vs Phase14 1
EXP-P16-2	Phase 16.2	PickPlace RGB-D Visi	SAC/HRL (Obs=RGBD)	20 eval eps	99.9-100% det, 14.81mm err; SAC 60% grasp/15%; HRL
EXP-P16-3	Phase 16.3	DynamicPickPlace	Tracker+Predictive	5x20 eval eps	100% linear/moving-tgt, 95% sine/disturb (100% gra
EXP-P16-4	Phase 16.4	ContactAssemblyEnv	Compliant Impedance	30 eval eps	Compliant 53.33% vs Phase15 Scripted 50.0% (pos er
EXP-P16-6	Phase 16.6	Autonomous MultiSort	RGBD+HRL+Safety	20 eval eps	25.0% Obj-1 placed, 0% full 2-obj, 229 replans
EXP-P17-1	Phase 17.1	MuJoCo PickPlace	Frozen PyBullet SAC/	5x20 eval eps	Zero-shot 40% contact/5% grasp -> Calib 50%/15%/5%
EXP-P17-2	Phase 17.2	Sim-to-Real Stress	SAC+SafetyMonitor	7x20 eval eps	100% 5mm/jitter/fric-mass, 90% 1-lag, 60% cam cali

Chapter 13: Technical Limitations, Failure Mode Taxonomy, and Scientific Conclusions

13.1 Remaining Open Bottlenecks

Five verified limitations bound current project readiness: (1) Multi-object autonomous RL (Phases 15.2.5C & 16.6) has achieved 25% Object-1 sorting but 0% autonomous end-to-end 2-object RL transfer vs 100% scripted; (2) PyBullet-to-MuJoCo zero-shot accuracy drops from 98% to 0% (5% calibrated); (3) Unprivileged RGB-D visual execution without visual fine-tuning achieves 15-20% vs 100% GT; (4) Learned peg-in-slot insertion remains unsolved (23.33% BC align, 0% full insert vs 53.33% compliant scripted); and (5) No physical Franka hardware execution was attempted.

13.2 Key Scientific Conclusions

Across 17 phases, breakthroughs consistently came from coupling reward geometry with physical mechanics: fingertip friction parity (Ph 6), dense credit shaping to break static HER equilibria (Ph 11-12), inference force/slip shields (+41 pp zero-retraining boost in Ph 14), unbiased action-progress masking in Behavioral Cloning (Ph 15.2.5B7), non-occluded camera geometry (Ph 16.2), and cross-engine falsification (Ph 17.1).

Chapter 14: Future Scope — Phase 18 Architecture Roadmap [UNIMPLEMENTED FUTURE WORK]

14.1 Strict Scope Boundary Statement

Notice: Phase 18 is NOT implemented, executed, or empirically tested anywhere in this repository. All proposals below define future research directions building upon the verified Phase 1-17 artifacts.

14.2 Proposed Phase 18 Engineering Workstreams

18A Dual-Engine MuJoCo/PyBullet Co-Training to close the 98%->5% solver compliance gap; 18B Closed-Loop Visuomotor Fine-Tuning over noisy 15 mm unprojected RGB-D inputs; 18C Long-Horizon Multi-Object BC -> Option-Critic Curriculum; 18D Physical System Identification & Dry-Run libfranka FCI Calibration; and 18E 6-Axis Wrist Wrench Compliant Peg-in-Hole Insertion.

Chapter 15: Annotated Bibliography and Peer-Reviewed References

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Appendix A: Reproducibility Checklists and Repository Artifact Inventory

A.1 Deterministic Environment & Reproducibility Command Specification

Reproduce the verified 122-test suite cleanly inside the project Conda environment with Intel OpenMP coexistence enabled:

set KMP_DUPLICATE_LIB_OK=TRUE && .conda-env\python.exe -m pytest -v

A.2.0 Historical Record: Phase 0 -- Project Initialization & Infrastructure Scaffold

Period: 2026-10-03 (git commit 3a345f3) | Status: COMPLETE (SOURCE_VERIFIED). Objective: Establish Python 3.11/PyBullet/SB3 workspace, YAML ConfigDict engine, directory layout, and smoke import/config tests. Key Components: pyproject.toml, requirements.txt, src/utils/config.py, configs/default.yaml. Outcome: 18/18 initial dependency and ConfigDict tests passed (later rolled into 37/37 Phase 1 suite). Test Evidence: tests/test_imports.py (13 tests), tests/test_config.py (5 tests). Notes: None logged

A.2.1 Historical Record: Phase 1 -- Franka Panda Kinematics, PyBullet Foundation & Cartesian Controller

Period: 2026-10-03 (git commits a5cc06b, 7ad9848) | Status: COMPLETE (SOURCE_VERIFIED). Objective: Implement BaseRobot & FrankaPanda URDF kinematic model, PyBulletSimulation 240 Hz engine, damped least-squares IK, and workspace clamping. Key Components: BaseRobot, FrankaPanda, PyBulletSimulation, PandaController, demo_panda. Outcome: Waypoints reached within 0.7 mm to 4.5 mm position error; gripper sweep 0.00-8.00 cm verified. Test Evidence: 37/37 tests passing in 4.71s (docs/results.md). Notes: Missing per-step wall-clock timestamp logs for early demo trials beyond summary table.

A.2.2 Historical Record: Phase 2 -- Goal-Conditioned Manipulation Environment Abstraction & Reach Benchmark

Period: Oct 2026 (pre-commit 008cd30; recovered from docs/ & tests/) | Status: COMPLETE (REPORT_VERIFIED). Objective: Design BasePandaEnv and Gymnasium Dict-space PandaReachEnv (sparse -1/0 reward, 0.05m threshold, safety bounds). Key Components: BasePandaEnv, PandaReachEnv, VisualTarget. Outcome: Deterministic seeding, step boundaries (50 steps), and 0.05m GoalEnv Dict contract verified. Test Evidence: tests/test_environment.py, docs/reach_benchmark.md. Notes: Separate Phase 2 vs Phase 3 standalone git commits squashed/not split in repo log (marked partially consolidated).

A.2.3 Historical Record: Phase 3 -- SAC + HER RL Training, Evaluation & Callback Infrastructure

Period: Oct 2026 (recovered from src/rl/, logs/reach_sac, checkpoints/reach_sac) | Status: COMPLETE (LOG_VERIFIED). Objective: Integrate Stable-Baselines3 SAC with HerReplayBuffer (future, k=4) on PandaReachEnv across seeds. Key Components: train.py, evaluate.py, SuccessRateCallback. Outcome: 10k SAC Reach: actor_loss -9.127 -> +19.518, critic_loss 0.042 -> 0.002, rollout/success_rate 0.0 -> 0.010, eval 0.0 (10k smoke). Test Evidence: checkpoints/reach_sac/evaluations.npz; logs/reach_sac/SAC_3 TensorBoard scalars. Notes: logs/reach_sac/SAC_1 and SAC_2 contain 0 scalar events (aborted/empty logs before SAC_3).

A.2.4 Historical Record: Phase 4 -- Object Grasping Abstraction & PandaPickEnv

Period: Oct 2026 (recovered from src/objects/cube.py, src/environments/pick_env.py) | Status: COMPLETE (SOURCE_VERIFIED). Objective: Introduce rigid-body Cube manipulation primitive and elevation goal task PandaPickEnv (target Z=0.2m). Key Components: Cube, PandaPickEnv, configs/pick.yaml. Outcome: 10k Pick smoke: actor_loss -14.331 -> +20.005, critic_loss 0.020 -> 0.000, eval reward -100.0, eval success 0.0%. Test Evidence: tests/test_pick_env.py (2 tests), checkpoints/pick_sac/evaluations.npz. Notes: Explicit chronological split document between Phase 4 (Pick) and Phase 5 is NOT_RECOVERABLE from git history; inferred cleanly from config/env artifacts.

A.2.5 Historical Record: Phase 5 -- Reach/Pick Exploration & Early Controller Baseline Scripts

Period: Oct-Nov 2026 | Status: PARTIALLY_COMPLETE (HISTORICALLY_REPORTED). Objective: Validate scripted kinematic reachability and prepare transition from altitude-only Pick to 3D PickPlace. Key Components: PandaPickEnv, PandaReachEnv, scripted diagnostic reach/pick checks. Outcome: Scripted baselines solved Reach (6 steps) and Pick (67 steps), isolating kinematics from RL exploration gap. Test Evidence: docs/phase9_diagnostic_report.md verification entries; exact standalone Phase 5 report file NOT_RECOVERABLE.. Notes: Standalone Phase 5 markdown report absent from docs/; milestone behavior verified via Phase 6 & 9 logs.

A.2.6 Historical Record: Phase 6 -- Pick & Place Environment, HER Comparative Smoke & Finger Friction Bug Fix

Period: Oct-Nov 2026 (experiments/pick_place/smoke_results.md) | Status: COMPLETE (LOG_VERIFIED). Objective: Implement 3D target PandaPickPlaceEnv and verify E1 (No HER) vs E2 (SAC+HER k=4) replay mechanics. Key Components: PandaPickPlaceEnv, E1_no_her, E2_her, Panda finger lateral/spinning friction patch. Outcome: E2_her dropped actor_loss ~22.2 -> 2.59 vs E1 +17.77; collisions 7.7 (E1) -> 0.0 (E2); scripted P-controller solved task at step 74 after finger friction fix. Test Evidence: experiments/pick_place/E1_no_her & E2_her evaluations.npz + TB scalars + smoke_results.md. Notes: CRITICAL PHYSICS BUG: Default PyBullet finger friction caused cube slip on lift; fixed in src/robots/panda.py (lateralFriction=2.0, spinningFriction=0.1 on joints 9 & 10).

A.2.7 Historical Record: Phase 7 -- Color-Conditioned Dynamic Sorting Benchmark (PandaSortEnv)

Period: Oct-Nov 2026 (docs/experiments.md EXP-P7-SORT) | Status: COMPLETE (LOG_VERIFIED). Objective: Implement 3-bin colored SortingBin + PandaSortEnv routing colored cubes to matching dynamic bins. Key Components: SortingBin, PandaSortEnv, configs/sort.yaml. Outcome: Scripted sorting succeeded at Step 79; 10k SAC+HER actor_loss -24.686 -> +3.710, reward -150.0, 0% eval success at 10k smoke. Test Evidence: tests/test_sort_env.py (3 tests), experiments/sort/models/evaluations.npz, logs/sort/SAC_2. Notes: logs/sort/SAC_1 aborted empty (0 scalars) before SAC_2 succeeded.

A.2.8 Historical Record: Phase 8 -- Sequential Multi-Object Sorting & Orientation-Aware Assembly Foundations

Period: Oct-Nov 2026 (docs/experiments.md EXP-P8-MULTI_SORT, EXP-P9-ASSEMBLY) | Status: COMPLETE (LOG_VERIFIED). Objective: Extend sorting to N=2 objects (PandaMultiSortEnv, 31D obs, 6D goal) and introduce 5-DOF orientation-aware PandaAssemblyEnv. Key Components: PandaMultiSortEnv, Peg, Receptacle, PandaAssemblyEnv. Outcome: MultiSort scripted succeeded step 209 (10k RL -600.0, 0%); Assembly scripted succeeded step 143 (10k RL -200.0, 0%). Test Evidence: tests/test_multi_sort_env.py (3 tests), tests/test_assembly_env.py (3 tests), evaluations.npz + TB logs. Notes: Lazy PyBullet sim.client_id=-1 during PandaAssemblyEnv.__init__ (resolved in Phase 9).

A.2.9 Historical Record: Phase 9 -- Full-System Diagnostic Audit & Lazy Simulation Connection Bug Fix

Period: Nov 2026 (docs/phase9_diagnostic_report.md) | Status: COMPLETE (REPORT_VERIFIED). Objective: Audit all subsystems, scripted trajectories, HER reward bounds, and resolve initialization bugs prior to scaling. Key Components: System-wide audit suite, PandaAssemblyEnv.reset() lazy client fix. Outcome: 52/52 tests passed in 6.47s; 100% scripted reachability confirmed. Test Evidence: docs/phase9_diagnostic_report.md (52/52 pytest pass). Notes: Fixed PandaAssemblyEnv.__init__ client_id=-1 crash by lazily passing sim.client_id inside reset().

A.2.10 Historical Record: Phase 10 -- Deterministic Domain Randomization Engine (DR-0 to DR-4)

Period: Nov 2026 (docs/phase10_dr_report.md) | Status: COMPLETE (REPORT_VERIFIED). Objective: Implement seeded DomainRandomizer covering object/target pos, Shoemake yaw, mass (0.05-0.20 kg), and friction (0.5-1.5). Key Components: DomainRandomizer, BasePandaEnv DR hook, pick_place_dr*.yaml. Outcome: Scripted success: R0 90%, R1 85%, R2 100%, R3 85%, R4 75% (slip under randomized mass/friction). Test Evidence: 56/56 tests passing including tests/test_randomization.py (4 tests). Notes: 25% scripted heuristic drop on R4 physics proved RL feedback control necessity.

A.2.11 Historical Record: Phase 11 -- 100k Sparse DR Matrix & Diagnostic Isolation Series (11.5-11.10b)

Period: Nov 2026 (docs/phase11_*.md, results/phase11_*.csv) | Status: COMPLETE (LOG_VERIFIED). Objective: Train 25 sparse 100k models (DR0-DR4 x 5 seeds), diagnose 0% sparse trap, isolate transport vs acquisition via dense reward & curriculum. Key Components: PandaPickPlaceDenseRewardEnv, PandaCurriculumEnv, PandaTransportDiagnosticEnv. Outcome: Proved isolated transport reaches 100% in 40k steps, whereas sparse 100k suffered HER static-resting-goal trap (0%). Test Evidence: results/phase11_final_summary.csv + 11_6..11_10b metrics CSVs. Notes: HER Trap: 0% object motion caused HER to sample unmoved resting goals; Stage 1 curriculum failed at 50k.

A.2.12 Historical Record: Phase 12 -- 1M-Step Dense SAC+HER Scaling Breakthrough

Period: Nov 2026 (docs/phase12_1m_dense_scaling_report.md) | Status: COMPLETE (LOG_VERIFIED). Objective: Extend dense Seed 42 from 225k to 1,000,000 steps to test compute-limited hypothesis. Key Components: train_phase12.py, checkpoint_225k -> 1M resume. Outcome: S-curve breakthrough: 0% (300k) -> 43% (400k) -> 99% (500k) -> 100% (600k..1M), 0.014m final dist, +502.2 reward. Test Evidence: results/phase12_1m/checkpoint_metrics.csv & failure_taxonomy.csv. Notes: Single-seed DR-0 only; OOD generalization unverified until Phase 13.

A.2.13 Historical Record: Phase 13 -- Dense Domain Randomization (DR1/DR2/DR4) & Causal OOD Failure Diagnostics (13.5)

Period: 2026-11-03 to 2026-11-04 (docs/phase13_*.md) | Status: COMPLETE (LOG_VERIFIED). Objective: Train Dense DR1/DR2/DR4 policies (500k-1M) and isolate OOD Held-out collapse via D1-D5 controlled sweeps. Key Components: train_phase13_dr.py, run_phase13_5_diagnostics.py, DR4 Seed42/43 500k checkpoints. Outcome: D5 isolated Stage-3 vertical jerk (0.14 m/s^2) causing normal force collapse 68.4 -> 0.0 N (30.5% GRASP_BUT_NO_LIFT). Test Evidence: results/phase13_5/d1..d5 CSVs and fast_track matched/heldout logs. Notes: DR4 Seed 43 dropped to 33.0% final OOD due to corner misses and lift slip.

A.2.14 Historical Record: Phase 14 -- Zero-Retraining Adaptive Control Suite (14-Lite -> 14.3 -> 14.4)

Period: 2026-11-10 (results/phase14_*, docs/phase14_closure_and_phase15_readiness.md) | Status: COMPLETE (LOG_VERIFIED). Objective: Resolve OOD slip, workspace-edge acquisition, and transport overshoot purely via inference controllers without SAC retraining. Key Components: Controller B (AccelSmoothing), Controller C (SlipAware), Controller D (DualStage), Controller E (AdaptiveTransport). Outcome: Held-out DR4 jumped 57.0% (A) -> 89.5% (D) -> 98.0% (E_ADAPTIVE_TRANSPORT); Seed 43: 33% -> 99%; DR0: 100%. Test Evidence: results/phase14_4/phase14_4_confirmation_results.json. Notes: Relies on oracle cube_pos and noiseless PyBullet getContactPoints force telemetry.

A.2.15 Historical Record: Phase 15 -- Sim-to-Real Sensitivity (15.1), Multi-Object Sorting (15.2) & Orientation Assembly (15.3)

Period: 2026-11-10 to 2026-11-11 (results/phase15_*) | Status: PARTIALLY_COMPLETE (SOURCE_VERIFIED). Objective: Benchmark Phase 14 under noise/latency/unprivileged sensors; build dense MultiSort, PreGrasp BC+SAC, robustness suite & 5-DOF Assembly. Key Components: ObservationNoiseWrapper, LatencyWrapper, UnprivilegedContactWrapper, PreGraspController, BCDecayingSAC, PandaSinglePreGraspEnv, PandaAssemblyDenseEnv, 28D->43D adapter. Outcome: Phase 2 Two-Object RL transfer BLOCKED on >=85% RL lift gate; 111/111 integration pytest passing in 29.47s. Test Evidence: tests/test_phase15_integration.py & 111/111 full suite pass. Notes: Premature gripper closure, HER unlifted-goal gradient flip, BC-to-SAC offline-to-online forgetting (90% -> 0-10%).

A.2.16 Historical Record: Phase 16 -- Hierarchical RL, Oblique RGB-D Vision, Dynamic Tracking, Contact Assembly & Deployment

Period: 2026-11-11 to 2026-11-12 (results/phase16_17/) | Status: COMPLETE (SOURCE_VERIFIED). Objective: Implement Discrete-SAC+HER HRL (16.1), 128x128 RGB-D 6D localization (16.2), dynamic intercept tracking (16.3), compliant assembly (16.4), decoupled runtime (16.5), autonomous 2-obj sorting (16.6). Key Components: SkillLibrary, HierarchicalManipulatorAgent, SimulatedRGBDCamera, VisionObjectLocalizer, DynamicObjectTracker, PandaContactAssemblyEnv, CompliantImpedanceAssemblyController, DeploymentRuntime. Outcome: HRL beat Flat SAC by +30 pp at 12k; predictive dynamic hit 95-100%; compliant scripted improved assembly 50%->53.33%; 122/122 tests pass. Test Evidence: tests/test_deployment_runtime.py, test_hrl_phase16.py, test_vision_and_dynamic_phase16.py, test_phase16_17_envs.py. Notes: Overhead self-occlusion (18.65% det fixed to 100%), PandaPickPlaceEnv z=0 default spawn when randomizer disabled, HRL hover +0.35 reward hack.

A.2.17 Historical Record: Phase 17 -- MuJoCo Cross-Simulator Validation, Sim-to-Real Readiness & Hardware Contract

Period: 2026-11-12 (results/phase16_17/) | Status: PARTIALLY_COMPLETE (SOURCE_VERIFIED). Objective: Validate frozen PyBullet policies inside MuJoCo 4.0.1 (17.1), execute 7-condition stress + E-stop tests (17.2), and specify ROS-2/Franka safety interfaces (17.3). Key Components: MuJoCoPandaPickPlaceEnv, SafetySupervisor, StubLiveFrankaDriver, ROS2TopicContract. Outcome: MuJoCo zero-shot 40% bilateral contact/5% grasp -> calibrated 50%/15%/5% PickPlace; Stress: 100% noise/jitter/mass-fric, 90% 1-step lag, 60% RGB-D calib bias, 20% 2-step lag; E-stop 32.41 us. Test Evidence: results/phase16_17/phase17_cross_simulator_results.json & phase17_safety_assessment.json. Notes: 9.4mm tabletop rest Z & elliptic slip gap PyBullet->MuJoCo; 17.3 live robot execution NOT_ATTEMPTED.

Appendix B: Historical TensorBoard & Evaluations NPZ Forensic Ledger

B.1 All 61 Archived TensorBoard Scalar Runs (Phases 2-15)

Table B.1: Complete Registry of All 61 Archived TensorBoard Runs

Log Directory	Max Step	Actor Loss	Critic Loss	Ent Coef	CPU FPS
experiments/phase13_dense_dr/dr1/seed42/logs/SAC_0	1,000,000	-106.363	1.195	0.034	26.000
experiments/phase13_dense_dr/dr1/seed43/logs/SAC_0	1,000,000	-104.216	4.040	0.028	21.000
experiments/phase13_dense_dr/dr1/seed44/logs/SAC_0	500,400	-89.780	0.221	0.033	14.000
experiments/phase13_dense_dr/dr1/seed45/logs/SAC_0	500,400	-91.694	0.119	0.023	15.000
experiments/phase13_dense_dr/dr2/seed42/logs/SAC_0	500,000	-92.855	3.646	0.036	19.000
experiments/phase13_dense_dr/dr2/seed43/logs/SAC_0	500,000	-85.336	5.218	0.088	18.000
experiments/phase13_dense_dr/dr4/seed42/logs/SAC_0	500,000	-96.356	3.303	0.050	15.000
experiments/phase13_dense_dr/dr4/seed43/logs/SAC_0	500,000	-86.009	13.122	0.021	15.000
experiments/pick_place/E1_no_her/logs/SAC_1	10,000	17.767	0.019	0.003	21.000
experiments/pick_place/E2_her/logs/SAC_1	10,000	2.594	0.012	0.005	22.000
experiments/pick_place/E2_her/logs/SAC_2	10,000	3.061	0.013	0.005	35.000
experiments/pnp_curriculum_seed42/logs/SAC_0	50,000	-78.466	0.666	0.021	31.000
experiments/pnp_dr0_dense_seed42/logs/SAC_1	1,000,000	-104.833	2.408	0.028	33.000
experiments/pnp_dr0_seed42/logs/SAC_1	100,000	-1756.392	154204.406	3.902	30.000
experiments/pnp_dr0_seed43/logs/SAC_1	100,000	3.240	0.007	0.001	30.000
experiments/pnp_dr0_seed44/logs/SAC_1	100,000	3.723	0.003	0.001	39.000
experiments/pnp_dr0_seed45/logs/SAC_1	100,000	4.024	0.010	0.000	34.000
experiments/pnp_dr0_seed46/logs/SAC_1	100,000	2.200	0.004	0.003	31.000
experiments/pnp_dr0_smoke_seed42/logs/SAC_1	10,000	3.950	0.003	0.004	32.000
experiments/pnp_dr1_seed42/logs/SAC_1	100,000	4.089	0.002	0.000	35.000
experiments/pnp_dr1_seed43/logs/SAC_1	100,000	3.359	0.002	0.001	36.000
experiments/pnp_dr1_seed44/logs/SAC_1	100,000	3.377	0.011	0.001	38.000
experiments/pnp_dr1_seed45/logs/SAC_1	100,000	-0.915	1.902	0.002	35.000
experiments/pnp_dr1_seed46/logs/SAC_1	100,000	3.754	0.003	0.001	31.000
experiments/pnp_dr2_seed42/logs/SAC_1	100,000	3.898	0.009	0.001	45.000
experiments/pnp_dr2_seed43/logs/SAC_1	100,000	4.098	0.004	0.000	37.000
experiments/pnp_dr2_seed44/logs/SAC_1	100,000	-0.765	0.013	0.004	36.000
experiments/pnp_dr2_seed45/logs/SAC_1	100,000	3.189	0.005	0.001	38.000
experiments/pnp_dr2_seed46/logs/SAC_1	100,000	3.948	0.017	0.001	34.000
experiments/pnp_dr3_seed42/logs/SAC_1	100,000	3.865	0.005	0.000	34.000
experiments/pnp_dr3_seed43/logs/SAC_1	100,000	3.824	0.003	0.001	35.000
experiments/pnp_dr3_seed44/logs/SAC_1	100,000	3.742	0.008	0.001	35.000
experiments/pnp_dr3_seed45/logs/SAC_1	100,000	3.430	0.005	0.001	34.000
experiments/pnp_dr3_seed46/logs/SAC_1	100,000	3.829	0.002	0.001	33.000
experiments/pnp_dr4_seed42/logs/SAC_1	100,000	4.037	0.099	0.001	34.000
experiments/pnp_dr4_seed43/logs/SAC_1	100,000	3.879	0.003	0.001	40.000
experiments/pnp_dr4_seed44/logs/SAC_1	100,000	3.980	0.002	0.000	34.000
experiments/pnp_dr4_seed45/logs/SAC_1	100,000	3.971	0.017	0.000	29.000
experiments/pnp_dr4_seed46/logs/SAC_1	100,000	3.365	0.004	0.001	27.000
experiments/pnp_transport_seed42/logs/SAC_0	50,000	0.488	0.000	0.001	28.000
logs/assembly/SAC_1	10,000	3.264	0.019	0.003	30.000
logs/assembly_phase15_3/seed42/SAC_1	5,000	-4.246	0.523	0.020	31.000
logs/dr_baseline/SAC_1	2,000	-8.720	0.022	0.174	54.000

Log Directory	Max Step	Actor Loss	Critic Loss	Ent Coef	CPU FPS
logs/multi_sort/SAC_1	10,000	7.383	0.012	0.004	33.000
logs/multi_sort_phase15_4/seed42/SAC_1	50,000	-12.406	0.176	0.020	25.000
logs/multi_sort_phase15_5B/seed42/SAC_1	20,000	-6.327	0.002	0.004	22.000
logs/multi_sort_phase15_5B1/seed42/SAC_1	20,000	-9.709	0.156	0.008	15.000
logs/multi_sort_phase15_5B2/seed42/SAC_1	20,000	-7.739	0.085	0.020	28.000
logs/multi_sort_phase15_5B3/seed42/SAC_1	5,000	-20.202	0.119	0.255	33.000
logs/multi_sort_phase15_5B4/seed42/SAC_1	10,000	-7.946	0.137	0.015	32.000
logs/multi_sort_phase15_5B5/seed42/SAC_1	10,000	-5.261	0.087	0.015	30.000
logs/multi_sort_phase15_5B6/seed42/SAC_1	10,000	-3.257	0.096	0.015	35.000
logs/multi_sort_phase15_5B7/seed42/SAC_1	10,000	-3.703	0.093	0.015	34.000
logs/multi_sort_phase15_5B8/seed42/SAC_1	10,000	-6.117	0.897	0.027	40.000
logs/multi_sort_phase15_dense/seed42/SAC_1	10,000	-20.980	2.921	0.072	14.000
logs/pick_sac/SAC_1	10,000	20.005	0.000	0.000	41.000
logs/reach_sac/SAC_1	0	-	-	-	Aborted / Empty
logs/reach_sac/SAC_2	0	-	-	-	Aborted / Empty
logs/reach_sac/SAC_3	10,000	19.518	0.002	0.002	49.000
logs/sort/SAC_1	0	-	-	-	Aborted / Empty
logs/sort/SAC_2	10,000	3.710	0.001	0.002	42.000

B.2 All 44 Evaluation NPZ Checkpoint Archives

Table B.2: Complete Registry of All 44 evaluations.npz Archives

Experiment Path	Evals	Last Step	Mean Rew (Start -> End)	Peak Success
checkpoints/pick_sac	4	10,000	-100.0 -> -100.0	0.0%
checkpoints/reach_sac	4	10,000	-50.0 -> -50.0	0.0%
experiments/assembly	4	10,000	-200.0 -> -200.0	0.0%
experiments/assembly_phase15_3/seed42	2	5,000	-8.3 -> -6.5	0.0%
experiments/multi_sort	4	10,000	-600.0 -> -600.0	0.0%
experiments/multi_sort_phase15_4/seed42	4	50,000	-27.0 -> -24.5	0.0%
experiments/multi_sort_phase15_5B1/seed42	4	20,000	-6.2 -> -9.4	0.0%
experiments/multi_sort_phase15_5B2/seed42	6	20,000	-5.1 -> -2.9	0.0%
experiments/multi_sort_phase15_5B3/seed42	2	5,000	-2.8 -> -2.2	0.0%
experiments/multi_sort_phase15_5B4/seed42	4	10,000	-8.0 -> -5.6	0.0%
experiments/multi_sort_phase15_5B5/seed42	4	10,000	-4.0 -> -6.6	0.0%
experiments/multi_sort_phase15_5B6/seed42	4	10,000	-3.1 -> -5.6	10.0%
experiments/multi_sort_phase15_5B7/seed42	4	10,000	3.9 -> -7.7	20.0%
experiments/multi_sort_phase15_5B8/seed42	4	10,000	0.7 -> -5.9	0.0%
experiments/multi_sort_phase15_5B/seed42	4	20,000	-2.6 -> -1.3	0.0%
experiments/multi_sort_phase15_dense/seed42	4	10,000	-6.0 -> -5.5	0.0%
experiments/pick_place/E1_no_her	4	10,000	-100.0 -> -100.0	0.0%
experiments/pick_place/E2_her	4	10,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr0_seed42	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr0_seed43	40	100,000	-90.0 -> -100.0	10.0%
experiments/pnp_dr0_seed44	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr0_seed45	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr0_seed46	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr0_smoke_seed42	4	10,000	-100.0 -> -100.0	0.0%

Experiment Path	Evals	Last Step	Mean Rew (Start -> End)	Peak Success
experiments/pnp_dr1_seed42	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr1_seed43	40	100,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr1_seed44	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr1_seed45	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr1_seed46	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr2_seed42	40	100,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr2_seed43	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr2_seed44	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr2_seed45	40	100,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr2_seed46	40	100,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr3_seed42	40	100,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr3_seed43	40	100,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr3_seed44	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr3_seed45	40	100,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr3_seed46	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr4_seed42	40	100,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr4_seed43	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr4_seed44	40	100,000	-100.0 -> -100.0	0.0%
experiments/pnp_dr4_seed45	40	100,000	-100.0 -> -100.0	10.0%
experiments/pnp_dr4_seed46	40	100,000	-100.0 -> -100.0	10.0%
experiments/sort	4	10,000	-150.0 -> -150.0	0.0%

Appendix C: Phase 13 Seed-Level DR Checkpoints & Failure Taxonomy Ledger

C.1 All Phase 13 DR1, DR2, and DR4 Seed Checkpoint Progressions (60 Evaluations)

Table C.1: Phase 13 Seed-Resolved Evaluation Checkpoints (DR1/DR2/DR4)

Run	Step	Regime	Reach	Grasp	Lift	Success	Final Dist
DR1/seed42	250000	Nominal_DR0	63%	50%	7%	7%	0.2694 m
DR1/seed42	250000	Matched_DR1	65%	49%	11%	11%	0.2689 m
DR1/seed42	250000	Heldout_Stress	0%	0%	0%	0%	0.5698 m
DR1/seed42	500000	Nominal_DR0	100%	100%	100%	100%	0.0154 m
DR1/seed42	500000	Matched_DR1	100%	100%	100%	100%	0.0152 m
DR1/seed42	500000	Heldout_Stress	99%	95%	44%	43%	0.1762 m
DR1/seed42	750000	Nominal_DR0	100%	100%	100%	100%	0.0101 m
DR1/seed42	750000	Matched_DR1	100%	100%	100%	100%	0.0104 m
DR1/seed42	750000	Heldout_Stress	100%	100%	95%	95%	0.0207 m
DR1/seed42	1000000	Nominal_DR0	100%	100%	100%	100%	0.0109 m
DR1/seed42	1000000	Matched_DR1	100%	100%	100%	100%	0.0112 m
DR1/seed42	1000000	Heldout_Stress	100%	100%	96%	96%	0.0260 m
DR1/seed43	250000	Nominal_DR0	31%	16%	5%	5%	0.2323 m
DR1/seed43	250000	Matched_DR1	36%	21%	5%	7%	0.2398 m
DR1/seed43	250000	Heldout_Stress	1%	1%	0%	0%	0.5688 m
DR1/seed43	500000	Nominal_DR0	100%	99%	99%	99%	0.0185 m
DR1/seed43	500000	Matched_DR1	100%	100%	100%	100%	0.0134 m
DR1/seed43	500000	Heldout_Stress	100%	100%	82%	82%	0.0974 m
DR1/seed43	750000	Nominal_DR0	100%	100%	100%	100%	0.0141 m
DR1/seed43	750000	Matched_DR1	100%	100%	100%	100%	0.0132 m
DR1/seed43	750000	Heldout_Stress	100%	100%	72%	72%	0.1052 m
DR1/seed43	1000000	Nominal_DR0	100%	100%	100%	100%	0.0195 m
DR1/seed43	1000000	Matched_DR1	100%	100%	100%	100%	0.0200 m
DR1/seed43	1000000	Heldout_Stress	100%	100%	51%	51%	0.1401 m
DR1/seed44	250000	Nominal_DR0	1%	0%	0%	0%	0.2419 m
DR1/seed44	250000	Matched_DR1	1%	1%	0%	2%	0.2458 m
DR1/seed44	250000	Heldout_Stress	12%	1%	0%	0%	0.5668 m
DR1/seed44	500000	Nominal_DR0	100%	100%	100%	100%	0.0173 m
DR1/seed44	500000	Matched_DR1	100%	100%	100%	100%	0.0175 m
DR1/seed44	500000	Heldout_Stress	100%	100%	94%	94%	0.0418 m
DR1/seed45	250000	Nominal_DR0	100%	96%	83%	83%	0.0669 m
DR1/seed45	250000	Matched_DR1	100%	96%	81%	82%	0.0681 m
DR1/seed45	250000	Heldout_Stress	100%	94%	53%	52%	0.1532 m
DR1/seed45	500000	Nominal_DR0	100%	100%	100%	100%	0.0194 m
DR1/seed45	500000	Matched_DR1	100%	100%	100%	100%	0.0187 m
DR1/seed45	500000	Heldout_Stress	100%	100%	79%	79%	0.0564 m
DR2/seed42	250000	Nominal_DR0	22%	8%	0%	2%	0.2354 m
DR2/seed42	250000	Matched_DR2	22%	11%	1%	2%	0.2184 m
DR2/seed42	250000	Heldout_Stress	1%	0%	0%	0%	0.4820 m
DR2/seed42	500000	Nominal_DR0	100%	100%	100%	100%	0.0323 m
DR2/seed42	500000	Matched_DR2	100%	100%	98%	99%	0.0349 m
DR2/seed42	500000	Heldout_Stress	100%	100%	80%	80%	0.0813 m
DR2/seed43	250000	Nominal_DR0	67%	55%	15%	9%	0.2089 m

Run	Step	Regime	Reach	Grasp	Lift	Success	Final Dist
DR2/seed43	250000	Matched_DR2	71%	54%	6%	9%	0.2210 m
DR2/seed43	250000	Heldout_Stress	60%	57%	2%	0%	0.4613 m
DR2/seed43	500000	Nominal_DR0	100%	100%	98%	98%	0.0194 m
DR2/seed43	500000	Matched_DR2	100%	100%	99%	99%	0.0171 m
DR2/seed43	500000	Heldout_Stress	100%	100%	95%	95%	0.0386 m
DR4/seed42	250000	Nominal_DR0	96%	80%	26%	27%	0.2415 m
DR4/seed42	250000	Matched_DR4	95%	82%	30%	32%	0.2102 m
DR4/seed42	250000	Heldout_Stress	0%	0%	0%	0%	0.5698 m
DR4/seed42	500000	Nominal_DR0	100%	100%	100%	100%	0.0142 m
DR4/seed42	500000	Matched_DR4	100%	100%	100%	100%	0.0156 m
DR4/seed42	500000	Heldout_Stress	100%	100%	82%	81%	0.0713 m
DR4/seed43	250000	Nominal_DR0	29%	25%	1%	2%	0.2436 m
DR4/seed43	250000	Matched_DR4	34%	25%	2%	3%	0.2469 m
DR4/seed43	250000	Heldout_Stress	20%	17%	0%	0%	0.5764 m
DR4/seed43	500000	Nominal_DR0	99%	99%	96%	96%	0.0399 m
DR4/seed43	500000	Matched_DR4	100%	99%	98%	98%	0.0363 m
DR4/seed43	500000	Heldout_Stress	93%	84%	41%	41%	0.3054 m

C.2 Non-Zero Episode Failure Taxonomy Distribution (152 Entries)

Table C.2: Phase 13 Multi-Seed Failure Category Distribution

Run	Step	Regime	Failure Mode	Count	Fraction
DR1/seed42	250000	Nominal_DR0	SUCCESS	7	7.0%
DR1/seed42	250000	Nominal_DR0	NO_REACH	37	37.0%
DR1/seed42	250000	Nominal_DR0	REACH_BUT_NO_GRASP	13	13.0%
DR1/seed42	250000	Nominal_DR0	GRASP_BUT_NO_LIFT	43	43.0%
DR1/seed42	250000	Matched_DR1	SUCCESS	11	11.0%
DR1/seed42	250000	Matched_DR1	NO_REACH	34	34.0%
DR1/seed42	250000	Matched_DR1	REACH_BUT_NO_GRASP	16	16.0%
DR1/seed42	250000	Matched_DR1	GRASP_BUT_NO_LIFT	38	38.0%
DR1/seed42	250000	Matched_DR1	LIFT_BUT_DROPPED	1	1.0%
DR1/seed42	250000	Heldout_Stress	NO_REACH	100	100.0%
DR1/seed42	500000	Nominal_DR0	SUCCESS	100	100.0%
DR1/seed42	500000	Matched_DR1	SUCCESS	100	100.0%
DR1/seed42	500000	Heldout_Stress	SUCCESS	43	43.0%
DR1/seed42	500000	Heldout_Stress	NO_REACH	1	1.0%
DR1/seed42	500000	Heldout_Stress	REACH_BUT_NO_GRASP	4	4.0%
DR1/seed42	500000	Heldout_Stress	GRASP_BUT_NO_LIFT	51	51.0%
DR1/seed42	500000	Heldout_Stress	LIFT_AND_HELD_BUT_WRONG_DIRECTION	1	1.0%
DR1/seed42	750000	Nominal_DR0	SUCCESS	100	100.0%
DR1/seed42	750000	Matched_DR1	SUCCESS	100	100.0%
DR1/seed42	750000	Heldout_Stress	SUCCESS	95	95.0%
DR1/seed42	750000	Heldout_Stress	GRASP_BUT_NO_LIFT	5	5.0%
DR1/seed42	1000000	Nominal_DR0	SUCCESS	100	100.0%
DR1/seed42	1000000	Matched_DR1	SUCCESS	100	100.0%
DR1/seed42	1000000	Heldout_Stress	SUCCESS	96	96.0%
DR1/seed42	1000000	Heldout_Stress	GRASP_BUT_NO_LIFT	4	4.0%
DR1/seed43	250000	Nominal_DR0	SUCCESS	5	5.0%
DR1/seed43	250000	Nominal_DR0	NO_REACH	69	69.0%

Run	Step	Regime	Failure Mode	Count	Fraction
DR1/seed43	250000	Nominal_DR0	REACH_BUT_NO_GRASP	15	15.0%
DR1/seed43	250000	Nominal_DR0	GRASP_BUT_NO_LIFT	10	10.0%
DR1/seed43	250000	Nominal_DR0	LIFT_BUT_DROPPED	1	1.0%
DR1/seed43	250000	Matched_DR1	SUCCESS	7	7.0%
DR1/seed43	250000	Matched_DR1	NO_REACH	63	63.0%
DR1/seed43	250000	Matched_DR1	REACH_BUT_NO_GRASP	14	14.0%
DR1/seed43	250000	Matched_DR1	GRASP_BUT_NO_LIFT	12	12.0%
DR1/seed43	250000	Matched_DR1	LIFT_BUT_DROPPED	1	1.0%
DR1/seed43	250000	Matched_DR1	LIFT_AND_HELD_BUT_WRONG_DIRECTION	3	3.0%
DR1/seed43	250000	Heldout_Stress	NO_REACH	99	99.0%
DR1/seed43	250000	Heldout_Stress	GRASP_BUT_NO_LIFT	1	1.0%
DR1/seed43	500000	Nominal_DR0	SUCCESS	99	99.0%
DR1/seed43	500000	Nominal_DR0	REACH_BUT_NO_GRASP	1	1.0%
DR1/seed43	500000	Matched_DR1	SUCCESS	100	100.0%
DR1/seed43	500000	Heldout_Stress	SUCCESS	82	82.0%
DR1/seed43	500000	Heldout_Stress	GRASP_BUT_NO_LIFT	18	18.0%
DR1/seed43	750000	Nominal_DR0	SUCCESS	100	100.0%
DR1/seed43	750000	Matched_DR1	SUCCESS	100	100.0%
DR1/seed43	750000	Heldout_Stress	SUCCESS	72	72.0%
DR1/seed43	750000	Heldout_Stress	GRASP_BUT_NO_LIFT	28	28.0%
DR1/seed43	1000000	Nominal_DR0	SUCCESS	100	100.0%
DR1/seed43	1000000	Matched_DR1	SUCCESS	100	100.0%
DR1/seed43	1000000	Heldout_Stress	SUCCESS	51	51.0%
DR1/seed43	1000000	Heldout_Stress	GRASP_BUT_NO_LIFT	49	49.0%
DR1/seed44	250000	Nominal_DR0	NO_REACH	99	99.0%
DR1/seed44	250000	Nominal_DR0	REACH_BUT_NO_GRASP	1	1.0%
DR1/seed44	250000	Matched_DR1	SUCCESS	2	2.0%
DR1/seed44	250000	Matched_DR1	NO_REACH	98	98.0%
DR1/seed44	250000	Heldout_Stress	NO_REACH	88	88.0%
DR1/seed44	250000	Heldout_Stress	REACH_BUT_NO_GRASP	11	11.0%
DR1/seed44	250000	Heldout_Stress	GRASP_BUT_NO_LIFT	1	1.0%
DR1/seed44	500000	Nominal_DR0	SUCCESS	100	100.0%
DR1/seed44	500000	Matched_DR1	SUCCESS	100	100.0%
DR1/seed44	500000	Heldout_Stress	SUCCESS	94	94.0%
DR1/seed44	500000	Heldout_Stress	GRASP_BUT_NO_LIFT	6	6.0%
DR1/seed45	250000	Nominal_DR0	SUCCESS	83	83.0%
DR1/seed45	250000	Nominal_DR0	REACH_BUT_NO_GRASP	4	4.0%
DR1/seed45	250000	Nominal_DR0	GRASP_BUT_NO_LIFT	13	13.0%
DR1/seed45	250000	Matched_DR1	SUCCESS	82	82.0%
DR1/seed45	250000	Matched_DR1	REACH_BUT_NO_GRASP	4	4.0%
DR1/seed45	250000	Matched_DR1	GRASP_BUT_NO_LIFT	13	13.0%
DR1/seed45	250000	Matched_DR1	LIFT_BUT_DROPPED	1	1.0%
DR1/seed45	250000	Heldout_Stress	SUCCESS	52	52.0%
DR1/seed45	250000	Heldout_Stress	REACH_BUT_NO_GRASP	6	6.0%
DR1/seed45	250000	Heldout_Stress	GRASP_BUT_NO_LIFT	41	41.0%
DR1/seed45	250000	Heldout_Stress	LIFT_BUT_DROPPED	1	1.0%
DR1/seed45	500000	Nominal_DR0	SUCCESS	100	100.0%
DR1/seed45	500000	Matched_DR1	SUCCESS	100	100.0%
DR1/seed45	500000	Heldout_Stress	SUCCESS	79	79.0%

Run	Step	Regime	Failure Mode	Count	Fraction
DR1/seed45	500000	Heldout_Stress	GRASP_BUT_NO_LIFT	21	21.0%
DR2/seed42	250000	Nominal_DR0	SUCCESS	2	2.0%
DR2/seed42	250000	Nominal_DR0	NO_REACH	78	78.0%
DR2/seed42	250000	Nominal_DR0	REACH_BUT_NO_GRASP	13	13.0%
DR2/seed42	250000	Nominal_DR0	GRASP_BUT_NO_LIFT	7	7.0%
DR2/seed42	250000	Matched_DR2	SUCCESS	2	2.0%
DR2/seed42	250000	Matched_DR2	NO_REACH	78	78.0%
DR2/seed42	250000	Matched_DR2	REACH_BUT_NO_GRASP	11	11.0%
DR2/seed42	250000	Matched_DR2	GRASP_BUT_NO_LIFT	9	9.0%
DR2/seed42	250000	Heldout_Stress	NO_REACH	99	99.0%
DR2/seed42	250000	Heldout_Stress	REACH_BUT_NO_GRASP	1	1.0%
DR2/seed42	500000	Nominal_DR0	SUCCESS	100	100.0%
DR2/seed42	500000	Matched_DR2	SUCCESS	99	99.0%
DR2/seed42	500000	Matched_DR2	GRASP_BUT_NO_LIFT	1	1.0%
DR2/seed42	500000	Heldout_Stress	SUCCESS	80	80.0%
DR2/seed42	500000	Heldout_Stress	GRASP_BUT_NO_LIFT	20	20.0%
DR2/seed43	250000	Nominal_DR0	SUCCESS	9	9.0%
DR2/seed43	250000	Nominal_DR0	NO_REACH	33	33.0%
DR2/seed43	250000	Nominal_DR0	REACH_BUT_NO_GRASP	12	12.0%
DR2/seed43	250000	Nominal_DR0	GRASP_BUT_NO_LIFT	40	40.0%
DR2/seed43	250000	Nominal_DR0	LIFT_BUT_DROPPED	4	4.0%
DR2/seed43	250000	Nominal_DR0	LIFT_AND_HELD_BUT_WRONG_DIRECTION	2	2.0%
DR2/seed43	250000	Matched_DR2	SUCCESS	9	9.0%
DR2/seed43	250000	Matched_DR2	NO_REACH	28	28.0%
DR2/seed43	250000	Matched_DR2	REACH_BUT_NO_GRASP	17	17.0%
DR2/seed43	250000	Matched_DR2	GRASP_BUT_NO_LIFT	44	44.0%
DR2/seed43	250000	Matched_DR2	LIFT_BUT_DROPPED	1	1.0%
DR2/seed43	250000	Matched_DR2	LIFT_AND_HELD_BUT_WRONG_DIRECTION	1	1.0%
DR2/seed43	250000	Heldout_Stress	NO_REACH	40	40.0%
DR2/seed43	250000	Heldout_Stress	REACH_BUT_NO_GRASP	3	3.0%
DR2/seed43	250000	Heldout_Stress	GRASP_BUT_NO_LIFT	55	55.0%
DR2/seed43	250000	Heldout_Stress	LIFT_AND_HELD_BUT_WRONG_DIRECTION	2	2.0%
DR2/seed43	500000	Nominal_DR0	SUCCESS	98	98.0%
DR2/seed43	500000	Nominal_DR0	GRASP_BUT_NO_LIFT	2	2.0%
DR2/seed43	500000	Matched_DR2	SUCCESS	99	99.0%
DR2/seed43	500000	Matched_DR2	GRASP_BUT_NO_LIFT	1	1.0%
DR2/seed43	500000	Heldout_Stress	SUCCESS	95	95.0%
DR2/seed43	500000	Heldout_Stress	GRASP_BUT_NO_LIFT	5	5.0%
DR4/seed42	250000	Nominal_DR0	SUCCESS	27	27.0%
DR4/seed42	250000	Nominal_DR0	NO_REACH	3	3.0%
DR4/seed42	250000	Nominal_DR0	REACH_BUT_NO_GRASP	16	16.0%
DR4/seed42	250000	Nominal_DR0	GRASP_BUT_NO_LIFT	51	51.0%
DR4/seed42	250000	Nominal_DR0	LIFT_BUT_DROPPED	3	3.0%
DR4/seed42	250000	Matched_DR4	SUCCESS	32	32.0%
DR4/seed42	250000	Matched_DR4	NO_REACH	5	5.0%
DR4/seed42	250000	Matched_DR4	REACH_BUT_NO_GRASP	13	13.0%
DR4/seed42	250000	Matched_DR4	GRASP_BUT_NO_LIFT	49	49.0%
DR4/seed42	250000	Matched_DR4	LIFT_BUT_DROPPED	1	1.0%
DR4/seed42	250000	Heldout_Stress	NO_REACH	100	100.0%

Run	Step	Regime	Failure Mode	Count	Fraction
DR4/seed42	500000	Nominal_DR0	SUCCESS	100	100.0%
DR4/seed42	500000	Matched_DR4	SUCCESS	100	100.0%
DR4/seed42	500000	Heldout_Stress	SUCCESS	81	81.0%
DR4/seed42	500000	Heldout_Stress	GRASP_BUT_NO_LIFT	18	18.0%
DR4/seed42	500000	Heldout_Stress	LIFT_AND_HELD_BUT_WRONG_DIRECTION	1	1.0%
DR4/seed43	250000	Nominal_DR0	SUCCESS	2	2.0%
DR4/seed43	250000	Nominal_DR0	NO_REACH	71	71.0%
DR4/seed43	250000	Nominal_DR0	REACH_BUT_NO_GRASP	4	4.0%
DR4/seed43	250000	Nominal_DR0	GRASP_BUT_NO_LIFT	23	23.0%
DR4/seed43	250000	Matched_DR4	SUCCESS	3	3.0%
DR4/seed43	250000	Matched_DR4	NO_REACH	66	66.0%
DR4/seed43	250000	Matched_DR4	REACH_BUT_NO_GRASP	9	9.0%
DR4/seed43	250000	Matched_DR4	GRASP_BUT_NO_LIFT	21	21.0%
DR4/seed43	250000	Matched_DR4	LIFT_BUT_DROPPED	1	1.0%
DR4/seed43	250000	Heldout_Stress	NO_REACH	80	80.0%
DR4/seed43	250000	Heldout_Stress	REACH_BUT_NO_GRASP	3	3.0%
DR4/seed43	250000	Heldout_Stress	GRASP_BUT_NO_LIFT	17	17.0%
DR4/seed43	500000	Nominal_DR0	SUCCESS	96	96.0%
DR4/seed43	500000	Nominal_DR0	NO_REACH	1	1.0%
DR4/seed43	500000	Nominal_DR0	GRASP_BUT_NO_LIFT	3	3.0%
DR4/seed43	500000	Matched_DR4	SUCCESS	98	98.0%
DR4/seed43	500000	Matched_DR4	REACH_BUT_NO_GRASP	1	1.0%
DR4/seed43	500000	Matched_DR4	GRASP_BUT_NO_LIFT	1	1.0%
DR4/seed43	500000	Heldout_Stress	SUCCESS	41	41.0%
DR4/seed43	500000	Heldout_Stress	NO_REACH	7	7.0%
DR4/seed43	500000	Heldout_Stress	REACH_BUT_NO_GRASP	9	9.0%
DR4/seed43	500000	Heldout_Stress	GRASP_BUT_NO_LIFT	43	43.0%

Appendix D: Complete Configuration Snapshots & Phase 15 Ablation Records

D.1 All 30 Repository YAML Configuration Specifications (configs/*.yaml)

Table D.1: Complete YAML Configuration Registry Across All Phases

YAML Path	Top Keys	Max Steps / Task	LR	Gamma	Budget
assembly.yaml	['env', 'rl', 'eval', 'paths']	200	0.001	0.95	10000
assembly_phase15_3.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	120	0.0003	0.98	5000
default.yaml	['sim', 'robot', 'task', 'rl']	-	0.001	0.98	-
multi_sort.yaml	['env', 'rl', 'eval', 'paths']	300	0.001	0.95	10000
multi_sort_phase15.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	300	0.0003	0.98	10000
multi_sort_phase15_4.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	300	0.0003	0.98	10000
multi_sort_phase15_5A.yaml	['env', 'rl', 'paths']	300	0.0003	0.98	20000
multi_sort_phase15_5B.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	80	0.0003	0.98	20000
multi_sort_phase15_5B1.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	80	0.0003	0.98	10000
multi_sort_phase15_5B2.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	80	0.0003	0.98	5000
multi_sort_phase15_5B3.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	80	0.0003	0.98	5000
multi_sort_phase15_5B4.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	80	0.0003	0.98	10000
multi_sort_phase15_5B5.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	80	0.0003	0.98	10000
multi_sort_phase15_5B6.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	80	0.0003	0.98	10000
multi_sort_phase15_5B7.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	80	0.0003	0.98	10000
multi_sort_phase15_5B8.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	80	0.0001	0.98	10000
multi_sort_phase15_dense.yaml	['env', 'rl', 'eval', 'checkpointing', 'paths']	300	0.0003	0.98	10000
phase14_4_config.yaml	['controller', 'evaluation']	-	-	-	-
phase15_1_config.yaml	['experiment', 'checkpoints', 'observation_noise', 'latency', 'unprivileged_contact', 'readiness_thresholds']	-	-	-	-
pick.yaml	['env', 'rl', 'eval', 'paths']	100	0.001	0.95	10000
pick_place_baseline.yaml	['env', 'rl', 'eval', 'paths']	100	0.001	0.95	5000
pick_place_dr_physics.yaml	['env', 'rl', 'eval', 'paths']	100	0.001	0.95	5000
pick_place_dr_pose.yaml	['env', 'rl', 'eval', 'paths']	100	0.001	0.95	5000
pick_place_dr_position.yaml	['env', 'rl', 'eval', 'paths']	100	0.001	0.95	5000
pick_place_dr_position_target.yaml	['env', 'rl', 'eval', 'paths']	100	0.001	0.95	5000
pick_place_heldout.yaml	['env']	100	-	-	-
pick_place_her.yaml	['env', 'rl', 'eval', 'paths']	100	0.001	0.95	10000
pick_place_no_her.yaml	['env', 'rl', 'eval', 'paths']	100	0.001	0.95	10000
reach.yaml	['env', 'rl', 'eval', 'paths']	50	0.001	0.95	10000
robot/panda.yaml	['robot']	-	-	-	-
sort.yaml	['env', 'rl', 'eval', 'paths']	150	0.001	0.95	10000

D.2 Phase 15.2.5B-B8 Chronological Ablation Result File Index

Table D.2: Phase 15.2.5B-B8 Archived JSON Summary Records

Subphase	Artifact File	Recorded JSON Telemetry Summary
5B	phase15_2_5B_training_results.json	mode=smoke20k, eval_elapsed_seconds=70.8
5B1	phase15_2_5B1_training_results.json	eval_elapsed_seconds=138.48
5B2	phase15_2_5B2_training_results.json	eval_elapsed_seconds=107.85
5B3	phase15_2_5B3_100ep_results.json	
5B3	phase15_2_5B3_5k_diag.json	
5B4	phase15_2_5B4_training_results.json	
5B5	phase15_2_5B5_training_results.json	
5B6	phase15_2_5B6_training_results.json	
5B7	phase15_2_5B7_training_results.json	
5B8	phase15_2_5B8_training_results.json	

Appendix E: Core Source Module AST API Reference & Complete 587-File Inventory

E.1 Automated AST Symbol Index Across Source Modules (src/**/*.py)

Table E.1: Complete Source Tree API Index

Module Path	Lines	Top-Level Classes	Functions
src/__init__.py	3	-	-
src/controllers/__init__.py	10	-	-
src/controllers/assembly_controller.py	153	AssemblyStage, ScriptedAssemblyController	-
src/controllers/compliant_assembly_controller.py	110	CompliantImpedanceAssemblyController	-
src/controllers/pregrasp_controller.py	338	PreGraspStage, PreGraspConfig, PreGraspController	query_pybullet_telemetry
src/cross_sim/__init__.py	2	-	-
src/cross_sim/mujoco_panda_env.py	278	MuJoCoPandaPickPlaceEnv	-
src/deployment/__init__.py	2	-	-
src/deployment/hardware_interface.py	173	HardwareUnavailableError, ROS2TopicContract, FrankaState, CartesianDeltaCommand	get_hardware_interface_documentation
src/deployment/runtime.py	334	RuntimeConfig, ObsPreprocessor, CheckpointLoader, PolicyInferenceInterface	-
src/environments/__init__.py	1	-	-
src/environments/assembly_dense_env.py	314	AssemblyRewardConfig, PandaAssemblyDenseEnv	normalize_quaternion, quaternion_geodesic_error, yaw_from_quaternion, wrap_angle
src/environments/assembly_env.py	243	PandaAssemblyEnv	-
src/environments/base_env.py	88	BasePandaEnv	-
src/environments/contact_assembly_env.py	92	ContactState, PandaContactAssemblyEnv	-
src/environments/curriculum_env.py	93	PandaCurriculumEnv	-
src/environments/dense_multi_sort_env.py	616	SortingStage, MultiSortRewardConfig, PandaMultiSortDenseRewardEnv, MultiSortProximityRewardConfig	-
src/environments/dense_pick_place_env.py	108	PandaPickPlaceDenseRewardEnv	-
src/environments/dynamic_pick_place_env.py	140	PandaDynamicPickPlaceEnv	-
src/environments/multi_sort_env.py	292	PandaMultiSortEnv	-
src/environments/pick_env.py	174	PandaPickEnv	-
src/environments/pick_place_env.py	238	PandaPickPlaceEnv	-
src/environments/pregrasp_multi_sort_env.py	411	PreGraspRewardConfig, PandaMultiSortPreGraspEnv	-
src/environments/reach_env.py	177	PandaReachEnv	-
src/environments/single_pregrasp_env.py	372	SinglePreGraspRewardConfig, PandaSinglePreGraspEnv	-
src/environments/sort_env.py	234	PandaSortEnv	-
src/environments/transport_diagnostic_env.py	66	PandaTransportDiagnosticEnv	-
src/hrl/__init__.py	2	-	-

Module Path	Lines	Top-Level Classes	Functions
src/hrl/hrl_agent.py	333	MetaPolicyNet, MetaTwinQNet, SMDPReplayBuffer, HierarchicalManipulatorAgent	build_meta_state, _compute_smdp_option_reward
src/hrl/skills.py	248	SkillActorNet, SkillTerminationNet, SkillLibrary	extract_skill_features, compute_teacher_skill_target, train_skill_library_from_env_samples
src/metrics/__init__.py	1	-	-
src/objects/__init__.py	1	-	-
src/objects/bin.py	37	SortingBin	-
src/objects/cube.py	77	Cube	-
src/objects/parts.py	108	Receptacle, Peg	-
src/objects/targets.py	39	VisualTarget	-
src/perception/__init__.py	9	-	-
src/perception/dynamic_tracker.py	108	TrackerConfig, DynamicObjectTracker	-
src/perception/rgbd_camera.py	106	CameraConfig, SimulatedRGBDCamera	-
src/perception/vision_detector.py	144	VisualDetectionResult, VisionObjectLocalizer	-
src/rl/__init__.py	1	-	-
src/rl/adaptive_controllers.py	623	ControllerConfig, BaseLiftController, BaselineController, AccelerationSmoothingController	get_step_telemetry
src/rl/bc_sac.py	281	BCDecayingSAC	-
src/rl/callbacks.py	45	SuccessRateCallback	-
src/rl/evaluate.py	218	-	parse_args, load_config, make_env, main
src/rl/evaluate_phase11.py	118	-	load_config, run_eval, main
src/rl/sim_to_real_latency_wrapper.py	64	LatencyWrapper	-
src/rl/sim_to_real_observation_wrapper.py	90	ObservationNoiseWrapper	-
src/rl/train.py	366	-	parse_args, load_config, make_env, _validate_config
src/rl/transfer_utils.py	140	-	get_dict_concat_slices, transfer_single_to_multi_sac
src/rl/unprivileged_contact_wrapper.py	268	UnprivilegedAdaptiveController	strip_privileged_telemetry
src/robots/__init__.py	7	-	-
src/robots/base_robot.py	75	BaseRobot	-
src/robots/controller.py	205	PandaController	-
src/robots/demo_panda.py	99	-	run_demo, main
src/robots/panda.py	300	FrankaPanda	-
src/utils/__init__.py	5	-	-
src/utils/config.py	81	ConfigDict	deep_update, load_config
src/utils/randomization.py	117	DomainRandomizer	-
src/utils/simulation.py	135	PyBulletSimulation	-

E.2 Complete Tracked Repository File Manifest (587 Entries)

Table E.2: Master Repository File Inventory (587 Tracked Files)

#	Relative File Path	Category	Lines	Size
1	GSD-STYLE.md	root	274	6,981 B
2	PROJECT_RULES.md	root	316	9,464 B
3	README.md	root	132	10,313 B
4	audit_dr0.py	root	176	6,394 B
5	audit_dr1.py	root	176	6,411 B
6	audit_fast_track_dr1.py	root	89	4,652 B
7	audit_generic.py	root	159	6,201 B
8	audit_phase13_dr1.py	root	297	11,495 B
9	debug_init.py	root	35	1,123 B
10	diagnostic_11_5.py	root	111	3,981 B
11	diagnostic_11_6.py	root	103	3,628 B
12	diagnostic_11_7.py	root	218	8,875 B
13	diagnostic_11_8.py	root	348	14,247 B
14	diagnostic_dr_10k_smoke.py	root	75	2,341 B
15	diagnostic_dr_scripted.py	root	90	3,817 B
16	diagnostic_dr_smoke.py	root	25	769 B
17	diagnostic_env.py	root	16	411 B
18	diagnostic_physics_controller.py	root	56	1,597 B
19	draw_plots_pillow.py	root	83	4,211 B
20	eval_dense.py	root	42	1,389 B
21	generate_fast_track_report.py	root	110	10,574 B
22	generate_report_11_8.py	root	107	5,276 B
23	launch_fast_track.py	root	255	10,006 B
24	launch_phase11.py	root	83	3,076 B
25	launch_phase13.py	root	246	9,891 B
26	model_capabilities.yaml	root	108	2,943 B
27	orchestrate_phase11.py	root	181	7,400 B
28	plot_err.txt	root	0	0 B
29	plot_trajectories.py	root	83	4,355 B
30	pyproject.toml	root	52	1,132 B
31	requirements.txt	root	22	399 B
32	run_phase13_5_diagnostics.py	root	612	23,649 B
33	run_phase14_3.py	root	389	17,697 B
34	run_phase14_4.py	root	322	14,665 B
35	run_phase14_lite.py	root	541	22,201 B
36	run_phase15_1.py	root	441	19,919 B
37	stop_at_500k.py	root	40	1,415 B
38	test_debug.py	root	18	473 B
39	test_dense_reward.py	root	71	2,756 B
40	test_dr1_stats.py	root	54	2,038 B
41	test_svg.py	root	11	262 B
42	train_capacity.py	root	235	8,596 B
43	train_curriculum.py	root	316	11,491 B
44	train_dense.py	root	97	3,544 B
45	train_phase12.py	root	238	8,611 B
46	train_phase13_dr.py	root	613	26,341 B
47	train_transport.py	root	292	10,508 B
48	verify_500k.py	root	53	2,001 B
49	.agent/workflows/add-phase.md	.agent	96	2,033 B

#	Relative File Path	Category	Lines	Size
50	.agent/workflows/add-todo.md	.agent	69	1,629 B
51	.agent/workflows/audit-milestone.md	.agent	107	2,509 B
52	.agent/workflows/check-todos.md	.agent	80	1,987 B
53	.agent/workflows/complete-milestone.md	.agent	192	4,962 B
54	.agent/workflows/debug.md	.agent	262	6,517 B
55	.agent/workflows/discuss-phase.md	.agent	123	2,891 B
56	.agent/workflows/execute.md	.agent	415	11,659 B
57	.agent/workflows/help.md	.agent	96	3,534 B
58	.agent/workflows/insert-phase.md	.agent	109	2,760 B
59	.agent/workflows/install.md	.agent	177	5,613 B
60	.agent/workflows/list-phase-assumptions.md	.agent	82	2,163 B
61	.agent/workflows/map.md	.agent	419	9,849 B
62	.agent/workflows/new-milestone.md	.agent	155	3,674 B
63	.agent/workflows/new-project.md	.agent	368	9,694 B
64	.agent/workflows/pause.md	.agent	176	4,628 B
65	.agent/workflows/plan-milestone-gaps.md	.agent	116	2,475 B
66	.agent/workflows/plan.md	.agent	440	12,630 B
67	.agent/workflows/progress.md	.agent	90	2,545 B
68	.agent/workflows/remove-phase.md	.agent	139	2,995 B
69	.agent/workflows/research-phase.md	.agent	184	3,926 B
70	.agent/workflows/resume.md	.agent	131	3,844 B
71	.agent/workflows/sprint.md	.agent	192	4,623 B
72	.agent/workflows/update.md	.agent	201	5,703 B
73	.agent/workflows/verify.md	.agent	289	7,430 B
74	.agent/workflows/web-search.md	.agent	121	3,204 B
75	.agent/workflows/whats-new.md	.agent	80	2,617 B
76	.agents/agents/gsd-debugger.md	.agents	90	3,571 B
77	.agents/agents/gsd-executor.md	.agents	102	4,321 B
78	.agents/agents/gsd-planner.md	.agents	78	2,891 B
79	.agents/agents/gsd-researcher.md	.agents	94	3,581 B
80	.agents/agents/gsd-verifier.md	.agents	89	3,325 B
81	.agents/skills/codebase-mapper/SKILL.md	.agents	226	4,800 B
82	.agents/skills/context-compressor/SKILL.md	.agents	201	4,646 B
83	.agents/skills/context-fetch/SKILL.md	.agents	184	4,230 B
84	.agents/skills/context-health-monitor/SKILL.md	.agents	105	3,164 B
85	.agents/skills/debugger/SKILL.md	.agents	273	7,531 B
86	.agents/skills/empirical-validation/SKILL.md	.agents	97	2,689 B
87	.agents/skills/executor/SKILL.md	.agents	501	13,328 B
88	.agents/skills/plan-checker/SKILL.md	.agents	283	6,638 B
89	.agents/skills/planner/SKILL.md	.agents	485	13,262 B
90	.agents/skills/subagent-delegation/SKILL.md	.agents	156	7,007 B
91	.agents/skills/token-budget/SKILL.md	.agents	166	3,800 B
92	.agents/skills/verifier/SKILL.md	.agents	421	10,537 B
93	.gemini/GEMINI.md	.gemini	67	1,912 B
94	.gsd/examples/cross-platform.md	.gsd	99	2,229 B
95	.gsd/examples/multi-wave-workflow.md	.gsd	256	5,912 B
96	.gsd/examples/quick-reference.md	.gsd	73	2,486 B
97	.gsd/examples/workflow-example.md	.gsd	139	2,602 B
98	.gsd/templates/DEBUG.md	.gsd	123	3,165 B

#	Relative File Path	Category	Lines	Size
99	.gsd/templates/PLAN.md	.gsd	90	2,077 B
100	.gsd/templates/RESEARCH.md	.gsd	75	1,584 B
101	.gsd/templates/SUMMARY.md	.gsd	103	1,950 B
102	.gsd/templates/UAT.md	.gsd	168	2,976 B
103	.gsd/templates/VERIFICATION.md	.gsd	70	1,540 B
104	.gsd/templates/architecture.md	.gsd	67	2,568 B
105	.gsd/templates/context.md	.gsd	91	1,505 B
106	.gsd/templates/decisions.md	.gsd	37	622 B
107	.gsd/templates/discovery.md	.gsd	122	2,668 B
108	.gsd/templates/journal.md	.gsd	46	851 B
109	.gsd/templates/milestone.md	.gsd	91	1,848 B
110	.gsd/templates/phase-summary.md	.gsd	52	1,111 B
111	.gsd/templates/project.md	.gsd	124	2,058 B
112	.gsd/templates/requirements.md	.gsd	92	2,193 B
113	.gsd/templates/roadmap.md	.gsd	103	1,829 B
114	.gsd/templates/spec.md	.gsd	51	1,232 B
115	.gsd/templates/sprint.md	.gsd	57	1,138 B
116	.gsd/templates/stack.md	.gsd	62	1,431 B
117	.gsd/templates/state.md	.gsd	92	1,567 B
118	.gsd/templates/state_snapshot.md	.gsd	132	2,618 B
119	.gsd/templates/todo.md	.gsd	32	628 B
120	.gsd/templates/token_report.md	.gsd	79	1,389 B
121	.gsd/templates/user-setup.md	.gsd	116	1,989 B
122	adapters/CLAUDE.md	adapters	77	1,965 B
123	adapters/GEMINI.md	adapters	92	2,055 B
124	adapters/GPT_OSS.md	adapters	130	2,902 B
125	configs/assembly.yaml	configs	29	530 B
126	configs/assembly_phase15_3.yaml	configs	53	1,231 B
127	configs/default.yaml	configs	28	636 B
128	configs/multi_sort.yaml	configs	29	517 B
129	configs/multi_sort_phase15.yaml	configs	38	1,119 B
130	configs/multi_sort_phase15_4.yaml	configs	40	944 B
131	configs/multi_sort_phase15_5A.yaml	configs	31	691 B
132	configs/multi_sort_phase15_5B.yaml	configs	39	968 B
133	configs/multi_sort_phase15_5B1.yaml	configs	46	1,230 B
134	configs/multi_sort_phase15_5B2.yaml	configs	51	1,266 B
135	configs/multi_sort_phase15_5B3.yaml	configs	53	1,328 B
136	configs/multi_sort_phase15_5B4.yaml	configs	55	1,389 B
137	configs/multi_sort_phase15_5B5.yaml	configs	55	1,331 B
138	configs/multi_sort_phase15_5B6.yaml	configs	56	1,367 B
139	configs/multi_sort_phase15_5B7.yaml	configs	57	1,389 B
140	configs/multi_sort_phase15_5B8.yaml	configs	57	1,387 B
141	configs/multi_sort_phase15_dense.yaml	configs	41	1,134 B
142	configs/phase14_4_config.yaml	configs	43	1,854 B
143	configs/phase15_1_config.yaml	configs	68	1,891 B
144	configs/pick.yaml	configs	23	369 B
145	configs/pick_place_baseline.yaml	configs	28	494 B
146	configs/pick_place_dr_physics.yaml	configs	42	743 B
147	configs/pick_place_dr_pose.yaml	configs	34	611 B

#	Relative File Path	Category	Lines	Size
148	configs/pick_place_dr_position.yaml	configs	30	534 B
149	configs/pick_place_dr_position_target.yaml	configs	32	589 B
150	configs/pick_place_heldout.yaml	configs	28	625 B
151	configs/pick_place_her.yaml	configs	26	476 B
152	configs/pick_place_no_her.yaml	configs	24	434 B
153	configs/reach.yaml	configs	23	371 B
154	configs/sort.yaml	configs	28	477 B
155	configs/robot/panda.yaml	configs	47	1,719 B
156	docs/architecture.md	docs	33	2,190 B
157	docs/experiments.md	docs	12	1,462 B
158	docs/methodology.md	docs	29	1,986 B
159	docs/phase10_dr_report.md	docs	79	4,687 B
160	docs/phase11_10_curriculum_report.md	docs	60	3,673 B
161	docs/phase11_10b_acquisition_learning_capacity.md	docs	70	4,553 B
162	docs/phase11_5_learning_capacity_diagnostic.md	docs	79	4,759 B
163	docs/phase11_6_dense_reward_diagnostic.md	docs	49	3,157 B
164	docs/phase11_7_transport_stability.md	docs	27	967 B
165	docs/phase11_8_transport_accessibility_diagnostic.md	docs	42	1,852 B
166	docs/phase11_9_transport_only_diagnostic.md	docs	36	1,723 B
167	docs/phase11_batch1_dr0.md	docs	84	4,486 B
168	docs/phase11_batch2_dr1.md	docs	95	4,962 B
169	docs/phase11_batch_dr2.md	docs	6	118 B
170	docs/phase11_batch_dr3.md	docs	6	118 B
171	docs/phase11_batch_dr4.md	docs	6	118 B
172	docs/phase11_final_report.md	docs	53	2,564 B
173	docs/phase11_results.md	docs	61	2,263 B
174	docs/phase12_1m_dense_scaling_report.md	docs	89	6,349 B
175	docs/phase13_5_diagnostic_report.md	docs	209	15,270 B
176	docs/phase13_fast_track_robustness_report.md	docs	93	7,891 B
177	docs/phase14_closure_and_phase15_readiness.md	docs	124	17,306 B
178	docs/phase17_3_hardware_architecture.md	docs	36	2,541 B
179	docs/phase9_diagnostic_report.md	docs	49	4,143 B
180	docs/reach_benchmark.md	docs	31	1,701 B
181	docs/results.md	docs	40	1,876 B
182	experiments/demo_assembly_scripted.py	experiments	120	4,172 B
183	experiments/demo_multi_sort_scripted.py	experiments	101	3,723 B
184	experiments/demo_panda.py	experiments	6	133 B
185	experiments/demo_pick_place_scripted.py	experiments	86	3,157 B
186	experiments/demo_pick_scripted.py	experiments	71	2,330 B
187	experiments/demo_reach_random.py	experiments	31	983 B
188	experiments/demo_reach_scripted.py	experiments	47	1,681 B
189	experiments/demo_sort_random.py	experiments	22	637 B
190	experiments/demo_sort_scripted.py	experiments	92	3,436 B
191	experiments/assembly_phase15_3/seed42/demonstrations/assembly_demo_summary.json	experiments	8	258 B
192	experiments/multi_sort_phase15_5A/phase15_2_5A_diagnostics.json	experiments	8616	248,126 B
193	experiments/multi_sort_phase15_5A/phase15_2_5A_final_report.md	experiments	88	8,941 B
194	experiments/multi_sort_phase15_5A/scripted_50ep_raw.json	experiments	8515	245,406 B
195	experiments/multi_sort_phase15_5B/phase15_2_5B_C_combined_report.md	experiments	62	6,289 B

#	Relative File Path	Category	Lines	Size
196	experiments/multi_sort_phase15_5B/phase15_2_5B_final_report.md	experiments	69	7,162 B
197	experiments/multi_sort_phase15_5B/phase15_2_5B_training_results.json	experiments	6851	160,948 B
198	experiments/multi_sort_phase15_5B1/phase15_2_5B1_action_pathway_audit.md	experiments	38	4,825 B
199	experiments/multi_sort_phase15_5B1/phase15_2_5B1_demonstration_audit.md	experiments	48	4,214 B
200	experiments/multi_sort_phase15_5B1/phase15_2_5B1_final_report.md	experiments	85	9,885 B
201	experiments/multi_sort_phase15_5B1/phase15_2_5B1_reward_audit.md	experiments	65	5,600 B
202	experiments/multi_sort_phase15_5B1/phase15_2_5B1_training_results.json	experiments	12451	346,748 B
203	experiments/multi_sort_phase15_5B1/seed42/demonstrations/demo_dataset_summary.json	experiments	94	1,838 B
204	experiments/multi_sort_phase15_5B2/phase15_2_5B2_training_results.json	experiments	14119	373,038 B
205	experiments/multi_sort_phase15_5B2/seed42/demonstrations/balanced_demo_summary.json	experiments	28	801 B
206	experiments/multi_sort_phase15_5B4/phase15_2_5B4_training_results.json	experiments	6592	206,058 B
207	experiments/multi_sort_phase15_5B5/phase15_2_5B5_training_results.json	experiments	6592	206,161 B
208	experiments/multi_sort_phase15_5B6/phase15_2_5B6_training_results.json	experiments	6592	206,006 B
209	experiments/multi_sort_phase15_5B7/phase15_2_5B7_training_results.json	experiments	6592	205,973 B
210	experiments/multi_sort_phase15_5B8/phase15_2_5B8_training_results.json	experiments	5494	171,593 B
211	experiments/multi_sort_phase15_5C/phase15_2_5C_transfer_audit.md	experiments	48	3,661 B
212	experiments/phase13_dense_dr/dr1/seed42/config_snapshot.yaml	experiments	30	585 B
213	experiments/phase13_dense_dr/dr1/seed42/training_metadata.json	experiments	31	1,187 B
214	experiments/phase13_dense_dr/dr1/seed42/evaluations/eval_checkpoint_1000000.csv	experiments	301	56,035 B
215	experiments/phase13_dense_dr/dr1/seed42/evaluations/eval_checkpoint_250000.csv	experiments	301	58,274 B
216	experiments/phase13_dense_dr/dr1/seed42/evaluations/eval_checkpoint_500000.csv	experiments	301	56,377 B
217	experiments/phase13_dense_dr/dr1/seed42/evaluations/eval_checkpoint_750000.csv	experiments	301	55,725 B
218	experiments/phase13_dense_dr/dr1/seed42/metrics/checkpoint_metrics.csv	experiments	13	1,969 B
219	experiments/phase13_dense_dr/dr1/seed42/metrics/completed.json	experiments	8	188 B
220	experiments/phase13_dense_dr/dr1/seed42/metrics/failure_taxonomy.csv	experiments	85	3,975 B
221	experiments/phase13_dense_dr/dr1/seed42/metrics/resource_check_10k.json	experiments	12	332 B
222	experiments/phase13_dense_dr/dr1/seed42/metrics/training_progress.csv	experiments	401	26,257 B
223	experiments/phase13_dense_dr/dr1/seed43/config_snapshot.yaml	experiments	30	585 B
224	experiments/phase13_dense_dr/dr1/seed43/training_metadata.json	experiments	31	1,187 B
225	experiments/phase13_dense_dr/dr1/seed43/evaluations/eval_checkpoint_1000000.csv	experiments	301	56,481 B
226	experiments/phase13_dense_dr/dr1/seed43/evaluations/eval_checkpoint_250000.csv	experiments	301	58,180 B
227	experiments/phase13_dense_dr/dr1/seed43/evaluations/eval_checkpoint_500000.csv	experiments	301	55,858 B
228	experiments/phase13_dense_dr/dr1/seed43/evaluations/eval_checkpoint_750000.csv	experiments	301	55,982 B
229	experiments/phase13_dense_dr/dr1/seed43/metrics/checkpoint_metrics.csv	experiments	13	1,980 B
230	experiments/phase13_dense_dr/dr1/seed43/metrics/completed.json	experiments	8	189 B
231	experiments/phase13_dense_dr/dr1/seed43/metrics/failure_taxonomy.csv	experiments	85	3,976 B
232	experiments/phase13_dense_dr/dr1/seed43/metrics/resource_check_10k.json	experiments	12	332 B
233	experiments/phase13_dense_dr/dr1/seed43/metrics/training_progress.csv	experiments	401	26,193 B
234	experiments/phase13_dense_dr/dr1/seed44/config_snapshot.yaml	experiments	30	585 B
235	experiments/phase13_dense_dr/dr1/seed44/training_metadata.json	experiments	22	544 B
236	experiments/phase13_dense_dr/dr1/seed44/evaluations/eval_checkpoint_250000.csv	experiments	301	58,029 B

#	Relative File Path	Category	Lines	Size
237	experiments/phase13_dense_dr/dr1/seed44/evaluations/eval_checkpoint_500000.csv	experiments	301	55,659 B
238	experiments/phase13_dense_dr/dr1/seed44/metrics/checkpoint_metrics.csv	experiments	7	1,090 B
239	experiments/phase13_dense_dr/dr1/seed44/metrics/failure_taxonomy.csv	experiments	43	1,997 B
240	experiments/phase13_dense_dr/dr1/seed44/metrics/resource_check_10k.json	experiments	12	329 B
241	experiments/phase13_dense_dr/dr1/seed44/metrics/training_progress.csv	experiments	23	1,494 B
242	experiments/phase13_dense_dr/dr1/seed45/config_snapshot.yaml	experiments	30	585 B
243	experiments/phase13_dense_dr/dr1/seed45/training_metadata.json	experiments	22	544 B
244	experiments/phase13_dense_dr/dr1/seed45/evaluations/eval_checkpoint_250000.csv	experiments	301	56,712 B
245	experiments/phase13_dense_dr/dr1/seed45/evaluations/eval_checkpoint_500000.csv	experiments	301	55,847 B
246	experiments/phase13_dense_dr/dr1/seed45/metrics/checkpoint_metrics.csv	experiments	7	1,102 B
247	experiments/phase13_dense_dr/dr1/seed45/metrics/failure_taxonomy.csv	experiments	43	2,004 B
248	experiments/phase13_dense_dr/dr1/seed45/metrics/resource_check_10k.json	experiments	12	330 B
249	experiments/phase13_dense_dr/dr1/seed45/metrics/training_progress.csv	experiments	17	1,122 B
250	experiments/phase13_dense_dr/dr2/seed42/config_snapshot.yaml	experiments	30	616 B
251	experiments/phase13_dense_dr/dr2/seed42/training_metadata.json	experiments	27	945 B
252	experiments/phase13_dense_dr/dr2/seed42/evaluations/eval_checkpoint_250000.csv	experiments	301	58,090 B
253	experiments/phase13_dense_dr/dr2/seed42/evaluations/eval_checkpoint_500000.csv	experiments	301	55,790 B
254	experiments/phase13_dense_dr/dr2/seed42/metrics/checkpoint_metrics.csv	experiments	7	1,095 B
255	experiments/phase13_dense_dr/dr2/seed42/metrics/completed.json	experiments	8	187 B
256	experiments/phase13_dense_dr/dr2/seed42/metrics/failure_taxonomy.csv	experiments	43	2,001 B
257	experiments/phase13_dense_dr/dr2/seed42/metrics/resource_check_10k.json	experiments	12	332 B
258	experiments/phase13_dense_dr/dr2/seed42/metrics/training_progress.csv	experiments	201	12,990 B
259	experiments/phase13_dense_dr/dr2/seed43/config_snapshot.yaml	experiments	30	616 B
260	experiments/phase13_dense_dr/dr2/seed43/training_metadata.json	experiments	27	945 B
261	experiments/phase13_dense_dr/dr2/seed43/evaluations/eval_checkpoint_250000.csv	experiments	301	58,771 B
262	experiments/phase13_dense_dr/dr2/seed43/evaluations/eval_checkpoint_500000.csv	experiments	301	55,704 B
263	experiments/phase13_dense_dr/dr2/seed43/metrics/checkpoint_metrics.csv	experiments	7	1,106 B
264	experiments/phase13_dense_dr/dr2/seed43/metrics/completed.json	experiments	8	187 B
265	experiments/phase13_dense_dr/dr2/seed43/metrics/failure_taxonomy.csv	experiments	43	2,010 B
266	experiments/phase13_dense_dr/dr2/seed43/metrics/resource_check_10k.json	experiments	12	332 B
267	experiments/phase13_dense_dr/dr2/seed43/metrics/training_progress.csv	experiments	201	13,021 B
268	experiments/phase13_dense_dr/dr4/seed42/config_snapshot.yaml	experiments	40	849 B
269	experiments/phase13_dense_dr/dr4/seed42/training_metadata.json	experiments	27	945 B
270	experiments/phase13_dense_dr/dr4/seed42/evaluations/eval_checkpoint_250000.csv	experiments	301	57,995 B
271	experiments/phase13_dense_dr/dr4/seed42/evaluations/eval_checkpoint_500000.csv	experiments	301	55,878 B
272	experiments/phase13_dense_dr/dr4/seed42/metrics/checkpoint_metrics.csv	experiments	7	1,090 B
273	experiments/phase13_dense_dr/dr4/seed42/metrics/completed.json	experiments	8	187 B
274	experiments/phase13_dense_dr/dr4/seed42/metrics/failure_taxonomy.csv	experiments	43	2,006 B
275	experiments/phase13_dense_dr/dr4/seed42/metrics/resource_check_10k.json	experiments	12	332 B
276	experiments/phase13_dense_dr/dr4/seed42/metrics/training_progress.csv	experiments	201	13,028 B
277	experiments/phase13_dense_dr/dr4/seed43/config_snapshot.yaml	experiments	40	849 B
278	experiments/phase13_dense_dr/dr4/seed43/training_metadata.json	experiments	27	944 B

#	Relative File Path	Category	Lines	Size
279	experiments/phase13_dense_dr/dr4/seed43/evaluations/eval_checkpoint_250000.csv	experiments	301	58,317 B
280	experiments/phase13_dense_dr/dr4/seed43/evaluations/eval_checkpoint_500000.csv	experiments	301	56,376 B
281	experiments/phase13_dense_dr/dr4/seed43/metrics/checkpoint_metrics.csv	experiments	7	1,108 B
282	experiments/phase13_dense_dr/dr4/seed43/metrics/completed.json	experiments	8	186 B
283	experiments/phase13_dense_dr/dr4/seed43/metrics/failure_taxonomy.csv	experiments	43	2,010 B
284	experiments/phase13_dense_dr/dr4/seed43/metrics/resource_check_10k.json	experiments	12	331 B
285	experiments/phase13_dense_dr/dr4/seed43/metrics/training_progress.csv	experiments	201	13,027 B
286	experiments/pick_place/run_experiments.ps1	experiments	18	884 B
287	experiments/pick_place/smoke_results.md	experiments	30	2,234 B
288	experiments/pnp_dr0_seed42/config.yaml	experiments	25	528 B
289	experiments/pnp_dr0_seed42/metadata.json	experiments	8	199 B
290	experiments/pnp_dr0_seed43/config.yaml	experiments	25	528 B
291	experiments/pnp_dr0_seed43/metadata.json	experiments	8	199 B
292	experiments/pnp_dr0_seed44/config.yaml	experiments	25	528 B
293	experiments/pnp_dr0_seed44/metadata.json	experiments	8	199 B
294	experiments/pnp_dr0_seed45/config.yaml	experiments	25	528 B
295	experiments/pnp_dr0_seed45/metadata.json	experiments	8	199 B
296	experiments/pnp_dr0_seed46/config.yaml	experiments	25	528 B
297	experiments/pnp_dr0_seed46/metadata.json	experiments	8	199 B
298	experiments/pnp_dr0_smoke_seed42/config.yaml	experiments	25	538 B
299	experiments/pnp_dr0_smoke_seed42/metadata.json	experiments	7	170 B
300	experiments/pnp_dr1_seed42/config.yaml	experiments	27	570 B
301	experiments/pnp_dr1_seed42/metadata.json	experiments	8	199 B
302	experiments/pnp_dr1_seed43/config.yaml	experiments	27	570 B
303	experiments/pnp_dr1_seed43/metadata.json	experiments	8	199 B
304	experiments/pnp_dr1_seed44/config.yaml	experiments	27	570 B
305	experiments/pnp_dr1_seed44/metadata.json	experiments	8	199 B
306	experiments/pnp_dr1_seed45/config.yaml	experiments	27	570 B
307	experiments/pnp_dr1_seed45/metadata.json	experiments	8	199 B
308	experiments/pnp_dr1_seed46/config.yaml	experiments	27	570 B
309	experiments/pnp_dr1_seed46/metadata.json	experiments	8	199 B
310	experiments/pnp_dr2_seed42/config.yaml	experiments	29	613 B
311	experiments/pnp_dr2_seed42/metadata.json	experiments	8	199 B
312	experiments/pnp_dr2_seed43/config.yaml	experiments	29	613 B
313	experiments/pnp_dr2_seed43/metadata.json	experiments	8	199 B
314	experiments/pnp_dr2_seed44/config.yaml	experiments	29	613 B
315	experiments/pnp_dr2_seed44/metadata.json	experiments	8	199 B
316	experiments/pnp_dr2_seed45/config.yaml	experiments	29	613 B
317	experiments/pnp_dr2_seed45/metadata.json	experiments	8	199 B
318	experiments/pnp_dr2_seed46/config.yaml	experiments	29	613 B
319	experiments/pnp_dr2_seed46/metadata.json	experiments	8	199 B
320	experiments/pnp_dr3_seed42/config.yaml	experiments	31	659 B
321	experiments/pnp_dr3_seed42/metadata.json	experiments	8	199 B
322	experiments/pnp_dr3_seed43/config.yaml	experiments	31	659 B
323	experiments/pnp_dr3_seed43/metadata.json	experiments	8	199 B
324	experiments/pnp_dr3_seed44/config.yaml	experiments	31	659 B
325	experiments/pnp_dr3_seed44/metadata.json	experiments	8	199 B

#	Relative File Path	Category	Lines	Size
326	experiments/pnp_dr3_seed45/config.yaml	experiments	31	659 B
327	experiments/pnp_dr3_seed45/metadata.json	experiments	8	199 B
328	experiments/pnp_dr3_seed46/config.yaml	experiments	31	659 B
329	experiments/pnp_dr3_seed46/metadata.json	experiments	8	199 B
330	experiments/pnp_dr4_seed42/config.yaml	experiments	39	792 B
331	experiments/pnp_dr4_seed42/metadata.json	experiments	8	199 B
332	experiments/pnp_dr4_seed43/config.yaml	experiments	39	792 B
333	experiments/pnp_dr4_seed43/metadata.json	experiments	8	199 B
334	experiments/pnp_dr4_seed44/config.yaml	experiments	39	792 B
335	experiments/pnp_dr4_seed44/metadata.json	experiments	8	199 B
336	experiments/pnp_dr4_seed45/config.yaml	experiments	39	792 B
337	experiments/pnp_dr4_seed45/metadata.json	experiments	8	199 B
338	experiments/pnp_dr4_seed46/config.yaml	experiments	39	792 B
339	experiments/pnp_dr4_seed46/metadata.json	experiments	8	199 B
340	results/phase11_10_curriculum_distribution.csv	results	2	72 B
341	results/phase11_10_curriculum_metrics.csv	results	4	561 B
342	results/phase11_10b_acquisition_metrics.csv	results	8	324 B
343	results/phase11_6_dense_reward_metrics.csv	results	101	10,239 B
344	results/phase11_8_state_initialization_metrics.csv	results	183	24,199 B
345	results/phase11_9_transport_metrics.csv	results	101	7,451 B
346	results/phase11_9_transport_trajectories.csv	results	2001	409,468 B
347	results/phase11_dr0_episode_metrics.csv	results	1001	94,971 B
348	results/phase11_dr0_summary.csv	results	11	1,268 B
349	results/phase11_dr1_episode_metrics.csv	results	1001	94,976 B
350	results/phase11_dr1_seed_summary.csv	results	11	1,125 B
351	results/phase11_dr1_summary.csv	results	11	1,264 B
352	results/phase11_dr2_episode_metrics.csv	results	1001	94,974 B
353	results/phase11_dr2_seed_summary.csv	results	11	1,127 B
354	results/phase11_dr2_summary.csv	results	11	1,266 B
355	results/phase11_dr3_episode_metrics.csv	results	1001	94,954 B
356	results/phase11_dr3_seed_summary.csv	results	11	1,126 B
357	results/phase11_dr3_summary.csv	results	11	1,265 B
358	results/phase11_dr4_episode_metrics.csv	results	1001	94,974 B
359	results/phase11_dr4_seed_summary.csv	results	11	1,126 B
360	results/phase11_dr4_summary.csv	results	11	1,265 B
361	results/phase11_final_episode_metrics.csv	results	5001	492,493 B
362	results/phase11_final_seed_summary.csv	results	51	5,077 B
363	results/phase11_final_summary.csv	results	51	5,736 B
364	results/phase12_1m/checkpoint_metrics.csv	results	9	723 B
365	results/phase12_1m/failure_taxonomy.csv	results	18	451 B
366	results/phase13_5/d1_baseline_reproduction/combined_reproduction.csv	results	201	37,596 B
367	results/phase13_5/d1_baseline_reproduction/seed42_reproduction.csv	results	101	19,141 B
368	results/phase13_5/d1_baseline_reproduction/seed43_reproduction.csv	results	101	18,693 B
369	results/phase13_5/d2_mass_sensitivity/mass_sensitivity.csv	results	401	77,896 B
370	results/phase13_5/d2_mass_sensitivity/mass_summary.csv	results	5	549 B
371	results/phase13_5/d3_friction_sensitivity/friction_sensitivity.csv	results	501	97,082 B
372	results/phase13_5/d3_friction_sensitivity/friction_summary.csv	results	6	371 B
373	results/phase13_5/d4_contact_geometry/contact_geometry.csv	results	421	88,390 B
374	results/phase13_5/d4_contact_geometry/contact_geometry_summary.csv	results	8	808 B

#	Relative File Path	Category	Lines	Size
375	results/phase13_5/d5_lift_trajectories/grasp_but_no_lift_sample_1_seed42_ep5002.csv	results	102	40,568 B
376	results/phase13_5/d5_lift_trajectories/grasp_but_no_lift_sample_2_seed42_ep5004.csv	results	102	41,117 B
377	results/phase13_5/d5_lift_trajectories/grasp_but_no_lift_sample_3_seed42_ep5005.csv	results	102	41,561 B
378	results/phase13_5/d5_lift_trajectories/success_sample_1_seed42_ep5001.csv	results	102	44,332 B
379	results/phase13_5/d5_lift_trajectories/success_sample_2_seed42_ep5003.csv	results	102	44,858 B
380	results/phase13_5/d5_lift_trajectories/success_sample_3_seed42_ep5006.csv	results	102	45,588 B
381	results/phase13_dr1/audit_report.json	results	26	794 B
382	results/phase13_dr1/stage_42-43_10k_check.json	results	26	781 B
383	results/phase13_dr1/stage_44-45_10k_check.json	results	26	781 B
384	results/phase13_dr2/stage_10k_check.json	results	26	781 B
385	results/phase13_dr4/stage_10k_check.json	results	26	780 B
386	results/phase13_fast_track/dr1_seed44_45_screening.json	results	222	8,075 B
387	results/phase14_3/confirm_d_seed42.csv	results	301	90,125 B
388	results/phase14_3/confirm_d_seed43.csv	results	301	90,474 B
389	results/phase14_3/confirmation_episodes.csv	results	1201	381,184 B
390	results/phase14_3/confirmation_summary.csv	results	13	1,663 B
391	results/phase14_3/phase14_3_episodes.csv	results	241	72,149 B
392	results/phase14_3/phase14_3_report.md	results	115	10,954 B
393	results/phase14_3/phase14_3_summary.csv	results	13	1,594 B
394	results/phase14_4/confirm_e_seed42.csv	results	301	52,670 B
395	results/phase14_4/confirm_e_seed43.csv	results	301	52,586 B
396	results/phase14_4/confirmation_episodes.csv	results	1801	284,408 B
397	results/phase14_4/confirmation_summary.csv	results	28	3,492 B
398	results/phase14_4/phase14_4_confirmation_results.json	results	599	20,812 B
399	results/phase14_4/phase14_4_final_report.md	results	96	9,966 B
400	results/phase14_4/phase14_4_screening_results.json	results	203	6,977 B
401	results/phase14_4/screening_episodes.csv	results	121	20,841 B
402	results/phase14_lite/controller_comparison.csv	results	28	7,388 B
403	results/phase14_lite/lift_telemetry.csv	results	6001	1,927,178 B
404	results/phase14_lite/lift_telemetry_seed42.csv	results	3001	1,024,109 B
405	results/phase14_lite/lift_telemetry_seed43.csv	results	3001	963,408 B
406	results/phase14_lite/phase14_lite_report.md	results	133	13,367 B
407	results/phase14_lite/seed42_results.csv	results	901	293,812 B
408	results/phase14_lite/seed43_results.csv	results	901	293,084 B
409	results/phase14_lite/screening/controller_comparison.csv	results	28	8,944 B
410	results/phase14_lite/screening/lift_telemetry.csv	results	6001	1,969,259 B
411	results/phase14_lite/screening/seed42_results.csv	results	271	87,590 B
412	results/phase14_lite/screening/seed43_results.csv	results	271	87,481 B
413	results/phase15_1/phase15_1_episodes.csv	results	321	51,021 B
414	results/phase15_1/phase15_1_final_report.md	results	182	21,223 B
415	results/phase15_1/phase15_1_results.json	results	606	21,293 B
416	results/phase15_1/phase15_1_seed42.csv	results	161	25,844 B
417	results/phase15_1/phase15_1_seed43.csv	results	161	25,884 B
418	results/phase15_1/phase15_1_summary.csv	results	25	3,595 B
419	results/phase15_2_1/phase15_2_1_audit_report.md	results	108	12,148 B
420	results/phase15_2_1/phase15_2_1_environment_tests.md	results	50	4,383 B
421	results/phase15_2_1/phase15_2_1_reward_her_audit.md	results	56	6,047 B

#	Relative File Path	Category	Lines	Size
422	results/phase15_2_1/phase15_2_1_scripted_results.json	results	647	16,924 B
423	results/phase15_2_1/phase15_2_training_plan.md	results	85	6,552 B
424	results/phase15_2_2/phase15_2_2_final_report.md	results	65	6,385 B
425	results/phase15_2_2/phase15_2_2_reward_design.md	results	57	5,443 B
426	results/phase15_2_2/phase15_2_2_test_report.md	results	51	4,375 B
427	results/phase15_2_3/phase15_2_3_final_report.md	results	62	5,777 B
428	results/phase15_2_3/phase15_2_3_integration_report.md	results	32	3,154 B
429	results/phase15_2_3/phase15_2_3_results.json	results	3241	85,328 B
430	results/phase15_2_3/phase15_2_3_training_report.md	results	66	5,117 B
431	results/phase15_2_4/phase15_2_4_final_report.md	results	81	7,545 B
432	results/phase15_2_4/phase15_2_4_reward_comparison.md	results	41	3,579 B
433	results/phase15_2_4/phase15_2_4_training_results.json	results	6220	151,255 B
434	results/phase15_2_5A/phase15_2_5A_diagnostics.json	results	8616	248,126 B
435	results/phase15_2_5A/phase15_2_5A_final_report.md	results	88	8,941 B
436	results/phase15_2_5B/phase15_2_5B_C_combined_report.md	results	62	6,289 B
437	results/phase15_2_5B/phase15_2_5B_final_report.md	results	69	7,162 B
438	results/phase15_2_5B/phase15_2_5B_training_results.json	results	6851	160,948 B
439	results/phase15_2_5B1/phase15_2_5B1_action_pathway_audit.md	results	38	4,825 B
440	results/phase15_2_5B1/phase15_2_5B1_demonstration_audit.md	results	48	4,214 B
441	results/phase15_2_5B1/phase15_2_5B1_final_report.md	results	85	9,872 B
442	results/phase15_2_5B1/phase15_2_5B1_reward_audit.md	results	65	5,600 B
443	results/phase15_2_5B1/phase15_2_5B1_training_results.json	results	12451	346,748 B
444	results/phase15_2_5B2/phase15_2_5B2_training_results.json	results	14119	373,038 B
445	results/phase15_2_5B3/phase15_2_5B3_100ep_results.json	results	3438	85,967 B
446	results/phase15_2_5B3/phase15_2_5B3_5k_diag.json	results	4396	137,220 B
447	results/phase15_2_5B3/phase15_2_5B3_final_report.md	results	43	4,583 B
448	results/phase15_2_5B4/phase15_2_5B4_training_results.json	results	6592	206,058 B
449	results/phase15_2_5B5/phase15_2_5B5_training_results.json	results	6592	206,161 B
450	results/phase15_2_5B6/phase15_2_5B6_training_results.json	results	6592	206,006 B
451	results/phase15_2_5B7/phase15_2_5B7_training_results.json	results	6592	205,973 B
452	results/phase15_2_5B8/phase15_2_5B8_training_results.json	results	5494	171,593 B
453	results/phase15_2_5C/phase15_2_5C_transfer_audit.md	results	48	3,661 B
454	results/phase15_3_assembly/phase15_3_assembly_learned_results.json	results	1040	26,867 B
455	results/phase15_3_assembly/phase15_3_assembly_report.md	results	31	3,291 B
456	results/phase15_3_assembly/phase15_3_assembly_scripted_results.json	results	528	13,739 B
457	results/phase15_master/PHASE15_FINAL_TECHNICAL_REPORT.md	results	141	14,201 B
458	results/phase15_master/phase0_audit.md	results	48	3,671 B
459	results/phase15_phase3/phase15_3_robustness_report.md	results	39	3,198 B
460	results/phase15_phase3/phase15_phase3_robustness_results.json	results	367	10,377 B
461	results/phase16_17/PHASE16_FINAL_TECHNICAL_REPORT.md	results	53	5,276 B
462	results/phase16_17/PHASE17_DEPLOYMENT_READINESS_REPORT.md	results	50	4,542 B
463	results/phase16_17/phase16_0_baseline_audit.md	results	25	1,578 B
464	results/phase16_17/phase16_17_experiment_registry.json	results	31	2,452 B
465	results/phase16_17/phase16_17_failure_analysis.md	results	27	3,853 B
466	results/phase16_17/phase16_17_master_charts.py	results	275	11,951 B
467	results/phase16_17/phase16_1_hrl_results.json	results	1474	29,339 B
468	results/phase16_17/phase16_2_vision_results.json	results	58	1,566 B
469	results/phase16_17/phase16_3_dynamic_results.json	results	72	1,968 B
470	results/phase16_17/phase16_4_contact_assembly_results.json	results	49	1,652 B

#	Relative File Path	Category	Lines	Size
471	results/phase16_17/phase16_6_autonomous_results.json	results	17	377 B
472	results/phase16_17/phase16_benchmark_results.json	results	1672	38,372 B
473	results/phase16_17/phase16_completion_matrix.json	results	56	3,009 B
474	results/phase16_17/phase17_completion_matrix.json	results	28	1,716 B
475	results/phase16_17/phase17_cross_simulator_results.json	results	67	1,848 B
476	results/phase16_17/phase17_safety_assessment.json	results	57	1,485 B
477	scripts/eval_phase15_3_assembly.py	scripts	175	7,533 B
478	scripts/eval_phase15_5B1.py	scripts	249	11,797 B
479	scripts/eval_phase15_phase1_b3.py	scripts	131	5,669 B
480	scripts/eval_phase15_phase3_robustness.py	scripts	177	9,470 B
481	scripts/generate_phase15_2_charts.py	scripts	269	12,084 B
482	scripts/generate_phase15_3_assembly_demos.py	scripts	98	3,595 B
483	scripts/generate_phase15_5B1_demos.py	scripts	188	7,804 B
484	scripts/generate_phase15_5B2_balanced_demos.py	scripts	210	8,802 B
485	scripts/generate_phase15_master_charts.py	scripts	362	16,828 B
486	scripts/run_phase16_1_hrl.py	scripts	355	15,446 B
487	scripts/run_phase16_2_and_6_vision_auto.py	scripts	295	13,788 B
488	scripts/run_phase16_3_dynamic.py	scripts	139	5,962 B
489	scripts/run_phase16_4_contact_assembly.py	scripts	138	6,139 B
490	scripts/run_phase17_cross_sim_and_safety.py	scripts	291	13,405 B
491	scripts/run_phase3_complete.py	scripts	80	3,753 B
492	scripts/search_repo.ps1	scripts	95	2,253 B
493	scripts/setup_search.ps1	scripts	97	4,683 B
494	scripts/validate-agents.ps1	scripts	124	5,070 B
495	scripts/validate-all.ps1	scripts	62	2,800 B
496	scripts/validate-encoding.ps1	scripts	68	2,866 B
497	scripts/validate-skills.ps1	scripts	68	2,439 B
498	scripts/validate-templates.ps1	scripts	58	2,307 B
499	scripts/validate-workflows.ps1	scripts	59	2,309 B
500	src/__init__.py	src	3	77 B
501	src/controllers/__init__.py	src	10	251 B
502	src/controllers/assembly_controller.py	src	153	6,761 B
503	src/controllers/compliant_assembly_controller.py	src	110	5,217 B
504	src/controllers/pregrasp_controller.py	src	338	16,684 B
505	src/cross_sim/__init__.py	src	2	76 B
506	src/cross_sim/mujoco_panda_env.py	src	278	13,956 B
507	src/deployment/__init__.py	src	2	93 B
508	src/deployment/hardware_interface.py	src	173	7,626 B
509	src/deployment/runtime.py	src	334	13,786 B
510	src/environments/__init__.py	src	1	64 B
511	src/environments/assembly_dense_env.py	src	314	14,690 B
512	src/environments/assembly_env.py	src	243	9,642 B
513	src/environments/base_env.py	src	88	3,030 B
514	src/environments/contact_assembly_env.py	src	92	3,688 B
515	src/environments/curriculum_env.py	src	93	3,905 B
516	src/environments/dense_multi_sort_env.py	src	616	30,726 B
517	src/environments/dense_pick_place_env.py	src	108	4,861 B
518	src/environments/dynamic_pick_place_env.py	src	140	6,118 B
519	src/environments/multi_sort_env.py	src	292	11,645 B

#	Relative File Path	Category	Lines	Size
520	src/environments/pick_env.py	src	174	6,578 B
521	src/environments/pick_place_env.py	src	238	9,335 B
522	src/environments/pregrasp_multi_sort_env.py	src	411	20,425 B
523	src/environments/reach_env.py	src	177	6,912 B
524	src/environments/single_pregrasp_env.py	src	372	19,019 B
525	src/environments/sort_env.py	src	234	9,167 B
526	src/environments/transport_diagnostic_env.py	src	66	2,738 B
527	src/hrl/__init__.py	src	2	65 B
528	src/hrl/hrl_agent.py	src	333	13,252 B
529	src/hrl/skills.py	src	248	10,110 B
530	src/metrics/__init__.py	src	1	50 B
531	src/objects/__init__.py	src	1	64 B
532	src/objects/bin.py	src	37	1,231 B
533	src/objects/cube.py	src	77	2,538 B
534	src/objects/parts.py	src	108	5,486 B
535	src/objects/targets.py	src	39	1,253 B
536	src/perception/__init__.py	src	9	300 B
537	src/perception/dynamic_tracker.py	src	108	4,576 B
538	src/perception/rgbd_camera.py	src	106	4,571 B
539	src/perception/vision_detector.py	src	144	5,770 B
540	src/rl/__init__.py	src	1	69 B
541	src/rl/adaptive_controllers.py	src	623	30,115 B
542	src/rl/bc_sac.py	src	281	14,238 B
543	src/rl/callbacks.py	src	45	1,833 B
544	src/rl/evaluate.py	src	218	9,341 B
545	src/rl/evaluate_phase11.py	src	118	4,195 B
546	src/rl/sim_to_real_latency_wrapper.py	src	64	2,643 B
547	src/rl/sim_to_real_observation_wrapper.py	src	90	3,691 B
548	src/rl/train.py	src	366	17,067 B
549	src/rl/transfer_utils.py	src	140	6,928 B
550	src/rl/unprivileged_contact_wrapper.py	src	268	12,555 B
551	src/robots/__init__.py	src	7	256 B
552	src/robots/base_robot.py	src	75	2,212 B
553	src/robots/controller.py	src	205	7,081 B
554	src/robots/demo_panda.py	src	99	3,838 B
555	src/robots/panda.py	src	300	12,607 B
556	src/utils/__init__.py	src	5	132 B
557	src/utils/config.py	src	81	2,862 B
558	src/utils/randomization.py	src	117	4,244 B
559	src/utils/simulation.py	src	135	4,863 B
560	tests/test_assembly_dense_and_orientation.py	tests	73	2,849 B
561	tests/test_assembly_env.py	tests	42	1,326 B
562	tests/test_b1_demos_and_warmstart.py	tests	102	4,034 B
563	tests/test_bc_sac.py	tests	78	3,070 B
564	tests/test_config.py	tests	62	2,013 B
565	tests/test_controller.py	tests	115	3,712 B
566	tests/test_dense_multi_sort_env.py	tests	388	17,408 B
567	tests/test_deployment_runtime.py	tests	79	2,855 B
568	tests/test_environment.py	tests	90	2,932 B

#	Relative File Path	Category	Lines	Size
569	tests/test_hrl_phase16.py	tests	59	2,062 B
570	tests/test_imports.py	tests	67	2,000 B
571	tests/test_multi_sort_env.py	tests	43	1,399 B
572	tests/test_panda_robot.py	tests	157	5,874 B
573	tests/test_phase15_integration.py	tests	105	5,784 B
574	tests/test_phase15_wrappers.py	tests	120	5,331 B
575	tests/test_phase16_17_envs.py	tests	61	2,358 B
576	tests/test_phase16_integration_all.py	tests	27	1,301 B
577	tests/test_pick_env.py	tests	42	1,389 B
578	tests/test_pregrasp_controller.py	tests	90	3,396 B
579	tests/test_pregrasp_her_compatibility.py	tests	92	3,574 B
580	tests/test_pregrasp_reward.py	tests	107	5,773 B
581	tests/test_randomization.py	tests	69	2,355 B
582	tests/test_simulation.py	tests	55	1,963 B
583	tests/test_single_pregrasp_and_transfer.py	tests	101	4,155 B
584	tests/test_sort_env.py	tests	40	1,157 B
585	tests/test_vertical_action_pathway.py	tests	121	5,511 B
586	tests/test_vertical_potential_reward_her.py	tests	183	7,496 B
587	tests/test_vision_and_dynamic_phase16.py	tests	55	2,288 B

Appendix F: Phase 11.6 & Phase 11.9 Diagnostic Episode Archive

F.1 Phase 11.6 Initial Dense Reward Episode Evaluations (100 Episodes)

Table F.1: Phase 11.6 Dense Prototype Per-Episode Log (phase11_6_dense_reward_metrics.csv)

Split	Eval Seed	Grasp	Success	Final Dist	Reward	Colls
Nominal	1001	False	False	0.1114 m	-100.0	0
Nominal	1002	False	False	0.1407 m	-100.0	69
Nominal	1003	False	False	0.2982 m	-100.0	0
Nominal	1004	False	False	0.2748 m	-100.0	0
Nominal	1005	False	False	0.2403 m	-100.0	0
Nominal	1006	False	False	0.1557 m	-100.0	0
Nominal	1007	False	False	0.2025 m	-100.0	28
Nominal	1008	False	False	0.3332 m	-100.0	0
Nominal	1009	False	False	0.2669 m	-100.0	0
Nominal	1010	False	False	0.3531 m	-100.0	0
Nominal	1011	False	False	0.2729 m	-100.0	84
Nominal	1012	False	False	0.2584 m	-100.0	0
Nominal	1013	False	False	0.2419 m	-100.0	0
Nominal	1014	False	False	0.2218 m	-100.0	0
Nominal	1015	False	False	0.3127 m	-100.0	0
Nominal	1016	False	False	0.1410 m	-100.0	84
Nominal	1017	False	False	0.3606 m	-100.0	87
Nominal	1018	False	False	0.1818 m	-100.0	20
Nominal	1019	False	False	0.2297 m	-100.0	0
Nominal	1020	False	False	0.4336 m	-100.0	4
Nominal	1021	False	False	0.2124 m	-100.0	0
Nominal	1022	False	False	0.2735 m	-100.0	0
Nominal	1023	False	False	0.1495 m	-100.0	0
Nominal	1024	False	False	0.2492 m	-100.0	0
Nominal	1025	False	False	0.2700 m	-100.0	35
Nominal	1026	False	False	0.1634 m	-100.0	0
Nominal	1027	False	False	0.2852 m	-100.0	0
Nominal	1028	False	False	0.3067 m	-100.0	0
Nominal	1029	False	False	0.2963 m	-100.0	0
Nominal	1030	False	False	0.4667 m	-100.0	29
Nominal	1031	False	False	0.3645 m	-100.0	75
Nominal	1032	False	False	0.1801 m	-100.0	0
Nominal	1033	False	False	0.2178 m	-100.0	86
Nominal	1034	False	False	0.3196 m	-100.0	0
Nominal	1035	False	False	0.2227 m	-100.0	0
Nominal	1036	False	False	0.2895 m	-100.0	0
Nominal	1037	False	False	0.1962 m	-100.0	0
Nominal	1038	False	False	0.2322 m	-100.0	0
Nominal	1039	False	False	0.2117 m	-100.0	0
Nominal	1040	False	False	0.3101 m	-100.0	54
Nominal	1041	False	False	0.1949 m	-100.0	0
Nominal	1042	False	False	0.1876 m	-100.0	0
Nominal	1043	False	False	0.3195 m	-100.0	48
Nominal	1044	False	False	0.3136 m	-100.0	0

Split	Eval Seed	Grasp	Success	Final Dist	Reward	Colls
Nominal	1045	False	False	0.2435 m	-100.0	0
Nominal	1046	False	False	0.2504 m	-100.0	86
Nominal	1047	False	False	0.1872 m	-100.0	0
Nominal	1048	False	False	0.3160 m	-100.0	52
Nominal	1049	False	False	0.1912 m	-100.0	4
Nominal	1050	False	False	0.2119 m	-100.0	2
Nominal	1051	False	False	0.3117 m	-100.0	0
Nominal	1052	False	False	0.3020 m	-100.0	0
Nominal	1053	False	False	0.3233 m	-100.0	84
Nominal	1054	False	False	0.2607 m	-100.0	0
Nominal	1055	False	False	0.1587 m	-100.0	0
Nominal	1056	False	False	0.3227 m	-100.0	0
Nominal	1057	False	False	0.1974 m	-100.0	18
Nominal	1058	False	False	0.3039 m	-100.0	0
Nominal	1059	False	False	0.2314 m	-100.0	84
Nominal	1060	True	False	0.2541 m	-99.0	0
Nominal	1061	False	False	0.2473 m	-100.0	86
Nominal	1062	False	False	0.3462 m	-100.0	46
Nominal	1063	False	False	0.2139 m	-100.0	0
Nominal	1064	False	False	0.2194 m	-100.0	0
Nominal	1065	False	False	0.0850 m	-100.0	0
Nominal	1066	False	False	0.1748 m	-100.0	0
Nominal	1067	False	False	0.1955 m	-100.0	0
Nominal	1068	False	False	0.2453 m	-100.0	57
Nominal	1069	False	False	0.2371 m	-100.0	0
Nominal	1070	False	False	0.1775 m	-100.0	0
Nominal	1071	False	False	0.3738 m	-100.0	87
Nominal	1072	False	False	0.1322 m	-100.0	0
Nominal	1073	False	False	0.2421 m	-100.0	0
Nominal	1074	False	False	0.1139 m	-100.0	0
Nominal	1075	False	False	0.3809 m	-100.0	9
Nominal	1076	False	False	0.2686 m	-100.0	0
Nominal	1077	False	False	0.4429 m	-100.0	0
Nominal	1078	False	False	0.1866 m	-100.0	0
Nominal	1079	False	False	0.2496 m	-100.0	0
Nominal	1080	False	False	0.1657 m	-100.0	0
Nominal	1081	False	False	0.1199 m	-100.0	0
Nominal	1082	False	False	0.1061 m	-100.0	0
Nominal	1083	False	False	0.1879 m	-100.0	0
Nominal	1084	False	False	0.0578 m	-100.0	0
Nominal	1085	False	False	0.1264 m	-100.0	41
Nominal	1086	False	False	0.2846 m	-100.0	0
Nominal	1087	False	False	0.3088 m	-100.0	0
Nominal	1088	False	False	0.3627 m	-100.0	74
Nominal	1089	False	False	0.0968 m	-100.0	0
Nominal	1090	False	False	0.2545 m	-100.0	9
Nominal	1091	False	False	0.3629 m	-100.0	0
Nominal	1092	False	False	0.3018 m	-100.0	0
Nominal	1093	False	False	0.1999 m	-100.0	0

Split	Eval Seed	Grasp	Success	Final Dist	Reward	Colls
Nominal	1094	False	False	0.1640 m	-100.0	76
Nominal	1095	False	False	0.2551 m	-100.0	0
Nominal	1096	False	False	0.1591 m	-100.0	0
Nominal	1097	False	False	0.3195 m	-100.0	0
Nominal	1098	False	False	0.2462 m	-100.0	0
Nominal	1099	False	False	0.1650 m	-100.0	0
Nominal	1100	False	False	0.2178 m	-100.0	19

F.2 Phase 11.9 Isolated Pre-Grasped Transport Episodes (100 Episodes)

Table F.2: Phase 11.9 TransportDiagnosticEnv Per-Episode Log (phase11_9_transport_metrics.csv)

Ep #	Success	Init Dist	Final Dist	Hold Steps	Dropped
1	True	0.0827 m	0.0147 m	101	False
2	True	0.2300 m	0.0172 m	101	False
3	True	0.2611 m	0.0102 m	101	False
4	True	0.2706 m	0.0154 m	101	False
5	True	0.1685 m	0.0138 m	101	False
6	True	0.2618 m	0.0036 m	101	False
7	True	0.1885 m	0.0046 m	101	False
8	True	0.0994 m	0.0036 m	101	False
9	True	0.3410 m	0.0053 m	101	False
10	True	0.1620 m	0.0136 m	101	False
11	True	0.1555 m	0.0023 m	101	False
12	True	0.2482 m	0.0119 m	101	False
13	True	0.3576 m	0.0170 m	101	False
14	True	0.3533 m	0.0160 m	101	False
15	True	0.1038 m	0.0113 m	101	False
16	True	0.2175 m	0.0117 m	101	False
17	True	0.2179 m	0.0053 m	101	False
18	True	0.2659 m	0.0015 m	101	False
19	True	0.2232 m	0.0129 m	101	False
20	True	0.2885 m	0.0072 m	101	False
21	True	0.1320 m	0.0153 m	101	False
22	True	0.2144 m	0.0050 m	101	False
23	True	0.2880 m	0.0096 m	101	False
24	True	0.2343 m	0.0039 m	101	False
25	True	0.3925 m	0.0121 m	101	False
26	True	0.3118 m	0.0199 m	101	False
27	True	0.1384 m	0.0046 m	101	False
28	True	0.1012 m	0.0032 m	101	False
29	True	0.2577 m	0.0061 m	101	False
30	True	0.2230 m	0.0093 m	101	False
31	True	0.3138 m	0.0093 m	101	False
32	True	0.0814 m	0.0069 m	101	False
33	True	0.3149 m	0.0168 m	101	False
34	True	0.2086 m	0.0192 m	101	False
35	True	0.3676 m	0.0064 m	101	False
36	True	0.1314 m	0.0011 m	101	False
37	True	0.1911 m	0.0189 m	101	False

Ep #	Success	Init Dist	Final Dist	Hold Steps	Dropped
38	True	0.3130 m	0.0120 m	101	False
39	True	0.2303 m	0.0051 m	101	False
40	True	0.2889 m	0.0112 m	101	False
41	True	0.2285 m	0.0137 m	101	False
42	True	0.0967 m	0.0172 m	101	False
43	True	0.1932 m	0.0058 m	101	False
44	True	0.0451 m	0.0024 m	101	False
45	True	0.0450 m	0.0162 m	101	False
46	True	0.2545 m	0.0196 m	101	False
47	True	0.4826 m	0.0131 m	101	False
48	True	0.0894 m	0.0204 m	101	False
49	True	0.1861 m	0.0091 m	101	False
50	True	0.2285 m	0.0042 m	101	False
51	True	0.3591 m	0.0065 m	101	False
52	True	0.3407 m	0.0037 m	101	False
53	True	0.1733 m	0.0069 m	101	False
54	True	0.2148 m	0.0028 m	101	False
55	True	0.1344 m	0.0185 m	101	False
56	True	0.1312 m	0.0052 m	101	False
57	True	0.1800 m	0.0112 m	101	False
58	True	0.2346 m	0.0160 m	101	False
59	True	0.1605 m	0.0076 m	101	False
60	True	0.2042 m	0.0056 m	101	False
61	True	0.3752 m	0.0075 m	101	False
62	True	0.1323 m	0.0090 m	101	False
63	True	0.3696 m	0.0044 m	101	False
64	True	0.2959 m	0.0054 m	101	False
65	True	0.3640 m	0.0027 m	101	False
66	True	0.1334 m	0.0085 m	101	False
67	True	0.1570 m	0.0033 m	101	False
68	True	0.1843 m	0.0106 m	101	False
69	True	0.3034 m	0.0108 m	101	False
70	True	0.0477 m	0.0133 m	101	False
71	True	0.1018 m	0.0121 m	101	False
72	True	0.1641 m	0.0127 m	101	False
73	True	0.0858 m	0.0040 m	101	False
74	True	0.4525 m	0.0192 m	101	False
75	True	0.0808 m	0.0126 m	101	False
76	True	0.1575 m	0.0125 m	101	False
77	True	0.3440 m	0.0130 m	101	False
78	True	0.2174 m	0.0068 m	101	False
79	True	0.3096 m	0.0114 m	101	False
80	True	0.0831 m	0.0038 m	101	False
81	True	0.1949 m	0.0101 m	101	False
82	True	0.2600 m	0.0107 m	101	False
83	True	0.1993 m	0.0063 m	101	False
84	True	0.1992 m	0.0061 m	101	False
85	True	0.2679 m	0.0062 m	101	False
86	True	0.1880 m	0.0102 m	101	False

Ep #	Success	Init Dist	Final Dist	Hold Steps	Dropped
87	True	0.1542 m	0.0054 m	101	False
88	True	0.1186 m	0.0036 m	101	False
89	True	0.1871 m	0.0130 m	101	False
90	True	0.2180 m	0.0044 m	101	False
91	True	0.3482 m	0.0049 m	101	False
92	True	0.2497 m	0.0156 m	101	False
93	True	0.1974 m	0.0042 m	101	False
94	True	0.4117 m	0.0052 m	101	False
95	True	0.2057 m	0.0066 m	101	False
96	True	0.0884 m	0.0049 m	101	False
97	True	0.3914 m	0.0063 m	101	False
98	True	0.1549 m	0.0070 m	101	False
99	True	0.1464 m	0.0045 m	101	False
100	True	0.2596 m	0.0031 m	101	False

Appendix G: Phase 15.1 Complete 320-Episode Perturbation Audit Ledger

G.1 All 320 Adversarial Noise, Latency & Force Sensitivity Evaluations

Table G.1: Phase 15.1 Complete Per-Episode Perturbation Outcomes (phase15_1_episodes.csv)

Condition	Train Seed	Eval Seed	Reach	Grasp	Lift	Transport
A0_CLEAN_BASELINE	42	1001	True	True	True	True
A0_CLEAN_BASELINE	42	1002	True	True	True	True
A0_CLEAN_BASELINE	42	1003	True	True	True	True
A0_CLEAN_BASELINE	42	1004	True	True	True	True
A0_CLEAN_BASELINE	42	1005	True	True	True	True
A0_CLEAN_BASELINE	42	1006	True	True	True	True
A0_CLEAN_BASELINE	42	1007	True	True	True	True
A0_CLEAN_BASELINE	42	1008	True	True	True	True
A0_CLEAN_BASELINE	42	1009	True	True	True	True
A0_CLEAN_BASELINE	42	1010	True	True	True	True
A0_CLEAN_BASELINE	42	1011	True	True	True	True
A0_CLEAN_BASELINE	42	1012	True	True	True	True
A0_CLEAN_BASELINE	42	1013	True	True	True	True
A0_CLEAN_BASELINE	42	1014	True	True	False	False
A0_CLEAN_BASELINE	42	1015	True	True	True	True
A0_CLEAN_BASELINE	42	1016	True	True	True	True
A0_CLEAN_BASELINE	42	1017	True	True	True	True
A0_CLEAN_BASELINE	42	1018	True	True	True	True
A0_CLEAN_BASELINE	42	1019	True	True	True	True
A0_CLEAN_BASELINE	42	1020	True	True	True	True
A1_NOISE_3MM	42	1001	True	True	True	True
A1_NOISE_3MM	42	1002	True	True	True	True
A1_NOISE_3MM	42	1003	True	True	True	True
A1_NOISE_3MM	42	1004	True	True	True	True
A1_NOISE_3MM	42	1005	True	True	True	True
A1_NOISE_3MM	42	1006	True	True	True	True
A1_NOISE_3MM	42	1007	True	True	True	True
A1_NOISE_3MM	42	1008	True	True	True	True
A1_NOISE_3MM	42	1009	True	True	True	True
A1_NOISE_3MM	42	1010	True	True	True	True
A1_NOISE_3MM	42	1011	True	True	True	True
A1_NOISE_3MM	42	1012	True	True	True	True
A1_NOISE_3MM	42	1013	True	True	True	True
A1_NOISE_3MM	42	1014	True	True	True	True
A1_NOISE_3MM	42	1015	True	True	True	True
A1_NOISE_3MM	42	1016	True	True	True	True
A1_NOISE_3MM	42	1017	True	True	True	True
A1_NOISE_3MM	42	1018	True	True	True	True
A1_NOISE_3MM	42	1019	True	True	True	True
A1_NOISE_3MM	42	1020	True	True	True	True
A2_NOISE_5MM	42	1001	True	True	True	True
A2_NOISE_5MM	42	1002	True	True	True	True
A2_NOISE_5MM	42	1003	True	True	True	True

Condition	Train Seed	Eval Seed	Reach	Grasp	Lift	Transport
A2_NOISE_5MM	42	1004	True	True	True	True
A2_NOISE_5MM	42	1005	True	True	True	True
A2_NOISE_5MM	42	1006	True	True	True	True
A2_NOISE_5MM	42	1007	True	True	True	True
A2_NOISE_5MM	42	1008	True	True	True	True
A2_NOISE_5MM	42	1009	True	True	True	True
A2_NOISE_5MM	42	1010	True	True	True	True
A2_NOISE_5MM	42	1011	True	True	True	True
A2_NOISE_5MM	42	1012	True	True	True	True
A2_NOISE_5MM	42	1013	True	True	True	True
A2_NOISE_5MM	42	1014	True	True	True	True
A2_NOISE_5MM	42	1015	True	True	True	True
A2_NOISE_5MM	42	1016	True	True	True	True
A2_NOISE_5MM	42	1017	True	True	True	True
A2_NOISE_5MM	42	1018	True	True	True	True
A2_NOISE_5MM	42	1019	True	True	True	True
A2_NOISE_5MM	42	1020	True	True	True	True
B1_OBS_DELAY_1	42	1001	True	True	True	True
B1_OBS_DELAY_1	42	1002	True	True	True	True
B1_OBS_DELAY_1	42	1003	True	True	True	True
B1_OBS_DELAY_1	42	1004	True	True	True	True
B1_OBS_DELAY_1	42	1005	True	True	False	False
B1_OBS_DELAY_1	42	1006	True	True	True	True
B1_OBS_DELAY_1	42	1007	True	True	True	True
B1_OBS_DELAY_1	42	1008	True	True	False	False
B1_OBS_DELAY_1	42	1009	True	True	False	False
B1_OBS_DELAY_1	42	1010	True	True	True	True
B1_OBS_DELAY_1	42	1011	True	True	False	False
B1_OBS_DELAY_1	42	1012	True	True	True	True
B1_OBS_DELAY_1	42	1013	True	True	True	True
B1_OBS_DELAY_1	42	1014	True	True	True	True
B1_OBS_DELAY_1	42	1015	True	True	False	False
B1_OBS_DELAY_1	42	1016	True	False	False	False
B1_OBS_DELAY_1	42	1017	True	True	False	False
B1_OBS_DELAY_1	42	1018	True	True	False	False
B1_OBS_DELAY_1	42	1019	True	True	False	False
B1_OBS_DELAY_1	42	1020	True	True	True	True
B2_OBS_DELAY_2	42	1001	True	False	False	False
B2_OBS_DELAY_2	42	1002	True	True	False	False
B2_OBS_DELAY_2	42	1003	False	False	False	False
B2_OBS_DELAY_2	42	1004	True	False	False	False
B2_OBS_DELAY_2	42	1005	True	False	False	False
B2_OBS_DELAY_2	42	1006	True	True	False	False
B2_OBS_DELAY_2	42	1007	True	True	False	False
B2_OBS_DELAY_2	42	1008	True	False	False	False
B2_OBS_DELAY_2	42	1009	True	True	True	False
B2_OBS_DELAY_2	42	1010	True	False	False	False
B2_OBS_DELAY_2	42	1011	True	True	False	False
B2_OBS_DELAY_2	42	1012	True	False	False	False

Condition	Train Seed	Eval Seed	Reach	Grasp	Lift	Transport
B2_OBS_DELAY_2	42	1013	True	False	False	False
B2_OBS_DELAY_2	42	1014	False	False	False	False
B2_OBS_DELAY_2	42	1015	True	False	False	False
B2_OBS_DELAY_2	42	1016	True	False	False	False
B2_OBS_DELAY_2	42	1017	False	False	False	False
B2_OBS_DELAY_2	42	1018	False	False	False	False
B2_OBS_DELAY_2	42	1019	False	False	False	False
B2_OBS_DELAY_2	42	1020	False	False	False	False
B3_OBS_DELAY_4	42	1001	True	False	False	False
B3_OBS_DELAY_4	42	1002	True	False	False	False
B3_OBS_DELAY_4	42	1003	False	False	False	False
B3_OBS_DELAY_4	42	1004	False	False	False	False
B3_OBS_DELAY_4	42	1005	True	False	False	False
B3_OBS_DELAY_4	42	1006	False	False	False	False
B3_OBS_DELAY_4	42	1007	False	False	False	False
B3_OBS_DELAY_4	42	1008	False	False	False	False
B3_OBS_DELAY_4	42	1009	False	False	False	False
B3_OBS_DELAY_4	42	1010	False	False	False	False
B3_OBS_DELAY_4	42	1011	True	False	False	False
B3_OBS_DELAY_4	42	1012	False	False	False	False
B3_OBS_DELAY_4	42	1013	False	False	False	False
B3_OBS_DELAY_4	42	1014	False	False	False	False
B3_OBS_DELAY_4	42	1015	True	False	False	False
B3_OBS_DELAY_4	42	1016	False	False	False	False
B3_OBS_DELAY_4	42	1017	False	False	False	False
B3_OBS_DELAY_4	42	1018	False	False	False	False
B3_OBS_DELAY_4	42	1019	True	False	False	False
B3_OBS_DELAY_4	42	1020	False	False	False	False
B4_COMBINED_DELAY_1_1	42	1001	True	True	False	False
B4_COMBINED_DELAY_1_1	42	1002	True	True	False	False
B4_COMBINED_DELAY_1_1	42	1003	True	False	False	False
B4_COMBINED_DELAY_1_1	42	1004	True	False	False	False
B4_COMBINED_DELAY_1_1	42	1005	True	False	False	False
B4_COMBINED_DELAY_1_1	42	1006	True	True	False	False
B4_COMBINED_DELAY_1_1	42	1007	True	True	False	False
B4_COMBINED_DELAY_1_1	42	1008	True	False	False	False
B4_COMBINED_DELAY_1_1	42	1009	True	True	False	False
B4_COMBINED_DELAY_1_1	42	1010	True	False	False	False
B4_COMBINED_DELAY_1_1	42	1011	True	True	False	False
B4_COMBINED_DELAY_1_1	42	1012	True	True	False	False
B4_COMBINED_DELAY_1_1	42	1013	True	False	False	False
B4_COMBINED_DELAY_1_1	42	1014	False	False	False	False
B4_COMBINED_DELAY_1_1	42	1015	True	False	False	False
B4_COMBINED_DELAY_1_1	42	1016	True	True	False	False
B4_COMBINED_DELAY_1_1	42	1017	False	False	False	False
B4_COMBINED_DELAY_1_1	42	1018	False	False	False	False
B4_COMBINED_DELAY_1_1	42	1019	False	False	False	False
B4_COMBINED_DELAY_1_1	42	1020	True	False	False	False
C1_UNPRIVILEGED	42	1001	True	True	False	False

Condition	Train Seed	Eval Seed	Reach	Grasp	Lift	Transport
C1_UNPRIVILEGED	42	1002	True	True	True	True
C1_UNPRIVILEGED	42	1003	True	True	False	False
C1_UNPRIVILEGED	42	1004	True	True	False	False
C1_UNPRIVILEGED	42	1005	True	True	False	False
C1_UNPRIVILEGED	42	1006	True	True	False	False
C1_UNPRIVILEGED	42	1007	True	True	True	True
C1_UNPRIVILEGED	42	1008	True	True	False	False
C1_UNPRIVILEGED	42	1009	True	True	True	True
C1_UNPRIVILEGED	42	1010	True	True	False	False
C1_UNPRIVILEGED	42	1011	True	True	False	False
C1_UNPRIVILEGED	42	1012	True	True	True	True
C1_UNPRIVILEGED	42	1013	True	True	True	True
C1_UNPRIVILEGED	42	1014	True	True	False	False
C1_UNPRIVILEGED	42	1015	True	True	False	False
C1_UNPRIVILEGED	42	1016	True	True	True	True
C1_UNPRIVILEGED	42	1017	True	True	False	False
C1_UNPRIVILEGED	42	1018	True	True	True	True
C1_UNPRIVILEGED	42	1019	True	True	True	True
C1_UNPRIVILEGED	42	1020	True	True	True	True
A0_CLEAN_BASELINE	43	1001	True	True	True	True
A0_CLEAN_BASELINE	43	1002	True	True	True	True
A0_CLEAN_BASELINE	43	1003	True	True	True	True
A0_CLEAN_BASELINE	43	1004	True	True	True	True
A0_CLEAN_BASELINE	43	1005	True	True	True	True
A0_CLEAN_BASELINE	43	1006	True	True	True	True
A0_CLEAN_BASELINE	43	1007	True	True	True	True
A0_CLEAN_BASELINE	43	1008	True	True	True	True
A0_CLEAN_BASELINE	43	1009	True	True	True	True
A0_CLEAN_BASELINE	43	1010	True	True	True	True
A0_CLEAN_BASELINE	43	1011	True	True	True	True
A0_CLEAN_BASELINE	43	1012	True	True	True	True
A0_CLEAN_BASELINE	43	1013	True	True	True	True
A0_CLEAN_BASELINE	43	1014	True	True	True	True
A0_CLEAN_BASELINE	43	1015	True	True	True	True
A0_CLEAN_BASELINE	43	1016	True	True	True	True
A0_CLEAN_BASELINE	43	1017	True	True	True	True
A0_CLEAN_BASELINE	43	1018	True	True	True	True
A0_CLEAN_BASELINE	43	1019	True	True	True	True
A0_CLEAN_BASELINE	43	1020	True	True	True	True
A1_NOISE_3MM	43	1001	True	True	True	True
A1_NOISE_3MM	43	1002	True	True	True	True
A1_NOISE_3MM	43	1003	True	True	True	True
A1_NOISE_3MM	43	1004	True	True	True	True
A1_NOISE_3MM	43	1005	True	True	True	True
A1_NOISE_3MM	43	1006	True	True	True	True
A1_NOISE_3MM	43	1007	True	True	True	True
A1_NOISE_3MM	43	1008	True	True	True	True
A1_NOISE_3MM	43	1009	True	True	False	False
A1_NOISE_3MM	43	1010	True	True	True	True

Condition	Train Seed	Eval Seed	Reach	Grasp	Lift	Transport
A1_NOISE_3MM	43	1011	True	True	True	True
A1_NOISE_3MM	43	1012	True	True	True	True
A1_NOISE_3MM	43	1013	True	True	True	True
A1_NOISE_3MM	43	1014	True	True	True	True
A1_NOISE_3MM	43	1015	True	True	True	True
A1_NOISE_3MM	43	1016	True	True	True	True
A1_NOISE_3MM	43	1017	True	True	True	True
A1_NOISE_3MM	43	1018	True	True	True	True
A1_NOISE_3MM	43	1019	True	True	True	True
A1_NOISE_3MM	43	1020	True	True	True	True
A2_NOISE_5MM	43	1001	True	True	True	True
A2_NOISE_5MM	43	1002	True	True	True	True
A2_NOISE_5MM	43	1003	True	True	True	True
A2_NOISE_5MM	43	1004	True	True	False	False
A2_NOISE_5MM	43	1005	True	True	True	True
A2_NOISE_5MM	43	1006	True	True	True	True
A2_NOISE_5MM	43	1007	True	True	True	True
A2_NOISE_5MM	43	1008	True	True	True	True
A2_NOISE_5MM	43	1009	True	True	False	False
A2_NOISE_5MM	43	1010	True	True	True	True
A2_NOISE_5MM	43	1011	True	True	True	True
A2_NOISE_5MM	43	1012	True	True	True	True
A2_NOISE_5MM	43	1013	True	True	True	True
A2_NOISE_5MM	43	1014	True	True	True	True
A2_NOISE_5MM	43	1015	True	True	True	True
A2_NOISE_5MM	43	1016	True	True	True	True
A2_NOISE_5MM	43	1017	True	True	True	True
A2_NOISE_5MM	43	1018	True	True	True	True
A2_NOISE_5MM	43	1019	True	True	True	True
A2_NOISE_5MM	43	1020	True	True	True	True
B1_OBS_DELAY_1	43	1001	True	True	True	True
B1_OBS_DELAY_1	43	1002	True	True	True	True
B1_OBS_DELAY_1	43	1003	True	True	True	True
B1_OBS_DELAY_1	43	1004	True	True	True	True
B1_OBS_DELAY_1	43	1005	True	True	False	False
B1_OBS_DELAY_1	43	1006	True	True	True	True
B1_OBS_DELAY_1	43	1007	True	True	True	True
B1_OBS_DELAY_1	43	1008	True	True	True	True
B1_OBS_DELAY_1	43	1009	True	True	True	True
B1_OBS_DELAY_1	43	1010	True	True	True	True
B1_OBS_DELAY_1	43	1011	True	True	True	True
B1_OBS_DELAY_1	43	1012	True	True	True	True
B1_OBS_DELAY_1	43	1013	True	True	True	True
B1_OBS_DELAY_1	43	1014	True	True	False	False
B1_OBS_DELAY_1	43	1015	True	True	True	True
B1_OBS_DELAY_1	43	1016	True	True	True	True
B1_OBS_DELAY_1	43	1017	True	True	True	True
B1_OBS_DELAY_1	43	1018	True	True	True	True
B1_OBS_DELAY_1	43	1019	True	True	True	True

Condition	Train Seed	Eval Seed	Reach	Grasp	Lift	Transport
B1_OBS_DELAY_1	43	1020	True	True	True	True
B2_OBS_DELAY_2	43	1001	True	True	False	False
B2_OBS_DELAY_2	43	1002	True	False	False	False
B2_OBS_DELAY_2	43	1003	True	False	False	False
B2_OBS_DELAY_2	43	1004	False	False	False	False
B2_OBS_DELAY_2	43	1005	False	False	False	False
B2_OBS_DELAY_2	43	1006	True	True	False	False
B2_OBS_DELAY_2	43	1007	True	False	False	False
B2_OBS_DELAY_2	43	1008	True	False	False	False
B2_OBS_DELAY_2	43	1009	False	False	False	False
B2_OBS_DELAY_2	43	1010	True	False	False	False
B2_OBS_DELAY_2	43	1011	False	False	False	False
B2_OBS_DELAY_2	43	1012	True	True	False	False
B2_OBS_DELAY_2	43	1013	True	False	False	False
B2_OBS_DELAY_2	43	1014	True	True	False	False
B2_OBS_DELAY_2	43	1015	True	False	False	False
B2_OBS_DELAY_2	43	1016	True	False	False	False
B2_OBS_DELAY_2	43	1017	True	False	False	False
B2_OBS_DELAY_2	43	1018	True	True	False	False
B2_OBS_DELAY_2	43	1019	True	True	False	False
B2_OBS_DELAY_2	43	1020	True	True	True	False
B3_OBS_DELAY_4	43	1001	True	False	False	False
B3_OBS_DELAY_4	43	1002	True	False	False	False
B3_OBS_DELAY_4	43	1003	True	False	False	False
B3_OBS_DELAY_4	43	1004	False	False	False	False
B3_OBS_DELAY_4	43	1005	True	False	False	False
B3_OBS_DELAY_4	43	1006	False	False	False	False
B3_OBS_DELAY_4	43	1007	False	False	False	False
B3_OBS_DELAY_4	43	1008	True	False	False	False
B3_OBS_DELAY_4	43	1009	False	False	False	False
B3_OBS_DELAY_4	43	1010	True	False	False	False
B3_OBS_DELAY_4	43	1011	False	False	False	False
B3_OBS_DELAY_4	43	1012	False	False	False	False
B3_OBS_DELAY_4	43	1013	False	False	False	False
B3_OBS_DELAY_4	43	1014	True	True	False	False
B3_OBS_DELAY_4	43	1015	True	False	False	False
B3_OBS_DELAY_4	43	1016	False	False	False	False
B3_OBS_DELAY_4	43	1017	True	False	False	False
B3_OBS_DELAY_4	43	1018	True	True	False	False
B3_OBS_DELAY_4	43	1019	True	False	False	False
B3_OBS_DELAY_4	43	1020	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1001	True	True	False	False
B4_COMBINED_DELAY_1_1	43	1002	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1003	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1004	False	False	False	False
B4_COMBINED_DELAY_1_1	43	1005	False	False	False	False
B4_COMBINED_DELAY_1_1	43	1006	True	True	False	False
B4_COMBINED_DELAY_1_1	43	1007	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1008	True	False	False	False

Condition	Train Seed	Eval Seed	Reach	Grasp	Lift	Transport
B4_COMBINED_DELAY_1_1	43	1009	False	False	False	False
B4_COMBINED_DELAY_1_1	43	1010	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1011	False	False	False	False
B4_COMBINED_DELAY_1_1	43	1012	True	True	False	False
B4_COMBINED_DELAY_1_1	43	1013	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1014	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1015	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1016	True	True	True	False
B4_COMBINED_DELAY_1_1	43	1017	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1018	True	False	False	False
B4_COMBINED_DELAY_1_1	43	1019	True	True	False	False
B4_COMBINED_DELAY_1_1	43	1020	True	True	False	False
C1_UNPRIVILEGED	43	1001	True	True	False	False
C1_UNPRIVILEGED	43	1002	True	True	True	True
C1_UNPRIVILEGED	43	1003	True	True	False	False
C1_UNPRIVILEGED	43	1004	True	True	False	False
C1_UNPRIVILEGED	43	1005	True	True	False	False
C1_UNPRIVILEGED	43	1006	True	True	False	False
C1_UNPRIVILEGED	43	1007	True	True	True	True
C1_UNPRIVILEGED	43	1008	True	True	False	False
C1_UNPRIVILEGED	43	1009	True	True	False	False
C1_UNPRIVILEGED	43	1010	True	True	False	False
C1_UNPRIVILEGED	43	1011	True	True	False	False
C1_UNPRIVILEGED	43	1012	True	True	True	True
C1_UNPRIVILEGED	43	1013	True	True	True	True
C1_UNPRIVILEGED	43	1014	True	True	False	False
C1_UNPRIVILEGED	43	1015	True	True	False	False
C1_UNPRIVILEGED	43	1016	True	True	True	True
C1_UNPRIVILEGED	43	1017	True	True	False	False
C1_UNPRIVILEGED	43	1018	True	True	True	True
C1_UNPRIVILEGED	43	1019	True	True	True	True
C1_UNPRIVILEGED	43	1020	True	True	True	True

Appendix H: Phase 14.4 Held-Out & Confirmation Episode Telemetry Extract

H.1 Phase 14.4 Controller Confirmation Episodes (892 Logged Trials)

Table H.1: Phase 14.4 Individual Episode Physics & Slip Outcomes (892 Rows)

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Nominal_DR0	A_BASELINE	1001	True	True	0.0076	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1002	True	True	0.0168	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1003	True	True	0.0136	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1004	True	True	0.0144	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1005	True	True	0.0259	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1006	True	True	0.0167	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1007	True	True	0.0107	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1008	True	True	0.0145	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1009	True	True	0.0040	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1010	True	True	0.0150	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1011	True	True	0.0049	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1012	True	True	0.0095	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1013	True	True	0.0036	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1014	True	True	0.0147	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1015	True	True	0.0325	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1016	True	True	0.0067	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1017	True	True	0.0200	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1018	True	True	0.0102	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1019	True	True	0.0102	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1020	True	True	0.0227	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1021	True	True	0.0137	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1022	True	True	0.0074	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1023	True	True	0.0164	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1024	True	True	0.0241	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1025	True	True	0.0142	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1026	True	True	0.0148	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1027	True	True	0.0132	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1028	True	True	0.0079	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1029	True	True	0.0067	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1030	True	True	0.0182	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1031	True	True	0.0126	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1032	True	True	0.0236	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1033	True	True	0.0225	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1034	True	True	0.0220	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1035	True	True	0.0283	1	SUCCESS
42	Nominal_DR0	A_BASELINE	1036	True	True	0.0037	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1037	True	True	0.0154	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1038	True	True	0.0118	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1039	True	True	0.0130	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1040	True	True	0.0112	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1041	True	True	0.0041	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1042	True	True	0.0150	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1043	True	True	0.0152	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Nominal_DR0	A_BASELINE	1044	True	True	0.0132	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1045	True	True	0.0165	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1046	True	True	0.0044	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1047	True	True	0.0159	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1048	True	True	0.0247	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1049	True	True	0.0166	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1050	True	True	0.0069	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1051	True	True	0.0185	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1052	True	True	0.0181	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1053	True	True	0.0067	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1054	True	True	0.0089	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1055	True	True	0.0075	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1056	True	True	0.0127	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1057	True	True	0.0234	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1058	True	True	0.0178	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1059	True	True	0.0084	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1060	True	True	0.0285	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1061	True	True	0.0072	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1062	True	True	0.0325	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1063	True	True	0.0088	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1064	True	True	0.0128	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1065	True	True	0.0274	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1066	True	True	0.0157	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1067	True	True	0.0080	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1068	True	True	0.0280	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1069	True	True	0.0059	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1070	True	True	0.0085	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1071	True	True	0.0131	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1072	True	True	0.0270	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1073	True	True	0.0050	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1074	True	True	0.0053	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1075	True	True	0.0214	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1076	True	True	0.0062	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1077	True	True	0.0106	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1078	True	True	0.0177	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1079	True	True	0.0224	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1080	True	True	0.0068	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1081	True	True	0.0086	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1082	True	True	0.0199	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1083	True	True	0.0140	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1084	True	True	0.0281	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1085	True	True	0.0195	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1086	True	True	0.0110	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1087	True	True	0.0039	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1088	True	True	0.0120	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1089	True	True	0.0172	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1090	True	True	0.0177	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1091	True	True	0.0065	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1092	True	True	0.0182	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Nominal_DR0	A_BASELINE	1093	True	True	0.0103	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1094	True	True	0.0165	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1095	True	True	0.0175	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1096	True	True	0.0081	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1097	True	True	0.0125	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1098	True	True	0.0174	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1099	True	True	0.0032	0	SUCCESS
42	Nominal_DR0	A_BASELINE	1100	True	True	0.0111	0	SUCCESS
42	Matched_DR4	A_BASELINE	1001	True	True	0.0136	0	SUCCESS
42	Matched_DR4	A_BASELINE	1002	True	True	0.0180	0	SUCCESS
42	Matched_DR4	A_BASELINE	1003	True	True	0.0121	0	SUCCESS
42	Matched_DR4	A_BASELINE	1004	True	True	0.0034	0	SUCCESS
42	Matched_DR4	A_BASELINE	1005	True	True	0.0090	0	SUCCESS
42	Matched_DR4	A_BASELINE	1006	True	True	0.0136	0	SUCCESS
42	Matched_DR4	A_BASELINE	1007	True	True	0.0136	0	SUCCESS
42	Matched_DR4	A_BASELINE	1008	True	True	0.0177	0	SUCCESS
42	Matched_DR4	A_BASELINE	1009	True	True	0.0067	0	SUCCESS
42	Matched_DR4	A_BASELINE	1010	True	True	0.0072	0	SUCCESS
42	Matched_DR4	A_BASELINE	1011	True	True	0.0329	1	SUCCESS
42	Matched_DR4	A_BASELINE	1012	True	True	0.0126	0	SUCCESS
42	Matched_DR4	A_BASELINE	1013	True	True	0.0182	0	SUCCESS
42	Matched_DR4	A_BASELINE	1014	True	True	0.0126	0	SUCCESS
42	Matched_DR4	A_BASELINE	1015	True	True	0.0175	0	SUCCESS
42	Matched_DR4	A_BASELINE	1016	True	True	0.0175	0	SUCCESS
42	Matched_DR4	A_BASELINE	1017	True	True	0.0147	0	SUCCESS
42	Matched_DR4	A_BASELINE	1018	True	True	0.0087	1	SUCCESS
42	Matched_DR4	A_BASELINE	1019	True	True	0.0120	0	SUCCESS
42	Matched_DR4	A_BASELINE	1020	True	True	0.0038	0	SUCCESS
42	Matched_DR4	A_BASELINE	1021	True	True	0.0132	0	SUCCESS
42	Matched_DR4	A_BASELINE	1022	True	True	0.0050	0	SUCCESS
42	Matched_DR4	A_BASELINE	1023	True	True	0.0086	0	SUCCESS
42	Matched_DR4	A_BASELINE	1024	True	True	0.0130	0	SUCCESS
42	Matched_DR4	A_BASELINE	1025	True	True	0.0179	0	SUCCESS
42	Matched_DR4	A_BASELINE	1026	True	True	0.0188	0	SUCCESS
42	Matched_DR4	A_BASELINE	1027	True	True	0.0174	0	SUCCESS
42	Matched_DR4	A_BASELINE	1028	True	True	0.0057	0	SUCCESS
42	Matched_DR4	A_BASELINE	1029	True	True	0.0368	0	SUCCESS
42	Matched_DR4	A_BASELINE	1030	True	True	0.0169	0	SUCCESS
42	Matched_DR4	A_BASELINE	1031	True	True	0.0251	0	SUCCESS
42	Matched_DR4	A_BASELINE	1032	True	True	0.0164	0	SUCCESS
42	Matched_DR4	A_BASELINE	1033	True	True	0.0339	0	SUCCESS
42	Matched_DR4	A_BASELINE	1034	True	True	0.0221	0	SUCCESS
42	Matched_DR4	A_BASELINE	1035	True	True	0.0324	0	SUCCESS
42	Matched_DR4	A_BASELINE	1036	True	True	0.0265	0	SUCCESS
42	Matched_DR4	A_BASELINE	1037	True	True	0.0168	0	SUCCESS
42	Matched_DR4	A_BASELINE	1038	True	True	0.0172	0	SUCCESS
42	Matched_DR4	A_BASELINE	1039	True	True	0.0062	0	SUCCESS
42	Matched_DR4	A_BASELINE	1040	True	True	0.0220	0	SUCCESS
42	Matched_DR4	A_BASELINE	1041	True	True	0.0094	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Matched_DR4	A_BASELINE	1042	True	True	0.0135	0	SUCCESS
42	Matched_DR4	A_BASELINE	1043	True	True	0.0159	0	SUCCESS
42	Matched_DR4	A_BASELINE	1044	True	True	0.0115	0	SUCCESS
42	Matched_DR4	A_BASELINE	1045	True	True	0.0219	0	SUCCESS
42	Matched_DR4	A_BASELINE	1046	True	True	0.0113	0	SUCCESS
42	Matched_DR4	A_BASELINE	1047	True	True	0.0143	0	SUCCESS
42	Matched_DR4	A_BASELINE	1048	True	True	0.0184	0	SUCCESS
42	Matched_DR4	A_BASELINE	1049	True	True	0.0192	0	SUCCESS
42	Matched_DR4	A_BASELINE	1050	True	True	0.0153	0	SUCCESS
42	Matched_DR4	A_BASELINE	1051	True	True	0.0317	0	SUCCESS
42	Matched_DR4	A_BASELINE	1052	True	True	0.0149	0	SUCCESS
42	Matched_DR4	A_BASELINE	1053	True	True	0.0136	0	SUCCESS
42	Matched_DR4	A_BASELINE	1054	True	True	0.0128	0	SUCCESS
42	Matched_DR4	A_BASELINE	1055	True	True	0.0163	0	SUCCESS
42	Matched_DR4	A_BASELINE	1056	True	True	0.0175	0	SUCCESS
42	Matched_DR4	A_BASELINE	1057	True	True	0.0141	0	SUCCESS
42	Matched_DR4	A_BASELINE	1058	True	True	0.0147	0	SUCCESS
42	Matched_DR4	A_BASELINE	1059	True	True	0.0036	0	SUCCESS
42	Matched_DR4	A_BASELINE	1060	True	True	0.0147	0	SUCCESS
42	Matched_DR4	A_BASELINE	1061	True	True	0.0108	0	SUCCESS
42	Matched_DR4	A_BASELINE	1062	True	True	0.0076	0	SUCCESS
42	Matched_DR4	A_BASELINE	1063	True	True	0.0186	0	SUCCESS
42	Matched_DR4	A_BASELINE	1064	True	True	0.0128	0	SUCCESS
42	Matched_DR4	A_BASELINE	1065	True	True	0.0145	0	SUCCESS
42	Matched_DR4	A_BASELINE	1066	True	True	0.0063	0	SUCCESS
42	Matched_DR4	A_BASELINE	1067	True	True	0.0097	0	SUCCESS
42	Matched_DR4	A_BASELINE	1068	True	True	0.0076	0	SUCCESS
42	Matched_DR4	A_BASELINE	1069	True	True	0.0039	0	SUCCESS
42	Matched_DR4	A_BASELINE	1070	True	True	0.0149	0	SUCCESS
42	Matched_DR4	A_BASELINE	1071	True	True	0.0222	0	SUCCESS
42	Matched_DR4	A_BASELINE	1072	True	True	0.0226	0	SUCCESS
42	Matched_DR4	A_BASELINE	1073	True	True	0.0299	0	SUCCESS
42	Matched_DR4	A_BASELINE	1074	True	True	0.0170	0	SUCCESS
42	Matched_DR4	A_BASELINE	1075	True	True	0.0183	0	SUCCESS
42	Matched_DR4	A_BASELINE	1076	True	True	0.0196	0	SUCCESS
42	Matched_DR4	A_BASELINE	1077	True	True	0.0167	0	SUCCESS
42	Matched_DR4	A_BASELINE	1078	True	True	0.0106	0	SUCCESS
42	Matched_DR4	A_BASELINE	1079	True	True	0.0117	0	SUCCESS
42	Matched_DR4	A_BASELINE	1080	True	True	0.0163	0	SUCCESS
42	Matched_DR4	A_BASELINE	1081	True	True	0.0105	0	SUCCESS
42	Matched_DR4	A_BASELINE	1082	True	True	0.0185	0	SUCCESS
42	Matched_DR4	A_BASELINE	1083	True	True	0.0195	0	SUCCESS
42	Matched_DR4	A_BASELINE	1084	True	True	0.0115	0	SUCCESS
42	Matched_DR4	A_BASELINE	1085	True	True	0.0151	0	SUCCESS
42	Matched_DR4	A_BASELINE	1086	True	True	0.0148	0	SUCCESS
42	Matched_DR4	A_BASELINE	1087	True	True	0.0196	0	SUCCESS
42	Matched_DR4	A_BASELINE	1088	True	True	0.0183	0	SUCCESS
42	Matched_DR4	A_BASELINE	1089	True	True	0.0109	0	SUCCESS
42	Matched_DR4	A_BASELINE	1090	True	True	0.0153	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Matched_DR4	A_BASELINE	1091	True	True	0.0342	0	SUCCESS
42	Matched_DR4	A_BASELINE	1092	True	True	0.0239	0	SUCCESS
42	Matched_DR4	A_BASELINE	1093	True	True	0.0055	0	SUCCESS
42	Matched_DR4	A_BASELINE	1094	True	True	0.0136	0	SUCCESS
42	Matched_DR4	A_BASELINE	1095	True	True	0.0176	0	SUCCESS
42	Matched_DR4	A_BASELINE	1096	True	True	0.0197	0	SUCCESS
42	Matched_DR4	A_BASELINE	1097	True	True	0.0248	0	SUCCESS
42	Matched_DR4	A_BASELINE	1098	True	True	0.0216	0	SUCCESS
42	Matched_DR4	A_BASELINE	1099	True	True	0.0077	0	SUCCESS
42	Matched_DR4	A_BASELINE	1100	True	True	0.0073	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1001	True	True	0.0111	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1002	True	True	0.0106	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1003	True	False	0.2573	2	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1004	False	False	0.3147	0	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1005	True	True	0.0153	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1006	True	True	0.0225	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1007	True	True	0.0162	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1008	True	True	0.0140	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1009	True	True	0.0228	1	SUCCESS
42	Heldout_DR4	A_BASELINE	1010	True	True	0.0132	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1011	True	True	0.0101	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1012	True	True	0.0229	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1013	True	True	0.0247	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1014	True	False	0.3950	1	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1015	True	True	0.0169	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1016	True	True	0.0161	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1017	False	False	0.3255	1	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1018	True	True	0.0219	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1019	True	True	0.0229	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1020	True	True	0.0231	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1021	True	True	0.0122	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1022	True	True	0.0110	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1023	True	True	0.0174	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1024	True	False	0.3049	1	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1025	True	False	0.2720	0	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1026	True	True	0.0210	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1027	False	False	0.2789	1	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1028	True	True	0.0121	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1029	True	True	0.0094	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1030	True	True	0.0133	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1031	True	True	0.0138	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1032	True	True	0.0134	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1033	True	True	0.0161	2	SUCCESS
42	Heldout_DR4	A_BASELINE	1034	True	True	0.0197	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Heldout_DR4	A_BASELINE	1035	True	True	0.0114	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1036	True	True	0.0149	2	SUCCESS
42	Heldout_DR4	A_BASELINE	1037	False	False	0.2766	2	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1038	True	True	0.0101	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1039	True	True	0.0215	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1040	True	False	0.3011	4	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1041	True	True	0.0169	1	SUCCESS
42	Heldout_DR4	A_BASELINE	1042	True	True	0.0186	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1043	True	True	0.0269	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1044	True	False	0.2607	0	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1045	True	False	0.3376	1	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1046	True	True	0.0215	1	SUCCESS
42	Heldout_DR4	A_BASELINE	1047	True	True	0.0181	1	SUCCESS
42	Heldout_DR4	A_BASELINE	1048	True	True	0.0154	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1049	True	True	0.0121	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1050	False	False	0.2501	0	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1051	True	True	0.0151	2	SUCCESS
42	Heldout_DR4	A_BASELINE	1052	True	True	0.0250	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1053	True	True	0.0227	2	SUCCESS
42	Heldout_DR4	A_BASELINE	1054	True	True	0.2914	1	LIFT_AND_HELD_BUT_
42	Heldout_DR4	A_BASELINE	1055	True	True	0.0131	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1056	True	True	0.0206	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1057	False	False	0.3217	1	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1058	False	False	0.3539	1	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1059	True	True	0.0116	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1060	True	False	0.3452	3	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1061	True	True	0.0232	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1062	True	True	0.0186	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1063	True	True	0.0189	2	SUCCESS
42	Heldout_DR4	A_BASELINE	1064	True	True	0.0135	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1065	True	True	0.0120	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1066	True	True	0.0255	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1067	True	True	0.0142	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1068	True	True	0.0098	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1069	True	True	0.0238	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1070	True	True	0.0181	2	SUCCESS
42	Heldout_DR4	A_BASELINE	1071	True	False	0.3730	1	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1072	True	True	0.0145	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1073	True	True	0.0224	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1074	True	True	0.0231	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1075	True	True	0.0278	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1076	True	True	0.0126	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Heldout_DR4	A_BASELINE	1077	True	True	0.0138	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1078	True	True	0.0167	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1079	True	True	0.0175	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1080	True	True	0.0124	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1081	True	True	0.0136	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1082	True	True	0.0154	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1083	True	True	0.0127	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1084	True	True	0.0129	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1085	True	True	0.0119	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1086	True	True	0.0201	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1087	True	True	0.0123	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1088	True	True	0.0152	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1089	True	True	0.0156	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1090	True	True	0.0136	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1091	True	True	0.0161	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1092	True	True	0.0140	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1093	True	True	0.0136	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1094	True	True	0.0161	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1095	True	True	0.0165	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1096	True	False	0.2356	3	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1097	True	True	0.0234	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1098	True	False	0.2762	1	GRASP_BUT_NO_LIFT
42	Heldout_DR4	A_BASELINE	1099	True	True	0.0227	0	SUCCESS
42	Heldout_DR4	A_BASELINE	1100	True	True	0.0112	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1001	True	True	0.0349	9	SUCCESS
43	Nominal_DR0	A_BASELINE	1002	True	True	0.0198	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1003	True	True	0.0413	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1004	True	True	0.0339	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1005	True	True	0.0447	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1006	True	True	0.0378	3	SUCCESS
43	Nominal_DR0	A_BASELINE	1007	True	True	0.0304	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1008	True	True	0.0429	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1009	True	True	0.0351	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1010	True	True	0.0372	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1011	True	True	0.0176	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1012	True	True	0.0420	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1013	True	True	0.0251	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1014	True	True	0.0198	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1015	True	True	0.0291	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1016	True	True	0.0202	4	SUCCESS
43	Nominal_DR0	A_BASELINE	1017	True	True	0.0253	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1018	True	True	0.0393	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1019	True	True	0.0297	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1020	True	True	0.0211	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1021	True	True	0.0345	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1022	True	True	0.0310	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1023	True	True	0.0250	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
43	Nominal_DR0	A_BASELINE	1024	True	True	0.0411	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1025	True	True	0.0334	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1026	True	True	0.0322	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1027	True	True	0.0432	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1028	True	True	0.0304	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1029	True	True	0.0191	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1030	True	True	0.0452	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1031	True	True	0.0330	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1032	True	True	0.0182	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1033	True	True	0.0331	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1034	True	True	0.0224	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1035	True	True	0.0387	5	SUCCESS
43	Nominal_DR0	A_BASELINE	1036	True	True	0.0186	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1037	True	True	0.0353	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1038	True	True	0.0279	11	SUCCESS
43	Nominal_DR0	A_BASELINE	1039	True	True	0.0398	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1040	True	True	0.0356	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1041	True	True	0.0337	3	SUCCESS
43	Nominal_DR0	A_BASELINE	1042	True	True	0.0291	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1043	True	True	0.0260	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1044	True	False	0.2802	7	GRASP_BUT_NO_LIFT
43	Nominal_DR0	A_BASELINE	1045	True	True	0.0268	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1046	True	True	0.0196	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1047	True	True	0.0224	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1048	True	True	0.0263	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1049	True	True	0.0303	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1050	True	True	0.0347	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1051	True	True	0.0363	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1052	True	True	0.0265	2	SUCCESS
43	Nominal_DR0	A_BASELINE	1053	True	True	0.0243	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1054	True	True	0.0321	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1055	True	True	0.0323	9	SUCCESS
43	Nominal_DR0	A_BASELINE	1056	True	True	0.0388	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1057	True	True	0.0279	11	SUCCESS
43	Nominal_DR0	A_BASELINE	1058	True	True	0.0349	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1059	True	True	0.0368	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1060	True	True	0.0338	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1061	True	True	0.0265	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1062	True	True	0.0366	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1063	True	True	0.0166	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1064	True	True	0.0285	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1065	True	True	0.0403	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1066	True	True	0.0325	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1067	True	True	0.0203	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1068	True	True	0.0317	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1069	True	True	0.0300	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1070	True	True	0.0247	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1071	True	True	0.0216	1	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
43	Nominal_DR0	A_BASELINE	1072	True	True	0.0220	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1073	True	True	0.0227	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1074	True	True	0.0366	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1075	True	True	0.0175	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1076	True	True	0.0392	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1077	True	True	0.0560	0	LIFT_AND_HELD_BUT_
43	Nominal_DR0	A_BASELINE	1078	True	True	0.0287	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1079	True	True	0.0236	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1080	True	True	0.0307	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1081	True	True	0.0229	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1082	True	True	0.0232	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1083	True	True	0.0194	2	SUCCESS
43	Nominal_DR0	A_BASELINE	1084	True	True	0.0433	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1085	True	True	0.0264	3	SUCCESS
43	Nominal_DR0	A_BASELINE	1086	True	True	0.0248	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1087	True	False	0.2156	13	GRASP_BUT_NO_LIFT
43	Nominal_DR0	A_BASELINE	1088	False	False	0.3627	0	NO_REACH
43	Nominal_DR0	A_BASELINE	1089	True	True	0.0365	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1090	True	True	0.0422	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1091	True	True	0.0471	2	SUCCESS
43	Nominal_DR0	A_BASELINE	1092	True	True	0.0329	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1093	True	True	0.0359	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1094	True	True	0.0293	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1095	True	True	0.0202	1	SUCCESS
43	Nominal_DR0	A_BASELINE	1096	True	True	0.0208	2	SUCCESS
43	Nominal_DR0	A_BASELINE	1097	True	False	0.1840	2	GRASP_BUT_NO_LIFT
43	Nominal_DR0	A_BASELINE	1098	True	True	0.0435	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1099	True	True	0.0237	0	SUCCESS
43	Nominal_DR0	A_BASELINE	1100	True	True	0.0319	7	SUCCESS
43	Matched_DR4	A_BASELINE	1001	True	True	0.0219	1	SUCCESS
43	Matched_DR4	A_BASELINE	1002	False	False	0.2736	2	GRASP_BUT_NO_LIFT
43	Matched_DR4	A_BASELINE	1003	True	True	0.0269	2	SUCCESS
43	Matched_DR4	A_BASELINE	1004	True	True	0.0120	1	SUCCESS
43	Matched_DR4	A_BASELINE	1005	True	True	0.0238	0	SUCCESS
43	Matched_DR4	A_BASELINE	1006	True	True	0.0360	0	SUCCESS
43	Matched_DR4	A_BASELINE	1007	True	True	0.0438	0	SUCCESS
43	Matched_DR4	A_BASELINE	1008	True	True	0.0276	0	SUCCESS
43	Matched_DR4	A_BASELINE	1009	True	True	0.0208	0	SUCCESS
43	Matched_DR4	A_BASELINE	1010	True	True	0.0479	0	SUCCESS
43	Matched_DR4	A_BASELINE	1011	True	True	0.0329	2	SUCCESS
43	Matched_DR4	A_BASELINE	1012	True	True	0.0342	0	SUCCESS
43	Matched_DR4	A_BASELINE	1013	True	True	0.0459	0	SUCCESS
43	Matched_DR4	A_BASELINE	1014	True	True	0.0245	0	SUCCESS
43	Matched_DR4	A_BASELINE	1015	True	True	0.0369	0	SUCCESS
43	Matched_DR4	A_BASELINE	1016	True	True	0.0418	0	SUCCESS
43	Matched_DR4	A_BASELINE	1017	True	True	0.0320	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
43	Matched_DR4	A_BASELINE	1018	True	True	0.0236	1	SUCCESS
43	Matched_DR4	A_BASELINE	1019	True	True	0.0336	0	SUCCESS
43	Matched_DR4	A_BASELINE	1020	True	False	0.4742	0	REACH_BUT_NO_GRASP
43	Matched_DR4	A_BASELINE	1021	True	True	0.0402	0	SUCCESS
43	Matched_DR4	A_BASELINE	1022	True	True	0.0258	0	SUCCESS
43	Matched_DR4	A_BASELINE	1023	True	True	0.0338	0	SUCCESS
43	Matched_DR4	A_BASELINE	1024	True	True	0.0230	0	SUCCESS
43	Matched_DR4	A_BASELINE	1025	True	True	0.0428	1	SUCCESS
43	Matched_DR4	A_BASELINE	1026	True	True	0.0285	7	SUCCESS
43	Matched_DR4	A_BASELINE	1027	True	True	0.0175	0	SUCCESS
43	Matched_DR4	A_BASELINE	1028	True	True	0.0269	0	SUCCESS
43	Matched_DR4	A_BASELINE	1029	True	True	0.0290	0	SUCCESS
43	Matched_DR4	A_BASELINE	1030	True	True	0.0276	0	SUCCESS
43	Matched_DR4	A_BASELINE	1031	True	True	0.0314	3	SUCCESS
43	Matched_DR4	A_BASELINE	1032	True	True	0.0305	0	SUCCESS
43	Matched_DR4	A_BASELINE	1033	True	True	0.0326	0	SUCCESS
43	Matched_DR4	A_BASELINE	1034	True	True	0.0271	1	SUCCESS
43	Matched_DR4	A_BASELINE	1035	True	True	0.0334	0	SUCCESS
43	Matched_DR4	A_BASELINE	1036	True	True	0.0283	0	SUCCESS
43	Matched_DR4	A_BASELINE	1037	True	True	0.0377	0	SUCCESS
43	Matched_DR4	A_BASELINE	1038	True	True	0.0224	0	SUCCESS
43	Matched_DR4	A_BASELINE	1039	True	True	0.0251	0	SUCCESS
43	Matched_DR4	A_BASELINE	1040	True	True	0.0338	0	SUCCESS
43	Matched_DR4	A_BASELINE	1041	True	True	0.0367	0	SUCCESS
43	Matched_DR4	A_BASELINE	1042	True	True	0.0215	1	SUCCESS
43	Matched_DR4	A_BASELINE	1043	True	True	0.0217	0	SUCCESS
43	Matched_DR4	A_BASELINE	1044	True	True	0.0254	0	SUCCESS
43	Matched_DR4	A_BASELINE	1045	True	True	0.0466	0	SUCCESS
43	Matched_DR4	A_BASELINE	1046	True	True	0.0295	0	SUCCESS
43	Matched_DR4	A_BASELINE	1047	True	True	0.0182	2	SUCCESS
43	Matched_DR4	A_BASELINE	1048	True	True	0.0490	0	SUCCESS
43	Matched_DR4	A_BASELINE	1049	True	True	0.0209	2	SUCCESS
43	Matched_DR4	A_BASELINE	1050	True	True	0.0297	0	SUCCESS
43	Matched_DR4	A_BASELINE	1051	True	True	0.0433	0	SUCCESS
43	Matched_DR4	A_BASELINE	1052	True	True	0.0265	0	SUCCESS
43	Matched_DR4	A_BASELINE	1053	True	True	0.0244	0	SUCCESS
43	Matched_DR4	A_BASELINE	1054	True	True	0.0225	1	SUCCESS
43	Matched_DR4	A_BASELINE	1055	True	True	0.0331	0	SUCCESS
43	Matched_DR4	A_BASELINE	1056	True	True	0.0217	0	SUCCESS
43	Matched_DR4	A_BASELINE	1057	True	True	0.0222	3	SUCCESS
43	Matched_DR4	A_BASELINE	1058	True	True	0.0353	0	SUCCESS
43	Matched_DR4	A_BASELINE	1059	True	True	0.0309	3	SUCCESS
43	Matched_DR4	A_BASELINE	1060	True	True	0.0375	0	SUCCESS
43	Matched_DR4	A_BASELINE	1061	True	True	0.0414	0	SUCCESS
43	Matched_DR4	A_BASELINE	1062	True	True	0.0259	0	SUCCESS
43	Matched_DR4	A_BASELINE	1063	True	True	0.0447	2	SUCCESS
43	Matched_DR4	A_BASELINE	1064	True	True	0.0431	0	SUCCESS
43	Matched_DR4	A_BASELINE	1065	True	True	0.0310	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
43	Matched_DR4	A_BASELINE	1066	True	True	0.0240	0	SUCCESS
43	Matched_DR4	A_BASELINE	1067	True	True	0.0206	0	SUCCESS
43	Matched_DR4	A_BASELINE	1068	True	True	0.0260	0	SUCCESS
43	Matched_DR4	A_BASELINE	1069	True	True	0.0307	0	SUCCESS
43	Matched_DR4	A_BASELINE	1070	True	True	0.0383	0	SUCCESS
43	Matched_DR4	A_BASELINE	1071	True	True	0.0309	0	SUCCESS
43	Matched_DR4	A_BASELINE	1072	True	True	0.0269	0	SUCCESS
43	Matched_DR4	A_BASELINE	1073	True	True	0.0288	0	SUCCESS
43	Matched_DR4	A_BASELINE	1074	True	True	0.0218	1	SUCCESS
43	Matched_DR4	A_BASELINE	1075	True	True	0.0381	1	SUCCESS
43	Matched_DR4	A_BASELINE	1076	True	True	0.0297	0	SUCCESS
43	Matched_DR4	A_BASELINE	1077	True	True	0.0256	1	SUCCESS
43	Matched_DR4	A_BASELINE	1078	True	True	0.0237	0	SUCCESS
43	Matched_DR4	A_BASELINE	1079	True	True	0.0376	0	SUCCESS
43	Matched_DR4	A_BASELINE	1080	True	True	0.0170	2	SUCCESS
43	Matched_DR4	A_BASELINE	1081	True	True	0.0265	4	SUCCESS
43	Matched_DR4	A_BASELINE	1082	True	True	0.0281	1	SUCCESS
43	Matched_DR4	A_BASELINE	1083	True	True	0.0257	0	SUCCESS
43	Matched_DR4	A_BASELINE	1084	True	True	0.0315	0	SUCCESS
43	Matched_DR4	A_BASELINE	1085	True	True	0.0318	4	SUCCESS
43	Matched_DR4	A_BASELINE	1086	True	True	0.0192	0	SUCCESS
43	Matched_DR4	A_BASELINE	1087	True	True	0.0305	0	SUCCESS
43	Matched_DR4	A_BASELINE	1088	True	True	0.0283	1	SUCCESS
43	Matched_DR4	A_BASELINE	1089	True	True	0.0134	0	SUCCESS
43	Matched_DR4	A_BASELINE	1090	True	True	0.0223	4	SUCCESS
43	Matched_DR4	A_BASELINE	1091	True	True	0.0312	3	SUCCESS
43	Matched_DR4	A_BASELINE	1092	True	True	0.0260	0	SUCCESS
43	Matched_DR4	A_BASELINE	1093	True	True	0.0224	0	SUCCESS
43	Matched_DR4	A_BASELINE	1094	True	True	0.0248	2	SUCCESS
43	Matched_DR4	A_BASELINE	1095	True	True	0.0406	0	SUCCESS
43	Matched_DR4	A_BASELINE	1096	True	True	0.0350	0	SUCCESS
43	Matched_DR4	A_BASELINE	1097	True	True	0.0165	0	SUCCESS
43	Matched_DR4	A_BASELINE	1098	True	True	0.0199	0	SUCCESS
43	Matched_DR4	A_BASELINE	1099	True	True	0.0309	0	SUCCESS
43	Matched_DR4	A_BASELINE	1100	True	True	0.0158	1	SUCCESS
43	Heldout_DR4	A_BASELINE	1001	True	True	0.0271	7	SUCCESS
43	Heldout_DR4	A_BASELINE	1002	True	True	0.0446	8	SUCCESS
43	Heldout_DR4	A_BASELINE	1003	False	False	0.5724	0	REACH_BUT_NO_GRASP
43	Heldout_DR4	A_BASELINE	1004	True	False	0.2096	11	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1005	True	True	0.0273	17	SUCCESS
43	Heldout_DR4	A_BASELINE	1006	True	True	0.0541	8	LIFT_AND_HELD_BUT_
43	Heldout_DR4	A_BASELINE	1007	False	False	0.5916	0	REACH_BUT_NO_GRASP
43	Heldout_DR4	A_BASELINE	1008	True	True	0.0449	6	SUCCESS
43	Heldout_DR4	A_BASELINE	1009	False	False	0.5568	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1010	False	False	0.2626	25	GRASP_BUT_NO_LIFT

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
43	Heldout_DR4	A_BASELINE	1011	True	True	0.0410	7	SUCCESS
43	Heldout_DR4	A_BASELINE	1012	True	False	0.4871	8	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1013	False	False	0.5893	0	NO_REACH
43	Heldout_DR4	A_BASELINE	1014	True	False	0.6044	2	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1015	False	False	0.6214	0	NO_REACH
43	Heldout_DR4	A_BASELINE	1016	True	True	0.0389	4	SUCCESS
43	Heldout_DR4	A_BASELINE	1017	False	False	0.5626	2	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1018	True	True	0.0431	0	SUCCESS
43	Heldout_DR4	A_BASELINE	1019	True	True	0.0525	0	LIFT_AND_HELD_BUT_
43	Heldout_DR4	A_BASELINE	1020	True	True	0.0479	0	SUCCESS
43	Heldout_DR4	A_BASELINE	1021	True	False	0.3914	6	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1022	False	False	0.5767	0	REACH_BUT_NO_GRASP
43	Heldout_DR4	A_BASELINE	1023	True	False	0.2635	23	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1024	False	False	0.5416	2	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1025	False	False	0.5751	2	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1026	False	False	0.2228	23	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1027	True	True	0.0460	2	SUCCESS
43	Heldout_DR4	A_BASELINE	1028	True	True	0.0528	4	LIFT_AND_HELD_BUT_
43	Heldout_DR4	A_BASELINE	1029	True	True	0.0431	9	SUCCESS
43	Heldout_DR4	A_BASELINE	1030	True	True	0.0364	7	SUCCESS
43	Heldout_DR4	A_BASELINE	1031	True	False	0.5637	1	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1032	True	True	0.0529	0	LIFT_AND_HELD_BUT_
43	Heldout_DR4	A_BASELINE	1033	True	False	0.2914	6	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1034	True	True	0.0443	4	SUCCESS
43	Heldout_DR4	A_BASELINE	1035	True	False	0.3738	4	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1036	True	True	0.0483	3	SUCCESS
43	Heldout_DR4	A_BASELINE	1037	False	False	0.5857	1	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1038	True	True	0.0341	14	SUCCESS
43	Heldout_DR4	A_BASELINE	1039	True	False	0.3146	4	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1040	False	False	0.5315	5	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1041	True	True	0.0355	9	SUCCESS
43	Heldout_DR4	A_BASELINE	1042	False	False	0.5664	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1043	False	False	0.5596	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1044	True	True	0.0398	2	SUCCESS
43	Heldout_DR4	A_BASELINE	1045	False	False	0.5348	10	GRASP_BUT_NO_LIFT

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
43	Heldout_DR4	A_BASELINE	1046	True	False	0.6026	3	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1047	False	False	0.5484	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1048	False	False	0.5974	0	NO_REACH
43	Heldout_DR4	A_BASELINE	1049	False	False	0.5633	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1050	False	False	0.5771	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1051	True	False	0.2570	6	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1052	True	True	0.0451	2	SUCCESS
43	Heldout_DR4	A_BASELINE	1053	False	False	0.5721	0	NO_REACH
43	Heldout_DR4	A_BASELINE	1054	False	False	0.5693	1	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1055	False	False	0.6112	0	REACH_BUT_NO_GRASP
43	Heldout_DR4	A_BASELINE	1056	True	True	0.0478	3	SUCCESS
43	Heldout_DR4	A_BASELINE	1057	False	False	0.5499	1	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1058	True	True	0.0377	22	SUCCESS
43	Heldout_DR4	A_BASELINE	1059	True	True	0.0451	6	SUCCESS
43	Heldout_DR4	A_BASELINE	1060	False	False	0.5852	22	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1061	True	False	0.2804	23	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1062	True	True	0.0474	0	SUCCESS
43	Heldout_DR4	A_BASELINE	1063	True	False	0.5591	21	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1064	False	False	0.6130	0	REACH_BUT_NO_GRASP
43	Heldout_DR4	A_BASELINE	1065	False	False	0.5815	0	REACH_BUT_NO_GRASP
43	Heldout_DR4	A_BASELINE	1066	False	False	0.5726	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1067	True	False	0.3489	5	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1068	False	False	0.5940	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1069	True	True	0.0486	0	SUCCESS
43	Heldout_DR4	A_BASELINE	1070	True	True	0.0541	5	LIFT_AND_HELD_BUT_
43	Heldout_DR4	A_BASELINE	1071	True	True	0.0507	5	LIFT_AND_HELD_BUT_
43	Heldout_DR4	A_BASELINE	1072	True	False	0.2625	17	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1073	True	True	0.0451	2	SUCCESS
43	Heldout_DR4	A_BASELINE	1074	True	True	0.0465	0	SUCCESS
43	Heldout_DR4	A_BASELINE	1075	True	True	0.0510	6	LIFT_AND_HELD_BUT_
43	Heldout_DR4	A_BASELINE	1076	True	False	0.2216	3	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1077	True	True	0.0484	2	SUCCESS
43	Heldout_DR4	A_BASELINE	1078	True	True	0.0346	5	SUCCESS
43	Heldout_DR4	A_BASELINE	1079	False	False	0.5763	0	REACH_BUT_NO_GRASP
43	Heldout_DR4	A_BASELINE	1080	True	True	0.0422	4	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
43	Heldout_DR4	A_BASELINE	1081	False	False	0.5702	0	NO_REACH
43	Heldout_DR4	A_BASELINE	1082	True	False	0.3554	5	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1083	False	False	0.5678	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1084	False	False	0.5708	0	REACH_BUT_NO_GRASP
43	Heldout_DR4	A_BASELINE	1085	True	False	0.2499	19	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1086	True	True	0.0458	0	SUCCESS
43	Heldout_DR4	A_BASELINE	1087	True	False	0.2135	24	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1088	False	False	0.5757	0	REACH_BUT_NO_GRASP
43	Heldout_DR4	A_BASELINE	1089	False	False	0.5827	0	NO_REACH
43	Heldout_DR4	A_BASELINE	1090	True	False	0.2204	6	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1091	True	True	0.0405	3	SUCCESS
43	Heldout_DR4	A_BASELINE	1092	True	True	0.0505	2	LIFT_AND_HELD_BUT_
43	Heldout_DR4	A_BASELINE	1093	False	False	0.5678	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1094	True	True	0.0480	0	SUCCESS
43	Heldout_DR4	A_BASELINE	1095	False	False	0.5873	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1096	True	False	0.5091	0	GRASP_BUT_NO_LIFT
43	Heldout_DR4	A_BASELINE	1097	True	True	0.0308	5	SUCCESS
43	Heldout_DR4	A_BASELINE	1098	True	True	0.0426	6	SUCCESS
43	Heldout_DR4	A_BASELINE	1099	True	True	0.0320	10	SUCCESS
43	Heldout_DR4	A_BASELINE	1100	False	False	0.5832	0	NO_REACH
42	Heldout_DR4	D_DUAL_STAGE	1001	True	True	0.0179	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1002	True	True	0.0183	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1003	True	True	0.0202	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1004	True	True	0.0178	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1005	True	True	0.0219	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1006	True	True	0.0205	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1007	True	True	0.0126	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1008	True	True	0.0177	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1009	True	True	0.0123	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1010	True	True	0.0188	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1011	True	True	0.0131	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1012	True	True	0.0171	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1013	True	True	0.0106	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1014	True	False	0.3777	0	GRASP_BUT_NO_LIFT
42	Heldout_DR4	D_DUAL_STAGE	1015	True	True	0.0203	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1016	True	True	0.0245	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1017	True	True	0.0185	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1018	True	True	0.0192	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1019	True	True	0.0133	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1020	True	True	0.0125	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1021	True	True	0.0199	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Heldout_DR4	D_DUAL_STAGE	1022	True	True	0.0135	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1023	True	True	0.0219	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1024	True	True	0.0197	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1025	True	True	0.0167	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1026	True	True	0.0181	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1027	True	True	0.0171	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1028	True	True	0.0171	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1029	True	True	0.0244	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1030	True	True	0.0159	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1031	True	True	0.0115	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1032	True	True	0.0201	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1033	True	True	0.0155	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1034	True	True	0.0157	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1035	True	True	0.0154	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1036	True	True	0.0186	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1037	True	True	0.0169	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1038	True	True	0.0136	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1039	True	True	0.0205	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1040	True	True	0.0181	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1041	True	True	0.0188	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1042	True	True	0.0147	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1043	True	True	0.0132	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1044	True	True	0.0204	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1045	True	True	0.0196	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1046	True	True	0.0186	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1047	True	True	0.0183	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1048	True	True	0.0243	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1049	True	True	0.0136	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1050	True	True	0.0216	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1051	True	True	0.0142	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1052	True	True	0.0127	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1053	True	True	0.0215	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1054	True	False	0.3711	0	GRASP_BUT_NO_LIFT
42	Heldout_DR4	D_DUAL_STAGE	1055	True	True	0.0156	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1056	True	True	0.0131	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1057	True	True	0.0197	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1058	True	True	0.0210	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1059	True	True	0.0216	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1060	True	True	0.0129	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1061	True	True	0.0173	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1062	True	True	0.0209	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1063	True	True	0.0209	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1064	True	True	0.0173	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1065	True	True	0.0164	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1066	True	True	0.0186	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1067	True	True	0.0142	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1068	True	True	0.0200	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1069	True	True	0.0133	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Heldout_DR4	D_DUAL_STAGE	1070	True	True	0.0197	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1071	True	True	0.0166	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1072	True	True	0.0189	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1073	True	True	0.0137	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1074	True	True	0.0127	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1075	True	True	0.0242	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1076	True	True	0.0179	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1077	True	True	0.0207	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1078	True	True	0.0154	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1079	True	True	0.0240	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1080	True	True	0.0122	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1081	True	True	0.0147	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1082	True	True	0.0154	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1083	True	True	0.0158	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1084	True	True	0.0144	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1085	True	True	0.0192	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1086	True	True	0.0124	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1087	True	True	0.0181	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1088	True	True	0.0195	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1089	True	True	0.0092	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1090	True	True	0.0174	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1091	True	True	0.0166	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1092	True	True	0.0114	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1093	True	True	0.0246	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1094	True	True	0.0130	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1095	True	True	0.0182	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1096	True	True	0.0150	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1097	True	True	0.0170	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1098	True	True	0.0185	0	SUCCESS
42	Heldout_DR4	D_DUAL_STAGE	1099	True	False	0.3573	0	GRASP_BUT_NO_LIFT
42	Heldout_DR4	D_DUAL_STAGE	1100	True	True	0.0094	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1001	True	True	0.0167	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1002	True	True	0.0168	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1003	True	True	0.0136	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1004	True	True	0.0072	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1005	True	True	0.0123	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1006	True	True	0.0073	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1007	True	True	0.0233	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1008	True	True	0.0118	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1009	True	True	0.0169	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1010	True	True	0.0099	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1011	True	True	0.0049	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1012	True	True	0.0211	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1013	True	True	0.0036	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1014	True	True	0.0116	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1015	True	True	0.0325	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1016	True	True	0.0212	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1017	True	True	0.0078	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Nominal_DR0	D_DUAL_STAGE	1018	True	True	0.0194	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1019	True	True	0.0100	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1020	True	True	0.0227	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1021	True	True	0.0114	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1022	True	True	0.0238	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1023	True	True	0.0055	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1024	True	True	0.0239	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1025	True	True	0.0142	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1026	True	True	0.0148	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1027	True	True	0.0181	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1028	True	True	0.0031	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1029	True	True	0.0067	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1030	True	True	0.0083	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1031	True	True	0.0225	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1032	True	True	0.0236	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1033	True	True	0.0225	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1034	True	True	0.0288	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1035	True	True	0.0321	1	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1036	True	True	0.0082	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1037	True	True	0.0154	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1038	True	True	0.0118	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1039	True	True	0.0263	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1040	True	True	0.0112	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1041	True	True	0.0115	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1042	True	True	0.0156	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1043	True	True	0.0091	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1044	True	True	0.0138	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1045	True	True	0.0165	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1046	True	True	0.0169	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1047	True	True	0.0159	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1048	True	True	0.0247	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1049	True	True	0.0164	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1050	True	True	0.0069	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1051	True	True	0.0064	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1052	True	True	0.0126	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1053	True	True	0.0160	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1054	True	True	0.0079	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1055	True	True	0.0201	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1056	True	True	0.0109	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1057	True	True	0.0234	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1058	True	True	0.0131	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1059	True	True	0.0071	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1060	True	True	0.0230	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1061	True	True	0.0072	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1062	True	True	0.0339	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1063	True	True	0.0241	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1064	True	True	0.0128	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1065	True	True	0.0183	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1066	True	True	0.0157	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Nominal_DR0	D_DUAL_STAGE	1067	True	True	0.0049	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1068	True	True	0.0287	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1069	True	True	0.0195	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1070	True	True	0.0184	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1071	True	True	0.0136	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1072	True	True	0.0270	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1073	True	True	0.0050	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1074	True	True	0.0085	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1075	True	True	0.0149	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1076	True	True	0.0062	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1077	True	True	0.0228	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1078	True	True	0.0206	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1079	True	True	0.0256	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1080	True	True	0.0184	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1081	True	True	0.0126	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1082	True	True	0.0199	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1083	True	True	0.0140	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1084	True	True	0.0288	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1085	True	True	0.0185	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1086	True	True	0.0110	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1087	True	True	0.0099	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1088	True	True	0.0186	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1089	True	True	0.0260	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1090	True	True	0.0261	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1091	True	True	0.0097	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1092	True	True	0.0150	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1093	True	True	0.0241	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1094	True	True	0.0089	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1095	True	True	0.0159	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1096	True	True	0.0121	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1097	True	True	0.0125	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1098	True	True	0.0193	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1099	True	True	0.0028	0	SUCCESS
42	Nominal_DR0	D_DUAL_STAGE	1100	True	True	0.0111	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1001	True	True	0.0136	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1002	True	True	0.0125	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1003	True	True	0.0121	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1004	True	True	0.0042	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1005	True	True	0.0145	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1006	True	True	0.0083	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1007	True	True	0.0234	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1008	True	True	0.0127	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1009	True	True	0.0122	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1010	True	True	0.0149	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1011	True	True	0.0337	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1012	True	True	0.0247	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1013	True	True	0.0140	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1014	True	True	0.0143	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1015	True	True	0.0126	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Matched_DR4	D_DUAL_STAGE	1016	True	True	0.0122	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1017	True	True	0.0167	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1018	True	True	0.0087	1	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1019	True	True	0.0075	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1020	True	True	0.0132	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1021	True	True	0.0121	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1022	True	True	0.0164	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1023	True	True	0.0086	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1024	True	True	0.0124	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1025	True	True	0.0170	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1026	True	True	0.0172	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1027	True	True	0.0140	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1028	True	True	0.0057	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1029	True	True	0.0371	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1030	True	True	0.0133	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1031	True	True	0.0234	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1032	True	True	0.0071	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1033	True	True	0.0347	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1034	True	True	0.0247	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1035	True	True	0.0316	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1036	True	True	0.0243	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1037	True	True	0.0190	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1038	True	True	0.0173	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1039	True	True	0.0123	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1040	True	True	0.0204	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1041	True	True	0.0189	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1042	True	True	0.0119	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1043	True	True	0.0159	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1044	True	True	0.0134	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1045	True	True	0.0202	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1046	True	True	0.0128	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1047	True	True	0.0143	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1048	True	True	0.0270	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1049	True	True	0.0192	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1050	True	True	0.0147	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1051	True	True	0.0317	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1052	True	True	0.0140	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1053	True	True	0.0118	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1054	True	True	0.0123	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1055	True	True	0.0119	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1056	True	True	0.0075	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1057	True	True	0.0118	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1058	True	True	0.0139	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1059	True	True	0.0036	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1060	True	True	0.0149	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1061	True	True	0.0114	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1062	True	True	0.0076	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1063	True	True	0.0179	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1064	True	True	0.0126	0	SUCCESS

Seed	Regime	Ctrl	Eval Seed	Bilat	Lift	Final Dist	Slips	Failure Type
42	Matched_DR4	D_DUAL_STAGE	1065	True	True	0.0105	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1066	True	True	0.0063	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1067	True	True	0.0077	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1068	True	True	0.0076	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1069	True	True	0.0039	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1070	True	True	0.0149	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1071	True	True	0.0127	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1072	True	True	0.0105	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1073	True	True	0.0302	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1074	True	True	0.0061	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1075	True	True	0.0183	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1076	True	True	0.0168	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1077	True	True	0.0091	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1078	True	True	0.0118	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1079	True	True	0.0117	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1080	True	True	0.0121	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1081	True	True	0.0065	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1082	True	True	0.0190	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1083	True	True	0.0161	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1084	True	True	0.0118	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1085	True	True	0.0213	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1086	True	True	0.0135	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1087	True	True	0.0086	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1088	True	True	0.0083	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1089	True	True	0.0182	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1090	True	True	0.0123	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1091	True	True	0.0336	0	SUCCESS
42	Matched_DR4	D_DUAL_STAGE	1092	True	True	0.0066	0	SUCCESS